

# SHARING COORDINATES: IDENTIFIABILITY FROM OPTIMIZER RESPONSE

Author information pending

## ABSTRACT

Learned parameter sharing writes effective weights as  $\Theta = AB$ , with a simplex router  $A$  and jointly learned bases  $B$ . Static factors are non-unique, as established by simplex-structured matrix factorization theory. We ask when instantaneous optimizer response identifies the current factors. For strictly positive, full-column-rank softmax routers and full-row-rank bases with the same effective product, the complete response under fixed, known positive Euclidean block learning rates and temperature is identical if and only if the factors differ by a common permutation. We give an explicit inverse-stability bound on uniformly nondegenerate domains and extend it to small product perturbations. Under  $D - K \geq L$ ,  $K$  unit-norm queries designed from a noisy observed product control the diameter of the entire feasible factor set, with explicit constants and a stated product-noise threshold. The proof uniformly bounds basis components outside the observed subspace; no candidate-local assumption is needed. Audits of shared modules on nine pretrained language models from 160M to 8B parameters recover factors with error below 0.034 in all 1,080 declared sufficient-rank cases, including noisy observations. They also expose finite-precision failures and unreliable empirical rejection for simpler estimators. Recovered structural decisions do not outperform effective-parameter clustering in the tested setting. The results identify current coordinates relative to a specified optimizer metric. They neither guarantee an efficient recovery algorithm nor recover a historical sharing graph or an AdamW trajectory.

## 1 INTRODUCTION

When several layers share a learned bank of parameters, their effective weights can be written as  $\Theta = AB$ : each row of a router  $A$  selects a convex combination of the rows of a jointly learned basis matrix  $B$ . The router is useful for optimization and compression, but its entries are not uniquely determined by the effective weights. What additional observation would identify the *current* factors, rather than merely select a convenient representation?

Static ambiguity is established simplex-structured matrix factorization (SSMF) theory (Vu Thanh et al., 2023). For example, the invertible transform  $M_s = (1 - s)I + s\mathbf{1}\mathbf{1}^\top/K$ ,  $0 < s < 1$ , preserves the product under  $(A, B) \mapsto (AM_s, M_s^{-1}B)$  while moving each router row toward uniformity. This increases the entropy of nonuniform rows. It preserves coordinate ordering and argmax assignments; other feasible, nonuniform changes of basis can change the argmax partition (Appendix A). An effective model alone therefore need not determine coordinate-based summaries or decisions.

We study an additional observation supplied by a specified optimizer: the instantaneous response of the effective parameters to an arbitrary effective gradient. With  $A = \text{softmax}_{\text{row}}(Z/\tau)$ , fixed temperature, and known positive Euclidean learning rates for  $Z$  and  $B$ , this is a linear map  $\mathcal{R}_{A,B} : G \mapsto -\dot{\Theta}$ . It describes controlled interventions at a fixed checkpoint. The full map contains more information than a single task-gradient response at that checkpoint. It is a controlled observation distinct from a passive training trajectory and fixes the optimizer metric being compared.

Our central result is that this observation resolves the static ambiguity. For strictly positive, full-column-rank routers and full-row-rank bases with the same product, equal response operators imply a common basis permutation, and conversely. The proof recovers a Gram matrix and a family of router covariances from the response, then reduces the remaining ambiguity to simultaneous diagonalization. On uniformly nondegenerate domains, the same chain gives an explicit inverse-stability bound.

The complete-operator premise can also be replaced by finite, imperfect observations. If  $D - K \geq L$ , we design  $K$  unit-norm queries from a noisy observed product. Every pair of factors consistent with the product and response error budgets is close modulo permutation, with an explicit bound on the diameter of the *entire* feasible set. The true basis need not lie in the estimated product subspace, and candidates need not belong to one local solution branch. This is an information guarantee on a declared domain, not an efficient recovery algorithm or a data-derived certificate of that domain’s rank and interior margins.

The contributions are:

1. A model-specific characterization of the complete Euclidean-response stabilizer on a common product fiber, with rank and observation-boundary counterexamples (Section 4.1).
2. An explicit inverse bound for the original factors, a bridge allowing small product perturbations, and a whole-feasible-set diameter bound from noisy products and  $K$  designed response queries (Sections 4.2–4.3).
3. A frozen-protocol audit of shared modules on nine pretrained language models (160M–8B), across three adapter seeds and three gradient domains. Recovery, precision failures, transferred rejection rules, and a matched structural decision test distinguish coordinate identification from downstream usefulness (Section 5).

The response, diagonalization, robust-inverse, and probing tools have substantial precedents. Our contribution is their complete reduction for this observation model and the resulting finite-noisy guarantee, rather than a new general symmetry principle. The identified object is the current optimizer-dependent factorization. Neither the theorem nor the experiments recover the historical process that produced a model.

## 2 RELATED WORK

**Static factorization and coordinate dependence.** Under  $X = \Theta^\top$ ,  $W = B^\top$ , and  $H = A^\top$ , our static model is unconstrained SSF. The uniformizing path of Vu Thanh et al. (2023, Sec. IV-A) is exactly the one above after  $s = \delta/(1 + \delta)$ ; positive static recovery requires additional geometric conditions (Fu et al., 2018; Abdolali & Gillis, 2021). Low-rank factor gauges, ordinary-update dependence on coordinates, and gauge-sensitive merging are also established (Yen et al., 2025; Zheng & Wu, 2026; Gong & Xu, 2026). Our auxiliary ASLoRA audit concerns one documented first-action rule (Hu et al., 2024), not a new gauge or a general merging principle.

**Optimizer response and identifiability.** Jacobian–Gram operators and induced factor flows are standard (Jacot et al., 2018; Marcotte et al., 2026). Varre et al. (2026, Lem. D.2) already derive the single-router squared-covariance softmax flow; Euclidean factor isometries and compatible softmax permutation/shift updates are studied by Singh (2026); Lau & Su (2026). Function-level and conditional-density softmax-MoE identifiability and inverse bounds use different observations (Tran et al., 2025; Nguyen et al., 2023). Our question is a converse at two fixed factorizations of the same finite product: does equal complete response exclude every non-permutation gauge? It is metric-specific; natural-gradient or quotient geometries can instead remove chart dependence (van Oostrum et al., 2023; Mishra et al., 2014).

**The tools behind the inverse.** Orthogonal-to-permutation mechanisms, joint-diagonalizer uniqueness and sensitivity, and robust tensor uniqueness precede our final algebraic steps (Dorrell et al., 2026; Afsari, 2008; Bhaskara et al., 2014). In particular, Bhaskara et al. (2014, Thm. 5) already give a linear robust inverse when their conditioning parameters are fixed. We do not claim that exponent or global robust uniqueness as new. Our quantitative work tracks the stable reduction from response observations to diagonal data, eliminates scaling through the simplex constraints, and controls both original factors with explicit constants. Matrix–vector and structured operator probing are likewise established (Novak et al., 2022; Chiu & Demanet, 2012; Halikias & Townsend, 2024; Amsel et al., 2026). The noisy-product result combines a uniform bound outside the observed subspace with this response inverse; no optimality of  $K$  queries is claimed. Appendix F maps the closest results, the wider sharing literature, and unresolved full-text access limitations.

### 3 MODEL AND OBSERVATION

Let  $L$  be the number of layers,  $K$  the number of shared bases, and  $D$  the number of mixed parameters per layer. Write

$$\Theta = AB, \quad A \in \Delta_K^L, \quad B \in \mathbb{R}^{K \times D}, \quad (1)$$

where  $\Delta_K^L$  denotes nonnegative, row-stochastic  $L \times K$  matrices. The architecture uses the product row  $a_i B$  as parameters. The results do not automatically apply to mixtures of nonlinear expert *outputs*. Two factors are product-equivalent when

$$(A, B) \sim_{\Theta} (A', B') \iff AB = A'B'. \quad (2)$$

We measure factor discrepancy modulo the unavoidable basis labels:

$$d_{\text{perm}}((A, B), (A', B')) = \min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2}. \quad (3)$$

Here  $\mathfrak{S}_K$  is the set of permutation matrices and  $\|\mathcal{S}\|_{F \rightarrow F} = \sup_{\|G\|_F=1} \|\mathcal{S}(G)\|_F$ .

**Controlled Euclidean observation.** Set  $A = \text{softmax}_{\text{row}}(Z/\tau)$  with fixed  $\tau > 0$ . For a loss  $\mathcal{L} = \ell(\Theta)$  and known positive block rates, let

$$\dot{Z} = -\eta_Z \nabla_Z \mathcal{L}, \quad \dot{B} = -\eta_B \nabla_B \mathcal{L}, \quad (4)$$

$$\mathcal{R}_{A,B}(G) = -\dot{\Theta}. \quad (5)$$

where  $G = \nabla_{\Theta} \ell$ . Every  $G \in \mathbb{R}^{L \times D}$  is admissible, using the test loss  $\ell_G(\Theta) = \langle G, \Theta \rangle$ . The comparison fixes  $\tau, \eta_Z, \eta_B$  across charts. Direct factor penalties, adaptive preconditioning, and unknown optimizer state define different observation laws.

**Target and scope.** We identify the *current factors* under this law. An effective-parameter partition instead groups equal rows of  $\Theta$ ; a functional partition groups equal transition functions on a stated input distribution; a historical graph describes the process that generated the model. These are different targets. Likewise, a coordinate-derived action  $r$  is product invariant only if

$$r(A, B) = r(A', B') \quad \text{whenever } (A, B) \sim_{\Theta} (A', B'). \quad (6)$$

Identification relative to an optimizer does not itself imply this property, small action regret, or stable argmax decisions without a coordinate gap. The auxiliary decision experiments evaluate actions in a common canonical model rather than assign them a historical ground truth.

### 4 IDENTIFICATION AND STABILITY FROM OPTIMIZER RESPONSE

#### 4.1 EXACT RESPONSE IDENTIFIES THE CURRENT FACTORS

Writing router and gradient rows as columns  $a_i, g_i$ , and setting  $C_i = \text{Diag}(a_i) - a_i a_i^\top$ , the chain rule gives

$$[\mathcal{R}_{A,B}(G)]_i = \eta_B \sum_{j=1}^L \langle a_i, a_j \rangle g_j + \frac{\eta_Z}{\tau^2} B^\top C_i^2 B g_i. \quad (7)$$

The first term couples layers through the router Gram matrix; the second records how the softmax coordinates move. The question is whether these two forms of information remove the static ambiguity.

**Theorem 1** (Exact Euclidean-response stabilizer). *Let  $A, A' \in \Delta_K^L$  be strictly positive with column rank  $K$ , and let  $B, B' \in \mathbb{R}^{K \times D}$  have row rank  $K$ . Suppose  $AB = A'B'$ , and use the same  $\tau, \eta_Z, \eta_B > 0$  in both parameterizations. Then*

$$\mathcal{R}_{A,B}(G) = \mathcal{R}_{A',B'}(G) \quad \text{for every } G \in \mathbb{R}^{L \times D} \quad (8)$$

*if and only if there is a permutation matrix  $P$  such that*

$$A' = AP, \quad B' = P^\top B. \quad (9)$$

*The corresponding logits remain unidentified under the independent shifts  $z_i \mapsto z_i + c_i \mathbf{1}$ .*

**Proof idea.** Full rank and equal products imply  $A' = AM$ ,  $B' = M^{-1}B$ , with  $M\mathbf{1} = \mathbf{1}$ . Off-diagonal response blocks determine the off-diagonal router Gram entries. A diagonal block is a scalar identity plus  $\lambda B^\top C_i^2 B$ , where  $\lambda = \eta_Z/\tau^2$  and the latter matrix is positive semidefinite and singular, including when  $D = K$ . Its smallest eigenvalue therefore separates the scalar part. Equal responses give  $AA^\top = A'A'^\top$  and hence  $MM^\top = I$ . Cancelling the full-row-rank basis from the remaining blocks and taking the unique positive semidefinite square root yields  $C_i = MC'_i M^\top$ . The rank-one parts cancel, leaving  $\text{Diag}(a_i) = M \text{Diag}(a'_i) M^\top$  for every  $i$ . Because the router rows span  $\mathbb{R}^K$ , this forces  $M$  to be a signed permutation;  $M\mathbf{1} = \mathbf{1}$  makes every sign positive. The converse follows by substitution. Appendix B.2 gives the complete proof and the boundary examples.

For  $K \geq 3$ , Gram information alone is insufficient: non-permutation orthogonal transformations fixing  $\mathbf{1}$  can preserve positivity. The covariance blocks do the additional work. Dropping either rank assumption admits counterexamples, as does replacing the full response by one realized gradient. The theorem is a statement about a specified observation, not about all optimizers selecting the same chart.

#### 4.2 QUANTITATIVE INVERSE AND IMPERFECT PRODUCTS

**Corollary 1** (Stable response identification). *Fix  $K \geq 2$ ,  $L \geq K$ , and  $D \geq K$ . Restrict both factorizations to a uniformly nondegenerate set: every router entry is at least  $\alpha > 0$ , the smallest singular values of both routers and both basis matrices are at least  $s_A, s_B > 0$ , respectively, and their Frobenius norms are at most  $L_A, L_B$ . If  $AB = A'B'$  and both response operators use the same  $\tau, \eta_Z, \eta_B > 0$ , then a finite explicit constant  $C$ , depending only on  $K, L, \alpha, s_A, s_B, L_A, L_B, \tau, \eta_Z, \eta_B$ , satisfies*

$$\min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2} \leq C \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}. \quad (10)$$

Appendix B.3 gives a fully explicit, deliberately loose choice of  $C$ , a complete proof, and explicit families showing why the interior and two rank margins enter; the norm bounds control the nonlocal diameter. This is an inverse-stability statement, not a claim that its constant is tight in practical models.

**Why a linear inverse is possible.** Near equality of the response first controls the Gram and squared-covariance data. A general positive semidefinite square-root estimate would lose a square root in the error. Here the compared router covariances share the kernel  $\text{span}(\mathbf{1})$ , and the interior/rank margins give a uniform spectral gap on its orthogonal complement. A Sylvester equation therefore controls their difference linearly. The resulting approximate diagonal relations round to a permutation; a bounded-domain diameter handles large discrepancies. These steps give a global bound on the stated domain, not only near a preselected solution.

The assumptions have distinct roles. Appendix B.3 gives families with a vanishing router rank margin, basis rank margin, or interior margin for which a uniform linear inverse fails. In the last family both factors remain well-conditioned while the factor discrepancy is of order  $t$  and the response discrepancy is of order  $t^2$ . The constants are explicit but deliberately conservative.

**Small-product-perturbation bridge.** Equal products can be relaxed without assuming the two factors are already close. Write  $e_\Theta = \|AB - A'B'\|_F$  and  $e_R = \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}$ . For two factors in the same nondegenerate domain, if  $e_\Theta \leq \min\{\alpha s_B/2, s_A s_B/4\}$ , then Corollary 5 gives

$$d_{\text{perm}}((A, B), (A', B')) \leq C_* e_R + \kappa(1 + C_* L_R) e_\Theta. \quad (11)$$

Here  $\kappa = (s_B^{-2} + 4s_A^{-2})^{1/2}$ ;  $C_*$  is the explicit inverse constant on the enlarged domain  $(\alpha/2, s_A/2, s_B/2, 2L_A, 2L_B)$ , and  $L_R$  is its explicit forward Lipschitz bound. All constants are specified in Appendix C. The proof projects  $A'$  onto  $\text{col}(A)$  and constructs a nearby intermediate basis with product exactly  $AB$ . It applies the same-product inverse there and then returns to the original pair. Only the product discrepancy must be small; no candidate-neighborhood or common local-branch premise is imposed.

#### 4.3 FINITE OBSERVATIONS, INCLUDING A NOISY PRODUCT

The response is structured: its shared part is a layer Gram matrix, and each local block is supported in the product row space. This permits finite reconstruction when that space is known exactly.

**Corollary 2** (Finite structured response tomography). *Under the hypotheses of Theorem 1, suppose  $D - K \geq L$ . Let  $S = \text{row}(\Theta)$ , choose an orthonormal basis  $u_1, \dots, u_K$  of  $S$ , and choose orthonormal  $v_1, \dots, v_L \in S^\perp$ , all from  $\Theta$ . Define  $K$  effective-gradient probes by*

$$g_j^{(1)} = v_j + u_1, \quad g_j^{(r)} = u_r \quad (2 \leq r \leq K, 1 \leq j \leq L).$$

*Their  $K$  response matrices determine the complete operator  $\mathcal{R}_{A,B}$ . Hence two exact full-rank charts of the same  $\Theta$  agree on these probes if and only if they differ by a simultaneous basis permutation (with the same residual logit row shifts as above).*

Appendix B.4 gives the decoder and its component noise bounds with an exact product. Natural task gradients can replace these designed probes when the complementary shared system and every layer’s local system have full rank (Proposition 11). That condition must be checked at the checkpoint, not presumed from the number of gradients.

With a noisy product, its estimated row space generally excludes part of the true basis. The following result controls that missing part uniformly over all feasible factors. Its queries repeat the complement code in every probe, unlike Corollary 2.

**Corollary 3** (Global diameter under finite noisy observations). *Let  $L \geq K \geq 2$  and  $D - K \geq L$ . Fix the known positive rates and temperature, and restrict factors to row-stochastic routers with entries at least  $\alpha > 0$ , singular-value margins  $s_A, s_B > 0$ , and  $\|B\|_F \leq M$  for  $M > 0$ . From an arbitrary observed product  $\hat{\Theta}$ , choose its top- $K$  right-singular frame  $U$  and an orthonormal  $L$ -row complement frame  $W$ . There are exactly  $K$  unit-Frobenius probes  $G_p = (W + \mathbf{1}_L u_p^\top)/\sqrt{2L}$ , where  $u_p^\top$  is row  $p$  of  $U$ . For nonnegative error budgets, let  $\mathcal{F}$  consist of all domain members whose product residual is at most  $\varepsilon_\Theta$  and whose stacked response residual on these probes is at most  $\varepsilon_R$ . If  $\varepsilon_\Theta \leq \min\{\alpha s_B/4, s_A s_B/8\}$ , then*

$$\text{diam}_{d_{\text{perm}}}(\mathcal{F}) \leq C_\Theta \varepsilon_\Theta + C_R \varepsilon_R.$$

*Appendix D constructs the probes, expands every constant, and proves the bound for the entire feasible set in the original basis space. No true-subspace or local-branch premise is imposed.*

Here the residuals are  $\|AB - \hat{\Theta}\|_F$  and  $(\sum_{p=1}^K \|\mathcal{R}_{A,B}(G_p) - \hat{R}_p\|_F^2)^{1/2}$ . Each query returns a full  $L \times D$  response, not a scalar. The constants depend only on the dimensions, declared domain margins, and known optimizer parameters. Noise can be deterministic, adversarial, or dependent on the chosen probes, provided these budgets hold.

**Why the bound covers the whole feasible set.** Let  $\hat{P} = U^\top U$ . Every feasible rank- $K$  product competes with the best rank- $K$  approximation of  $\hat{\Theta}$ , so

$$\|B(I - \hat{P})\|_F \leq 2\varepsilon_\Theta/s_A \quad \text{for every } (A, B) \in \mathcal{F}. \quad (12)$$

Replacing  $B$  temporarily by  $B\hat{P}$  changes the complete response by at most  $4\lambda M \varepsilon_\Theta/s_A$ . For two such projected operators, multiplication of each response by  $W^\top$  recovers their shared Gram difference. Subtracting the recovered shared term and multiplying by  $U^\top$  then recovers their local blocks. This is a stable decoder on a common linear operator family. Adding back both projection errors gives, for any two feasible pairs,

$$\|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F} \leq 2h\varepsilon_R + \beta\varepsilon_\Theta, \quad (13)$$

where  $h = \sqrt{2L}(K^{-1/2} + 1 + \sqrt{L})$  and  $\beta = 8\lambda M(h\sqrt{K} + 1)/s_A$ . Their original products differ by at most  $2\varepsilon_\Theta$ ; Equation 11 then proves the diameter bound for the original, unprojected factors. Appendix D expands all constants and proves each step.

**What this guarantees.** If truth lies in the declared domain and satisfies the noise budgets, every feasible estimate is within the displayed bound of it, modulo labels. The theorem does not ensure that a solver finds a feasible estimate, infer domain membership from data, or provide numerically tight constants. It requires no smallness assumption on  $\varepsilon_R$ . For  $K = 1$ , the router is fixed and the product alone gives diameter at most  $2\varepsilon_\Theta/\sqrt{L}$  without response queries. The candidate-local numerical check in Proposition 12 is a separate, weaker guarantee about one feasible component; it must not be substituted for this global result.

Table 1: Joint-fit factor error with  $q = 4$ : maxima over three adaptation seeds, and (for natural queries) three text domains. Noise affects both product and response.  $D$  is the adapter basis dimension, not the backbone parameter count. Each reported maximum includes all declared cases.

| Model        | $L$ | $D$    | Natural, 0            | Natural, $10^{-4}$   | Designed, $10^{-4}$  |
|--------------|-----|--------|-----------------------|----------------------|----------------------|
| Pythia 160M  | 12  | 49152  | $6.5 \times 10^{-14}$ | $1.6 \times 10^{-4}$ | $1.4 \times 10^{-4}$ |
| Pythia 1B    | 16  | 131072 | $2.8 \times 10^{-13}$ | $5.7 \times 10^{-5}$ | $5.7 \times 10^{-5}$ |
| Pythia 2.8B  | 32  | 163840 | $8.3 \times 10^{-13}$ | $3.9 \times 10^{-5}$ | $3.8 \times 10^{-5}$ |
| Qwen3 0.6B   | 28  | 131072 | $6.6 \times 10^{-14}$ | $4.9 \times 10^{-5}$ | $4.9 \times 10^{-5}$ |
| Qwen3 1.7B   | 28  | 131072 | $9.2 \times 10^{-14}$ | $4.2 \times 10^{-5}$ | $4.2 \times 10^{-5}$ |
| Qwen3 4B     | 36  | 262144 | $1.0 \times 10^{-13}$ | $3.7 \times 10^{-5}$ | $3.7 \times 10^{-5}$ |
| Qwen3 8B     | 36  | 262144 | $2.6 \times 10^{-13}$ | $3.7 \times 10^{-5}$ | $3.6 \times 10^{-5}$ |
| SmolLM2 360M | 32  | 61440  | $7.8 \times 10^{-14}$ | $3.8 \times 10^{-5}$ | $3.8 \times 10^{-5}$ |
| SmolLM2 1.7B | 24  | 131072 | $6.4 \times 10^{-14}$ | $4.5 \times 10^{-5}$ | $4.4 \times 10^{-5}$ |

## 5 PRETRAINED-MODEL AUDITS AND THEIR BOUNDARIES

We test whether the response model remains numerically useful in trained modules attached to real pretrained language models, how noise and query choice affect recovery, and whether recovered coordinates improve a structural decision. The theory supplies the global quantifiers; these experiments do not test every feasible factorization or validate the theorem’s conservative constants. Earlier synthetic and byte-LM studies, including large errors and failed rejection rules, remain in Appendix H.

**Nine models, a fixed module, and held-out gradient domains.** We use three Pythia sizes (160M, 1B, 2.8B), four dense Qwen3 sizes (0.6B, 1.7B, 4B, 8B), and two SmolLM2 sizes (360M, 1.7B) (Biderman et al., 2023; Yang et al., 2025; Allal et al., 2025). Each frozen BF16 backbone receives a  $K = 4$  softmax-shared additive module on 64 fixed output channels of every attention output projection. Its bases are direct trainable matrices, not LoRA products. A fixed isometric embedding preserves the factor-response model (Proposition 13). We train only these modules for 256 AdamW updates on WikiText-2, using three seeds per model and the final checkpoint. The resulting 27 checkpoints are adapter replicates of nine fixed pretrained models, not 27 independent pretraining runs.

Natural queries are measured next-token-loss gradients from disjoint general-text, MBPP code, and GSM8K math documents (Merity et al., 2016; Austin et al., 2021; Cobbe et al., 2021). The code and math domains are not used for adaptation or calibration. Responses are controlled reverse-gradient/forward-JVP observations of the factor map in FP64, treating the measured BF16-backbone gradients as fixed inputs. They are not observed AdamW updates. The internal product and matrix-valued responses are available; this is not an output-only attack. The full protocol, model revisions, rejected access requests, development corrections, and data/source hashes are in Appendix G.

**Finite recovery across model sizes and query types.** We compare the repeated-complement design of Corollary 3, Gaussian queries, and three natural gradient families, with  $q \in \{1, 4, 8\}$ , four independent held-out directions, and relative noise  $0, 10^{-6}, 10^{-4}, 10^{-2}$  added to both product and responses. The noisy product subspace is re-estimated and orthogonal residuals are retained. Three estimators use observations only; permutation-aligned ground truth is supplied afterward for scoring.

All 1,080 cases with  $q \geq 4$  have sufficient observed ranks and return joint-fit factor error below 0.034; their median error is  $1.86 \times 10^{-5}$ . At  $q = 4$  without noise, the worst error across all models and query families is  $8.31 \times 10^{-13}$ . Table 1 reports maxima rather than selected seeds. All 540 cases with  $q = 1$  fail the sufficient local-rank check; joint fitting is not run and these cases count as rejected. This is the failure of the specified reconstruction condition, not a proof that every one-query estimator is impossible.

**Calibration transfers here, but does not certify recovery.** Four separate Pythia adapter checkpoints, with general-text validation queries only, fix acceptance thresholds before held-out recovery scoring. At error  $> 0.10$  defined as bad, the combined joint-fit rule accepts all 1,080 sufficient-rank cases with no observed bad output; residual-only accepts 868, also with none. This sample therefore

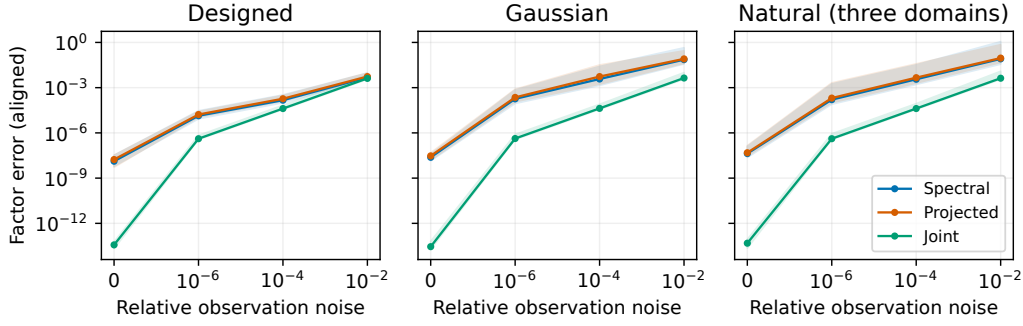


Figure 1: Recovery with four queries. Lines show medians and shaded bands the 10th–90th percentiles over all 27 checkpoints; the natural panel also includes all three domains. These are descriptive spreads across correlated cases, not confidence intervals. Noiseless results use controlled FP64 factor responses, not full-LLM FP64 inference.

cannot establish better bad-case detection by the combined rule. For projected spectral recovery, the combined rule accepts 43 bad outputs among 1,056, versus 20 among 1,037 for residual-only; its largest accepted error is 6.19. Raw spectral recovery has an all-reject calibrated policy under the strict simplex gate, even though many of its factor errors are small. Table 3 retains all denominators. The earlier synthetic study also retains a combined-rule accepted-bad fraction of 5.16%, above its 5% target. Neither calibration nor candidate-local diagnostics replace the theorem’s externally specified domain assumptions.

**Precision and structural usefulness remain limiting.** The controlled response formula agrees with autodiff within  $1.62 \times 10^{-16}$  relative error. Finite-step subtraction is less reliable: at step  $10^{-5}$ , FP32 errors range from 0.040 to 0.102, and BF16 errors from 1.00 to 1.95. Separately, shrinking the router-rank margin in learned factors amplifies noisy recovery error by roughly three orders of magnitude in the registered secondary stress test (Figure 2).

On nine declared checkpoints, balanced hard groups selected from recovered routers at noise  $10^{-4}$  and  $10^{-2}$  exactly reproduce the native router partitions. However, balanced clustering of the effective parameters also reproduces all nine. Equal-budget refitting and 64 recovery updates therefore provide no evidence that coordinate recovery improves this decision over the simpler effective-parameter control. Random balanced groups worsen general-text NLL after recovery at all nine checkpoints, while code/math differences have mixed signs. These are token-loss results, not generation accuracy or realized inference speedups. Detailed action results, repeated-action numerical controls, costs, and independent CPU replays are in Appendix G.

## 6 DISCUSSION AND LIMITATIONS

Static factors need not be unique, yet a specified optimizer can expose additional information about their current coordinates. For the finite softmax-router/shared-basis model, complete Euclidean response identifies the factors up to permutation on a common full-rank product fiber. The quantitative reduction controls original factor error on named nondegenerate domains. Its finite-noisy consequence bounds the whole feasible set using queries constructed from the observed product, without requiring the true basis to lie in an estimated subspace.

The guarantee has substantial conditions: known positive Euclidean block rates and temperature, internal access to the effective product and full matrix-valued responses, rank and interior margins, and, for the stated query construction,  $D - K \geq L$ . The product error must be below a domain-dependent threshold. The constants are conservative, query optimality is unproved, and an efficient globally feasible recovery algorithm is not supplied. Rank loss, proximity to the simplex boundary, and weak observation directions obstruct stronger uniform conclusions.

The numerical results illustrate these distinctions. Precise controlled responses permit accurate recovery, while finite arithmetic and noisy ill-conditioned problems can produce large errors and falsely accepted estimates. The empirical rejection rule and candidate-local check cannot replace the global theorem’s domain assumptions. No result here identifies an AdamW trajectory, a historical sharing graph, or a nonlinear output-mixture model. The pretrained-model study covers one fixed-rank, 64-channel additive module and three dense model families up to 8B; it does not establish universality across LLM architectures, sizes, or adapter parameterizations. Family comparisons confound architecture, pretraining data and, for the selected Qwen3 models, post-training. The simple effective-parameter control matches the recovered structural partitions, so this study does not establish an incremental decision or compression benefit. The useful positive conclusion is specific: finite optimizer-response information can resolve a static coordinate ambiguity, with an explicit account of the observations and margins required.

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## AI ASSISTANCE DISCLOSURE

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## A SECONDARY THEORETICAL STATEMENTS

This section preserves the auxiliary static, recovery, probe, and optimization statements referenced by the main text. They are separated from the central optimizer-response theorem because the static change of basis is SSMF prior art and the positive results use standard additional assumptions. Complete proofs and failure cases follow in Appendix B.

### A.1 STATIC ROUTER-COORDINATE CONSEQUENCES

The rows  $b_1, \dots, b_K$  are affinely independent when  $[B, \mathbf{1}]$  has row rank  $K$ .

**Proposition 1** (Local form of the SSMF gauge). *Assume  $K \geq 2$ ,  $A$  is strictly positive and has column rank  $K$ , and the rows of  $B$  are affinely independent. For every nonzero  $H \in \mathbb{R}^{K \times K}$  satisfying  $H\mathbf{1} = 0$ , and every sufficiently small  $t \neq 0$ ,*

$$M_t = I + tH, \quad A_t = AM_t, \quad B_t = M_t^{-1}B \quad (14)$$

*is a valid factorization with  $A_t \in \Delta_K^L$ , affinely independent bases, and  $A_t B_t = AB$ . It is not permutation equivalent to  $(A, B)$ . The local equivalence class therefore has at least  $K(K-1)$  continuous directions.*

The condition  $H\mathbf{1} = 0$  preserves row sums; positivity and invertibility survive a small perturbation; and  $[B_t, \mathbf{1}] = M_t^{-1}[B, \mathbf{1}]$  preserves affine independence. Full column rank of  $A$  rules out a hidden permutation.

For  $0 < s < 1$ , the gauge

$$M_s = (1 - s)I + s\mathbf{1}\mathbf{1}^\top / K \quad (15)$$

moves every router row toward uniform and strictly increases its entropy unless it was already uniform, while  $M_s^{-1}B$  keeps  $\Theta$  fixed. This uniformizing path preserves argmax assignments. They are not invariant under other feasible gauges, however. For example,

$$A = \begin{bmatrix} .1 & .9 \\ .3 & .7 \\ .4 & .6 \\ .9 & .1 \end{bmatrix}, \quad M = \begin{bmatrix} .9 & .1 \\ .4 & .6 \end{bmatrix} \quad (16)$$

gives argmax labels  $(2, 2, 2, 1)$  before and  $(2, 1, 1, 1)$  after multiplication by  $M$ . Yet  $(AM)(M^{-1}B) = AB$  for every  $B$ .

This ambiguity does not erase every partition statistic. Because  $M$  is invertible,  $a_i = a_j$  iff  $a_i M = a_j M$ ; affine independence of the rows of  $B$  already makes this equivalent to  $\theta_i = \theta_j$ . Indeed,  $qB = 0$  and  $q\mathbf{1} = 0$  imply  $q = 0$  when  $[B, \mathbf{1}]$  has row rank  $K$ .

**Proposition 2** (End-to-end impossibility). *Suppose an architecture and its observation law depend on  $(A, B)$  only through  $\Theta = AB$ . If two gauge-equivalent factorizations have different router summaries, no estimator based on end-to-end input–output observations can recover the correct summary in both worlds. This remains true with oracle access to the predictor on every input.*

This result concerns router coordinates; it neither equates effective-weight equality with functional equivalence nor asserts that either equals a planted generative graph. The latter is independently impossible under composition: a two-layer linear teacher with  $W_1 = W_2 = I$  (tied) and one with  $W_1 = 2I$ ,  $W_2 = I/2$  (untied) both implement  $x \mapsto x$ .

## A.2 POSITIVE RECOVERY UNDER ADDITIONAL INFORMATION

Call a factorization *separable* if every basis has an anchor layer  $i_j$  with  $a_{i_j} = e_j$ .

**Proposition 3** (Standard separable recovery). *If two  $K$ -separable factorizations of the same  $\Theta$  have affinely independent bases, they are permutation equivalent. Consequently, if routing is one-hot with discrete label  $c_i \in \{1, \dots, K\}$ , every basis is used, and the bases are affinely independent, then*

$$c_i = c_{i'} \iff \theta_i = \theta_{i'}, \quad (17)$$

*so the sharing partition is recoverable up to basis relabeling.*

This is standard separable-SSMF geometry specialized to layer rows, not a new simplex-identification result (Abdolali & Gillis, 2021). Separability makes  $\text{conv}\{\theta_i\}$  equal the simplex whose vertices are the bases; vertices are unique, and affine independence gives unique barycentric coordinates. The restriction to the separable model class is essential: an unrestricted soft factorization may place a larger simplex outside the observed convex hull.

**Theorem 2** (Stable noisy one-hot recovery). *Assume  $K \geq 2$ . Suppose  $c_i \in \{1, \dots, K\}$  and  $\hat{\theta}_i = b_{c_i} + e_i$ ,  $\|e_i\|_2 \leq \varepsilon$ , and  $\gamma = \min_{j \neq k} \|b_j - b_k\|_2 > 4\varepsilon$ . Connecting two layers when their observed parameters are at distance at most  $2\varepsilon$  recovers the exact one-hot partition. Each cluster mean estimates its basis to error at most  $\varepsilon$ .*

Parameter access is not always available. Shared-input probes instead target the clean functional partition defined in Section 3. Let  $Y_i(H)$  be the possibly noisy observation of layer  $i$ 's residual update on a common probe. The result below applies only when the distances induced by that observation law retain a positive gap for the clean target partition; layer-dependent noise can destroy this condition. Define

$$d_{ij}^2 = \mathbb{E}\|Y_i(H) - Y_j(H)\|_2^2, \quad \hat{d}_{ij}^2 = \frac{1}{n} \sum_{s=1}^n \|Y_i(H_s) - Y_j(H_s)\|_2^2. \quad (18)$$

**Theorem 3** (Finite-probe partition recovery). *Let  $L \geq 2$ , let  $0 < \eta < 1$ , and suppose the target partition has at least two classes. Let it have population within-class radius  $w = \max_{i \sim_f j} d_{ij}^2$  and between-class separation  $b = \min_{i \not\sim_f j} d_{ij}^2$ , with gap  $\delta = b - w > 0$ . Suppose the centered squared-distance observations satisfy, uniformly over layer pairs, the Bernstein tail*

$$\Pr\left(\left|\widehat{d}_{ij}^2 - d_{ij}^2\right| > u\right) \leq 2 \exp\left[-n \min\left\{\frac{u^2}{2\nu^2}, \frac{u}{2c}\right\}\right]. \quad (19)$$

If, for failure probability  $\eta$ ,

$$n \geq \max\left\{\frac{32\nu^2}{\delta^2}, \frac{8c}{\delta}\right\} \log \frac{2\binom{L}{2}}{\eta}, \quad (20)$$

then thresholding  $\widehat{d}_{ij}^2$  at  $(w + b)/2$  recovers the functional partition with probability at least  $1 - \eta$ . The same strict separation guarantees recovery by single, complete, or average agglomerative linkage when the number  $K_f$  of functional equivalence classes is known. In general  $K_f$  need not equal the number  $K$  of router bases.

The result follows by a union bound and certifies equivalence only under the specified probe distribution; behavior off its support remains unidentified.

### A.3 OPTIMIZATION AND INITIALIZATION EFFECTS

Let  $G = \nabla_{\Theta} \mathcal{L}$ .

**Proposition 4** (Factorization-induced dynamics). *For Euclidean gradient flow on unconstrained factors, without factor-specific regularizers,*

$$\dot{\Theta} = -G(B^\top B) - (AA^\top)G. \quad (21)$$

*Gauge-equivalent initializations therefore generally have different effective parameter trajectories despite identical initial  $\Theta$ , predictions, and loss.*

**Proposition 5** (Softmax saturation). *Let  $a_i = \text{softmax}(z_i/\tau)$  and suppose  $\max_j a_{ij} = 1 - \varepsilon$ . With  $g_i = \nabla_{\theta_i} \mathcal{L}$ ,*

$$\|\nabla_{z_i} \mathcal{L}\|_2 \leq \frac{2\varepsilon}{\tau} \|Bg_i\|_2. \quad (22)$$

*Thus the instantaneous logit gradient is attenuated as the current router row approaches the simplex boundary. This pointwise bound alone does not imply that an initial argmax graph remains fixed over a finite training trajectory; the loss gradients, basis norm, and probability margin can all change.*

These results justify testing gauge-equivalent initializations, final-initial router movement, and recovery under deliberately wrong initial graphs. They do not prove that gradient descent prefers a universal smooth, banded, or low-entropy structure. Nor do reconstruction loss or end-to-end accuracy by themselves identify router coordinates, the effective-weight equality partition, functional equivalence, or a latent generative graph.

## B THEORY PROOFS, EXTENSIONS, AND SCOPE

This appendix gives complete proofs for Section 4. It also records edge cases that are easy to lose when a soft router is visualized as a discrete sharing graph.

### B.1 STATIC SSMF GAUGE DETAILS

The non-identifiability conclusion and the symmetric uniformizing path are the unconstrained SSMF result of Vu Thanh et al. (2023, Section IV-A) under  $X = \Theta^\top$ ,  $W = B^\top$ , and  $H = A^\top$ ; see Appendix F. The proof below only records the direct local row-sum-preserving extension used to delimit which router summaries change.

*Proof of Proposition 1.* The constraint  $H\mathbf{1} = 0$  implies

$$M_t \mathbf{1} = (I + tH)\mathbf{1} = \mathbf{1}. \quad (23)$$

The determinant is continuous and  $\det(M_0) = 1$ , so  $M_t$  is invertible for all sufficiently small  $|t|$ . Strict positivity is also an open condition: because  $A$  has finitely many entries and  $A_{ij} > 0$ , there is a neighborhood of zero in which every entry of  $AM_t$  remains positive. Therefore

$$A_t \mathbf{1} = AM_t \mathbf{1} = A \mathbf{1} = \mathbf{1}, \quad A_t > 0, \quad (24)$$

and  $A_t \in \Delta_K^L$ . The effective parameters are unchanged because

$$A_t B_t = (AM_t)(M_t^{-1}B) = AB. \quad (25)$$

Equation 23 also gives  $M_t^{-1} \mathbf{1} = \mathbf{1}$ . Hence the augmented basis matrix satisfies

$$[B_t, \mathbf{1}] = [M_t^{-1}B, M_t^{-1} \mathbf{1}] = M_t^{-1}[B, \mathbf{1}]. \quad (26)$$

Left multiplication by an invertible matrix preserves row rank, so affine independence is preserved exactly, not merely infinitesimally.

It remains to rule out a permutation. If  $A_t = AP$  for a permutation matrix  $P$ , then  $A(M_t - P) = 0$ . Since  $A$  has column rank  $K$ , it has a left inverse, and therefore  $M_t = P$ . The identity is the only permutation matrix in a sufficiently small neighborhood of  $I$ . For nonzero  $t$  and nonzero  $H$ ,  $M_t \neq I$ , yielding a contradiction. Finally, the linear constraint  $H \mathbf{1} = 0$  supplies  $K$  independent equations on  $K^2$  entries, so its solution space has dimension  $K(K - 1)$ . Full column rank makes these local directions distinct at  $A$ .  $\square$

Any strictly positive router can be written as a softmax: for fixed  $\tau > 0$ , take  $z_{ij} = \tau \log A_{ij}$ , up to an arbitrary additive constant in each row. Thus softmax normalization does not eliminate the orbit.

**Entropy-changing gauges.** For  $M_s$  in equation 15, the eigenvalue along  $\mathbf{1}$  is one and every eigenvalue orthogonal to  $\mathbf{1}$  is  $1 - s$ ; hence it is invertible for  $0 < s < 1$ . Since

$$A \frac{\mathbf{1} \mathbf{1}^\top}{K} = \frac{\mathbf{1}_L \mathbf{1}^\top}{K} = U, \quad (27)$$

we have  $AM_s = (1 - s)A + sU$ . Strict concavity of Shannon entropy gives

$$H((1 - s)a_i + su) \geq (1 - s)H(a_i) + sH(u) > H(a_i) \quad (28)$$

for every nonuniform  $a_i$ , because  $u$  is the unique entropy maximizer. The effective parameters remain exactly unchanged after replacing  $B$  by  $M_s^{-1}B$ . As  $s \uparrow 1$ , the router can approach uniform while the basis norm may grow. A norm penalty can prefer one member of this orbit, but that is a gauge-fixing optimization convention rather than identification by the data.

**Verification of the argmax counterexample.** The matrix  $M$  in equation 16 is positive, row stochastic, and has determinant 0.5. Direct multiplication gives

$$AM = \begin{bmatrix} 0.45 & 0.55 \\ 0.55 & 0.45 \\ 0.60 & 0.40 \\ 0.85 & 0.15 \end{bmatrix}. \quad (29)$$

Thus two layer assignments flip. If  $B$  has two distinct rows, those rows are affinely independent; equation 26 shows that  $M^{-1}B$  remains so. The example therefore does not rely on collapsed bases.

*Proof of Proposition 2.* Let  $(A, B)$  and  $(A', B')$  be the two factorizations. Gauge equivalence gives  $AB = A'B' = \Theta$ . By assumption, for every input  $x$ ,

$$F(x; A, B) = F(x; AB) = F(x; A'B') = F(x; A', B'). \quad (30)$$

The complete input–output process, including its distribution under any design over inputs, is therefore identical in the two worlds. Every possibly randomized estimator has the same output distribution in both worlds. If the two target router summaries differ, the estimator cannot equal both with probability one. The argument already grants the estimator all function values, so increasing the sample size cannot repair the failure.  $\square$

The proposition assumes that factors enter the architecture only through the parameter product. If all nonlinear basis modules are executed separately and their outputs are mixed, an arbitrary linear change of basis need not remain in that model class. No claim for such output-space mixtures follows from this proof.

## B.2 OPTIMIZER-RESPONSE STABILIZER

We use the convention of Section 4.1. Set

$$\lambda_Z = \eta_Z / \tau^2, \quad C_i = \text{Diag}(a_i) - a_i a_i^\top, \quad H_i = B^\top C_i^2 B. \quad (31)$$

With rows of the input and output represented as column vectors, the  $(i, j)$  block of  $\mathcal{R}_{A,B}$  is

$$[\mathcal{R}_{A,B}]_{ij} = \eta_B (a_i^\top a_j) I_D + \mathbf{1}\{i = j\} \lambda_Z H_i. \quad (32)$$

*Proof of Theorem 1.* Because both factorizations have rank  $K$ , their column spaces equal the column space of their common product. Standard full-rank factorization algebra therefore gives a unique  $M \in \text{GL}(K)$  satisfying

$$A' = AM, \quad B' = M^{-1}B. \quad (33)$$

Both routers have unit row sums, so  $A(M\mathbf{1} - \mathbf{1}) = 0$ . Full column rank of  $A$  gives

$$M\mathbf{1} = \mathbf{1}. \quad (34)$$

Equality of the response on every  $G$  is equality of every block in equation 32. For  $i \neq j$ , it gives

$$a_i^\top a_j = (a'_i)^\top a'_j. \quad (35)$$

For a diagonal block, define  $H'_i = (B')^\top (C'_i)^2 B'$ . Its equality can be rearranged as

$$H_i - H'_i = c_i I_D, \quad c_i = -\frac{\eta_B}{\lambda_Z} (\|a_i\|_2^2 - \|a'_i\|_2^2) \quad (36)$$

for a scalar  $c_i$ . Every simplex covariance satisfies  $C_i \succeq 0$  and  $C_i \mathbf{1} = 0$ ; full row rank of  $B$  therefore gives  $H_i \succeq 0$  and  $\text{rank}(H_i) \leq K - 1 < D$ . The same holds for  $H'_i$ . If  $c_i > 0$ , then  $H_i = H'_i + c_i I_D$  would be positive definite; if  $c_i < 0$ , then  $H'_i = H_i - c_i I_D$  would be positive definite. Both contradict singularity, so  $c_i = 0$ . The remaining scalar part of the diagonal block then gives  $\|a_i\|_2^2 = \|a'_i\|_2^2$ . Together with equation 35,

$$AA^\top = A'A'^\top = AMM^\top A^\top. \quad (37)$$

Multiplication by a left inverse of  $A$  and its transpose yields

$$MM^\top = I_K. \quad (38)$$

Since  $H_i = H'_i$ , substituting equation 33 and cancelling  $B$  with a right inverse gives

$$C_i^2 = M^{-\top} (C'_i)^2 M^{-1}. \quad (39)$$

By equation 38, this is

$$C_i^2 = M(C'_i)^2 M^\top = (MC'_i M^\top)^2. \quad (40)$$

Both  $C_i$  and  $MC'_i M^\top$  are positive semidefinite. Uniqueness of the positive-semidefinite square root therefore gives

$$C_i = MC'_i M^\top. \quad (41)$$

In column convention,  $a'_i = M^\top a_i$ . Expanding equation 41, the two rank-one terms cancel and leave

$$\text{Diag}(a_i) = M \text{Diag}(a'_i) M^\top. \quad (42)$$

For distinct  $r, s$ , the  $(r, s)$  entry says

$$\sum_{\ell=1}^K M_{r\ell} M_{s\ell} a'_{i\ell} = 0 \quad \text{for every } i. \quad (43)$$

The rows of  $A'$  span  $\mathbb{R}^K$ , hence  $M_{r\ell} M_{s\ell} = 0$  for every  $\ell$ . Thus each column of the orthogonal  $M$  has exactly one nonzero entry, equal to  $+1$  or  $-1$ :  $M$  is a signed permutation. Equation 34 removes the negative signs, so  $M = P$  is a permutation matrix.

Conversely, if  $A' = AP$  and  $B' = P^\top B$ , then  $a'_i = P^\top a_i$ ,  $C'_i = P^\top C_i P$ , the router Gram matrix is unchanged, and

$$(B')^\top (C'_i)^2 B' = B^\top C_i^2 B. \quad (44)$$

Every block in equation 32 is therefore unchanged. This proves both directions.  $\square$

### B.3 A QUANTITATIVE INVERSE FOR THE RESPONSE

The exact stabilizer has a linear inverse on any uniformly interior, well-conditioned, norm-bounded set. The following constants are explicit so that the claim is checkable; they are not intended to be numerically tight. For a linear map between matrix spaces,  $\|\cdot\|_{F \rightarrow F}$  denotes the operator norm induced by the Frobenius norm.

**Theorem 4** (Quantitative optimizer-response stability). *Let  $K \geq 2$ ,  $L \geq K$ , and  $D \geq K$ . Let  $A, A' \in \mathbb{R}^{L \times K}$  be row-stochastic and let  $B, B' \in \mathbb{R}^{K \times D}$ . Suppose*

$$\min_{i,j} \{A_{ij}, A'_{ij}\} \geq \alpha > 0, \quad (45)$$

$$\sigma_{\min}(A), \sigma_{\min}(A') \geq s_A > 0, \quad \sigma_{\min}(B), \sigma_{\min}(B') \geq s_B > 0, \quad (46)$$

$$\|A\|_F, \|A'\|_F \leq L_A, \quad \|B\|_F, \|B'\|_F \leq L_B, \quad (47)$$

$$AB = A'B'. \quad (48)$$

Both response operators are defined with the same  $\tau, \eta_Z, \eta_B > 0$ . Put  $\lambda_Z = \eta_Z/\tau^2$  and

$$\delta = \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}. \quad (49)$$

Then

$$\min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2} \leq C\delta, \quad (50)$$

where the following is one finite, fully explicit choice. First define the factorization and Gram constants

$$L_M = \max\{1, L_A/s_A\}, \quad c_G = \frac{L}{\eta_B s_A^2}, \quad c_H = \frac{2}{\lambda_Z s_B^2}. \quad (51)$$

Next define the covariance and diagonal-algebra constants

$$\mu_* = \alpha \left(1 + \frac{1}{4L_M^2}\right), \quad c_J = \frac{L_M^2(c_G + c_H)}{\mu_*}, \quad c_D = \frac{\sqrt{LK}}{s_A} c_J. \quad (52)$$

Finally define the row-rounding and global constants

$$c_{\text{row}} = [Kc_D^2 + (c_G + Kc_D^2)^2]^{1/2}, \quad (53)$$

$$\delta_0 = \min \left\{ 1, \frac{1}{2c_G}, \frac{1}{2\sqrt{K} c_{\text{row}}} \right\}, \quad (54)$$

$$C_{\text{loc}} = \sqrt{L_A^2 + (L_M L_B)^2 \sqrt{K} c_{\text{row}}}, \quad C = \max \left\{ C_{\text{loc}}, \frac{2\sqrt{L_A^2 + L_B^2}}{\delta_0} \right\}. \quad (55)$$

*Proof.* Except where a Frobenius norm is displayed explicitly, all matrix norms in this proof are spectral norms. We divide the argument into four steps.

**Step 1: recovering the router Gram matrix and an almost-orthogonal gauge.** Injecting a vector into input block  $j$  and projecting the output onto block  $i$  are contractions for the Frobenius-induced operator norm. Therefore each block of  $\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}$  has spectral norm at most  $\delta$ . For  $i \neq j$ , equation 32 gives

$$|(AA^\top - A'A'^\top)_{ij}| \leq \frac{\delta}{\eta_B}. \quad (56)$$

For a diagonal block, write

$$E_i = t_i I_D + \lambda_Z (H_i - H'_i), \quad t_i = \eta_B (\|a_i\|_2^2 - \|a'_i\|_2^2), \quad (57)$$

where  $H'_i = (B')^\top (C'_i)^2 B'$ . Both  $H_i$  and  $H'_i$  are singular positive-semidefinite matrices: their ranks are at most  $K - 1 < D$ . If  $t_i \geq 0$ , evaluating  $E_i$  on a unit vector in  $\ker(H'_i)$  gives a quadratic form at least  $t_i$ . If  $t_i < 0$ , evaluating it on a unit vector in  $\ker(H_i)$  gives a quadratic form at most  $t_i$ . Since  $\|E_i\|_2 \leq \delta$ , in either case

$$|t_i| \leq \delta, \quad \|H_i - H'_i\|_2 \leq \frac{2\delta}{\lambda_Z}. \quad (58)$$

The diagonal entries of the router Gram difference consequently satisfy the same bound as equation 56, and hence

$$\|AA^\top - A'A'^\top\|_F \leq \frac{L\delta}{\eta_B}. \quad (59)$$

The common product has rank  $K$ , so full-rank factorization algebra gives a unique  $M \in \text{GL}(K)$  with

$$A' = AM, \quad B' = M^{-1}B, \quad M\mathbf{1} = \mathbf{1}. \quad (60)$$

Moreover,

$$M = A^\dagger A', \quad M^{-1} = (A')^\dagger A, \quad \|M\|_2, \|M^{-1}\|_2 \leq L_A/s_A \leq L_M. \quad (61)$$

Multiplying equation 59 on the left by  $A^\dagger$  and on the right by  $(A')^\top$  yields

$$\|I_K - MM^\top\|_2 \leq c_G\delta. \quad (62)$$

**Step 2: recovering the simplex covariance matrices.** Let  $Q = B^\top(BB^\top)^{-1}$ . Then  $BQ = I_K$  and  $\|Q\|_2 \leq 1/s_B$ . Substitute  $B' = M^{-1}B$  in  $H'_i$ , and multiply equation 58 by  $Q^\top$  and  $Q$ , to obtain

$$\|C_i^2 - M^{-\top}(C'_i)^2M^{-1}\|_2 \leq c_H\delta. \quad (63)$$

Set  $Y_i = M^\top C_i M$ . The identity

$$\begin{aligned} Y_i^2 - (C'_i)^2 &= M^\top C_i (MM^\top - I_K) C_i M \\ &\quad + M^\top \{C_i^2 - M^{-\top}(C'_i)^2M^{-1}\} M \end{aligned} \quad (64)$$

together with  $\|C_i\|_2 \leq 1$ , equation 61, equation 62, and equation 63 gives

$$\|Y_i^2 - (C'_i)^2\|_2 \leq L_M^2(c_G + c_H)\delta. \quad (65)$$

Strict interiority now supplies the spectral gap that is unavailable near the simplex boundary. If  $a$  is a probability vector with  $\min_j a_j \geq \alpha$ , then

$$C(a) = \text{Diag}(a) - aa^\top \succeq \alpha \Pi_{\mathbf{1}^\perp}. \quad (66)$$

Indeed, write  $a = \alpha\mathbf{1} + r$ , where  $r \geq 0$  and  $\mathbf{1}^\top r = 1 - K\alpha \leq 1$ . For  $x \perp \mathbf{1}$ , weighted Cauchy-Schwarz gives

$$x^\top C(a)x = \alpha\|x\|_2^2 + \sum_j r_j x_j^2 - \left( \sum_j r_j x_j \right)^2 \geq \alpha\|x\|_2^2.$$

Both  $Y_i$  and  $C'_i$  have kernel exactly  $\text{span}\{\mathbf{1}\}$ : for the former, use  $M^{-1}\mathbf{1} = \mathbf{1}$ . They are symmetric, so they preserve  $\mathbf{1}^\perp$ . For a unit  $x \perp \mathbf{1}$ , decompose  $Mx = c\mathbf{1} + y$ , with  $y \perp \mathbf{1}$ . Since  $x = c\mathbf{1} + M^{-1}y$ , orthogonality of  $x$  and equation 61 imply

$$|c|\sqrt{K} \leq L_M\|y\|_2, \quad 1 \leq |c|\sqrt{K} + L_M\|y\|_2, \quad (67)$$

and therefore  $\|y\|_2 \geq 1/(2L_M)$ . It follows from equation 66 that, on  $\mathbf{1}^\perp$ ,

$$Y_i \succeq \frac{\alpha}{4L_M^2} I, \quad C'_i \succeq \alpha I. \quad (68)$$

On this subspace put  $D_i = Y_i - C'_i$ . The Sylvester equation

$$Y_i D_i + D_i C'_i = Y_i^2 - (C'_i)^2 \quad (69)$$

has the integral solution

$$D_i = \int_0^\infty e^{-tY_i} \{Y_i^2 - (C'_i)^2\} e^{-tC'_i} dt. \quad (70)$$

The two lower bounds in equation 68 show that the norm of this inverse is at most  $1/\mu_*$ . Both sides vanish on  $\text{span}\{\mathbf{1}\}$ , so equation 65 proves on the whole space that

$$\|M^\top C_i M - C'_i\|_2 \leq c_J\delta. \quad (71)$$

The shared exact kernel is what permits a linear estimate here; a generic perturbation bound for square roots would only be Hölder of order one half.

**Step 3: approximately normalizing the diagonal algebra.** Because  $a'_i = M^\top a_i$ , expanding the two covariance matrices in equation 71 cancels their rank-one terms and gives

$$\|M^\top \text{Diag}(a_i)M - \text{Diag}(a'_i)\|_2 \leq c_J \delta. \quad (72)$$

Let  $\text{Off}(X)$  set the diagonal of  $X$  to zero, let  $E_{jj} = e_j e_j^\top$ , and define  $X_j = \text{Off}(M^\top E_{jj} M)$ . Taking the off-diagonal part is contractive in Frobenius norm, while  $\|X\|_F \leq \sqrt{K} \|X\|_2$ . Thus equation 72 implies

$$\left\| \sum_{j=1}^K A_{ij} X_j \right\|_F \leq \sqrt{K} c_J \delta \quad (1 \leq i \leq L). \quad (73)$$

Apply row  $j$  of  $A^\dagger$  to this system in the direct-sum Frobenius space. That row has Euclidean norm at most  $1/s_A$ ; Cauchy–Schwarz over the  $L$  rows gives

$$\|\text{Off}(M^\top E_{jj} M)\|_F \leq c_D \delta \quad (1 \leq j \leq K). \quad (74)$$

**Step 4: rounding the gauge to a permutation.** Write  $M^\top E_{jj} M = r_j r_j^\top$ , where  $r_j^\top$  is row  $j$  of  $M$ , and choose  $q$  maximizing  $|r_{jq}|$ . Suppose first that  $\delta \leq \delta_0$ . From equation 62,  $\|r_j\|_2^2 \geq 1/2$ , so  $r_{jq}^2 \geq 1/(2K)$ . The squared Frobenius norm in equation 74 contains  $2r_{jq}^2 \sum_{\ell \neq q} r_{j\ell}^2$ . Consequently

$$\sum_{\ell \neq q} r_{j\ell}^2 \leq K c_D^2 \delta^2, \quad (75)$$

$$|r_{jq}^2 - 1| \leq (c_G + K c_D^2) \delta. \quad (76)$$

The second inequality also uses  $\delta \leq 1$ . Equations equation 75–equation 76 show that row  $j$  is within  $c_{\text{row}} \delta$  of a signed coordinate vector.

Collect the selected signed coordinate rows into a matrix  $S$ . By equation 54,

$$\|M - S\|_2 \leq \|M - S\|_F \leq \sqrt{K} c_{\text{row}} \delta \leq \frac{1}{2}. \quad (77)$$

If two rows selected the same coordinate,  $S$  would be singular and  $\sigma_{\min}(M) \leq 1/2$ . This contradicts equation 62, which gives  $\sigma_{\min}(M) \geq 1/\sqrt{2}$ . The selected coordinates are therefore distinct. Each row of  $M$  sums to one by equation 60. Its distance from any negative coordinate vector is consequently at least  $2/\sqrt{K}$ , whereas its selected-row distance in equation 77 is at most  $1/(2\sqrt{K})$ . Every sign is positive, so  $S = P$  is a permutation matrix and

$$\|M - P\|_F \leq \sqrt{K} c_{\text{row}} \delta. \quad (78)$$

Now

$$\|A' - AP\|_F \leq L_A \|M - P\|_F.$$

Using  $M^{-1} - P^\top = M^{-1}(P - M)P^\top$  also gives

$$\|B' - P^\top B\|_F \leq L_M L_B \|M - P\|_F.$$

These two bounds and equation 78 prove equation 50 with  $C_{\text{loc}}$  whenever  $\delta \leq \delta_0$ . If  $\delta \geq \delta_0$ , choosing the identity permutation and using the parameter-set diameter gives

$$(\|A' - A\|_F^2 + \|B' - B\|_F^2)^{1/2} \leq 2\sqrt{L_A^2 + L_B^2} \leq \frac{2\sqrt{L_A^2 + L_B^2}}{\delta_0} \delta.$$

Taking the larger of the local and diameter constants completes the proof.  $\square$

**Why the quantitative assumptions appear.** The exact product constraint equation 48 cannot be replaced by response closeness alone:  $(A, B)$  and  $(A, -B)$  have identical responses but opposite effective matrices. The two singular-value margins keep the factorization gauge and the right inverse of  $B$  bounded. The interior margin creates the uniform covariance gap in equation 66; without it, the inverse can become arbitrarily ill-conditioned near a simplex face. Norm bounds make the nonlocal diameter argument finite. Finally, using the same temperature and block rates is part of specifying

the optimizer metric rather than a technical normalization. These assumptions are sufficient and transparent, not asserted to be jointly minimal.

Here are three explicit degenerating families showing that a uniform linear constant cannot ignore the interior or either rank margin. They also make clear that these are stability failures, not counterexamples to the exact theorem. Set  $K = L = D = 3$ ,  $U = \mathbf{1}\mathbf{1}^\top/3$ , and

$$Q = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & 2 \\ 2 & 2 & -1 \end{bmatrix}. \quad (79)$$

Then  $Q$  is a non-permutation orthogonal matrix and  $Q\mathbf{1} = \mathbf{1}$ .

*Router-rank margin.* For  $0 < \varepsilon \leq 1/4$ , let

$$A_\varepsilon = U + \varepsilon(I - U), \quad B_\varepsilon = I, \quad A'_\varepsilon = A_\varepsilon Q, \quad B'_\varepsilon = Q^\top.$$

Both routers are strictly positive, the basis singular values equal one, and all norms and entrywise interior margins remain uniformly bounded. However,  $\sigma_{\min}(A_\varepsilon) = \sigma_{\min}(A'_\varepsilon) = \varepsilon$ . The products and router Gram matrices agree exactly. At  $\varepsilon = 0$ , every router row is uniform and  $C_i = (I - U)/3$ , which commutes with  $Q$ ; hence the local response terms also agree. Define the permutation-orbit distance  $d_\varepsilon$  by

$$d_\varepsilon^2 = \min_P \{ \|A'_\varepsilon - A_\varepsilon P\|_F^2 + \|B'_\varepsilon - P^\top B_\varepsilon\|_F^2 \}.$$

Polynomial dependence on  $\varepsilon$  gives

$$\|\mathcal{R}_{A_\varepsilon, B_\varepsilon} - \mathcal{R}_{A'_\varepsilon, B'_\varepsilon}\|_{F \rightarrow F} = O(\varepsilon), \quad (80)$$

$$d_\varepsilon \longrightarrow \min_P \|Q - P^\top\|_F > 0. \quad (81)$$

Thus no uniform inverse constant survives as  $s_A \downarrow 0$ .

*Basis-rank margin.* Keep  $Q$  from equation 79, put

$$A = 0.3\mathbf{1}\mathbf{1}^\top + 0.1I, \quad A' = AQ, \quad B_\varepsilon = \mathbf{1}e_1^\top + \varepsilon I, \quad B'_\varepsilon = Q^\top B_\varepsilon.$$

The routers have fixed positive entries and fixed full rank; all norms remain bounded, while  $\sigma_{\min}(B_\varepsilon) \rightarrow 0$ . Because  $C_i \mathbf{1} = C'_i \mathbf{1} = 0$ , all cross terms with  $\mathbf{1}e_1^\top$  vanish and

$$B_\varepsilon^\top C_i^2 B_\varepsilon = \varepsilon^2 C_i^2, \quad (B'_\varepsilon)^\top (C'_i)^2 B'_\varepsilon = \varepsilon^2 Q(C_i)^2 Q^\top.$$

The Gram parts again agree exactly, so the response difference is  $O(\varepsilon^2)$ . The parameter distance tends to the strictly positive distance between  $A' = AQ$  and the finite permutation orbit of the full-rank  $A$ . Hence the constant must also deteriorate as  $s_B \downarrow 0$ .

*Interior margin.* Let

$$S = \frac{1}{\sqrt{3}} \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}, \quad Q_t = \exp(tS),$$

and, for sufficiently small  $t > 0$ , set

$$A_t = t\mathbf{1}\mathbf{1}^\top + (1 - 3t)I, \quad B_t = I, \quad A'_t = A_t Q_t, \quad B'_t = Q_t^\top.$$

Here  $S^\top = -S$  and  $S\mathbf{1} = 0$ , so  $Q_t$  is orthogonal and fixes  $\mathbf{1}$ . Moreover  $A'_t = I + t(\mathbf{1}\mathbf{1}^\top - 3I + S) + O(t^2)$ ; its off-diagonal first-order coefficients are  $1 \pm 1/\sqrt{3} > 0$ . Thus both routers are strictly positive for all sufficiently small  $t$ , while their minimum entry is  $\Theta(t)$ . Their smallest singular values are  $1 - 3t$ , and both basis singular values equal one, so every rank and norm margin stays uniform. The products and Gram responses agree exactly. Each row tends to a simplex vertex, so  $C(a_{t,i}) = O(t)$  and  $C(a'_{t,i}) = O(t)$ ; consequently the remaining squared-covariance response difference is  $O(t^2)$ . For the first row, direct expansion gives  $\text{tr}(C(a_{t,1})^2) = 10t^2 + O(t^3)$  and  $\text{tr}(C(a'_{t,1})^2) = 12t^2 + O(t^3)$ . Orthogonal conjugation preserves trace, so this block's difference has spectral norm at least  $(2/3)t^2 + O(t^3)$  before the fixed positive  $\lambda_Z$  multiplier. The full response gap is therefore  $\Theta(t^2)$ . On the other hand, the identity is the nearest permutation for small  $t$ , and  $\|Q_t - I\|_F = \sqrt{2}t + O(t^2)$ , so the parameter-orbit distance is  $\Theta(t)$ . The ratio of parameter distance to response error therefore diverges like  $1/t$ . This supplies the claimed simplex-face ill-conditioning while  $s_A, s_B, L_A, L_B$  remain controlled.

#### B.4 FINITE STRUCTURED RESPONSE TOMOGRAPHY

We now prove Corollary 2. Write

$$Q = AA^\top, \quad T_i = \lambda_Z B^\top C_i^2 B, \quad S = \text{row}(\Theta).$$

Under the rank hypotheses,  $\text{rank}(\Theta) = K$  and

$$S = \text{row}(B), \quad \text{range}(T_i) \subseteq S, \quad T_i x = 0 \text{ for every } x \in S^\perp. \quad (82)$$

Indeed, a left inverse of  $A$  gives  $B = A^\dagger \Theta$ , while  $\Theta = AB$  gives the reverse row-space inclusion; the remaining claims follow from the two outer factors  $B^\top$  and  $B$  in  $T_i$ .

Choose  $u_1, \dots, u_K$  and  $v_1, \dots, v_L$  as in the corollary and define the rows of the  $K$  probes by

$$g_j^{(1)} = v_j + u_1, \quad g_j^{(r)} = u_r \quad (2 \leq r \leq K). \quad (83)$$

Let  $y_i^{(r)} = [\mathcal{R}_{A,B}(G^{(r)})]_i$ . Then

$$Q_{ij} = \eta_B^{-1} \langle y_i^{(1)}, v_j \rangle, \quad (84)$$

$$T_i u_1 = y_i^{(1)} - \eta_B \sum_{j=1}^L Q_{ij} (v_j + u_1), \quad (85)$$

$$T_i u_r = y_i^{(r)} - \eta_B \left( \sum_{j=1}^L Q_{ij} \right) u_r, \quad 2 \leq r \leq K. \quad (86)$$

*Proof of Corollary 2.* Project the first response onto  $S^\perp$ . By equation 82,

$$\Pi_{S^\perp} y_i^{(1)} = \eta_B \sum_{j=1}^L Q_{ij} v_j.$$

Taking its inner product with  $v_j$  gives equation 84. Subtracting the now-known shared term from the unprojected response leaves  $T_i(v_i + u_1) = T_i u_1$ , proving equation 85. For probe  $r \geq 2$ , the same subtraction leaves  $T_i u_r$ , which proves equation 86.

For every  $x \in \mathbb{R}^D$ , Equation 82 gives

$$T_i x = \sum_{r=1}^K \langle u_r, x \rangle T_i u_r.$$

Thus Equations 84–86 recover  $Q$  and every complete local block  $T_i$ , and equation 7 then determines the whole response operator. Two exact charts of the same rank- $K$  product have the same  $S$ , so the same probes recover both operators. Equality on those probes is therefore equivalent to equality of the complete responses, and Theorem 1 gives the permutation conclusion and its converse. The  $Z$ -level row shifts remain unobserved.  $\square$

The complement-code solve in equation 84 and the  $S$ -coordinate solves use orthonormal systems and have condition number one. That statement concerns recovery of  $Q, T_1, \dots, T_L$  from response observations; it does not remove the rank and simplex-boundary conditioning in the inverse from these components back to the factor chart.

**Proposition 6** (Deterministic observation-noise propagation). *Suppose the observed response to probe  $r$  is  $Y^{(r)} + E^{(r)}$ , where  $\|E^{(r)}\|_F \leq \epsilon_r$ , and use Equations 84–86 for reconstruction. Then*

$$\|\hat{Q} - Q\|_F \leq \epsilon_1 / \eta_B. \quad (87)$$

*If  $T^{(S)}$  stacks the  $K \times K$  matrices representing every  $T_i$  in the  $u$  basis, then*

$$\|\hat{T}^{(S)} - T^{(S)}\|_F \leq \left[ (1 + \sqrt{L+1})^2 \epsilon_1^2 + \sum_{r=2}^K (\epsilon_r + \sqrt{L} \epsilon_1)^2 \right]^{1/2}. \quad (88)$$

*Proof.* Right multiplication by the orthonormal-row complement-code matrix is nonexpansive, which proves equation 87. The first probe has  $G^{(1)}(G^{(1)})^\top = I_L + \mathbf{1}\mathbf{1}^\top$ , hence  $\|G^{(1)}\|_2 = \sqrt{L+1}$ . The error in the reconstructed first local column is therefore at most

$$\epsilon_1 + \eta_B \|\hat{Q} - Q\|_F \|G^{(1)}\|_2 \leq (1 + \sqrt{L+1})\epsilon_1.$$

For every later probe, the erroneous shared subtraction contributes at most  $\eta_B \|(\hat{Q} - Q)\mathbf{1}\|_2 \leq \sqrt{L}\epsilon_1$ , in addition to  $\epsilon_r$ . Projection into the orthonormal  $u$  basis is nonexpansive; squaring and summing these  $K$  column bounds proves equation 88.  $\square$

This gives deterministic error radii, not a noisy “if and only if.” Two noisy reconstructions can be separated when their component gap exceeds the sum of the applicable radii. Biased, unknown, or optimizer-state-dependent observation error requires a different model.

**Constructive exact factor recovery.** The diagonal-algebra reduction also yields an explicit reconstruction once  $Q$  and the local blocks are known. Let  $U \in \mathbb{R}^{K \times D}$  contain the orthonormal row-space basis, factor  $Q = XX^\top$  with  $X$  full column rank, and put  $C = X^\dagger \Theta U^\top$ . There is an orthogonal  $O$  with  $X = AO$ , so  $C = O^\top BU^\top$  is invertible. Writing  $T_i^{(S)} = UT_iU^\top$  and  $x_i = X_{i,:}^\top$ , define

$$D_i = \left[ C^{-\top} (T_i^{(S)} / \lambda_Z) C^{-1} \right]^{1/2} + x_i x_i^\top = O^\top \text{diag}(a_i) O,$$

where the square root is the PSD one. A weighted sum of these symmetric matrices has distinct eigenvalues whenever the weighted sums of the columns of  $A$  are distinct. Full column rank makes those columns distinct, so continuous Gaussian weights satisfy this condition almost surely. If  $V$  is the resulting orthogonal eigenvector matrix, the diagonals of  $V^\top D_i V$  recover router rows up to one common permutation; eigenvector signs cancel. The basis then follows from  $\hat{B} = \hat{A}^\dagger \Theta$ . This is a constructive use of established joint-diagonalization algebra (Afsari, 2008; Bhaskara et al., 2014), not an additional uniqueness claim. Its numerical implementation symmetrizes observations and clips negative eigenvalues before taking square roots. Such clipping, and the conditioning of a chosen weighted eigensystem, require separate analysis under noise; the exact argument is not a stability guarantee for that estimator.

**Proposition 7** (Generic probes for the structured response). *Let  $p = D - K \geq 1$ , retain the exact product and rank assumptions above, and draw  $q$  effective-gradient matrices independently with iid standard Gaussian entries. If*

$$q \geq \max \left\{ K, \left\lceil \frac{L}{p} \right\rceil \right\}, \quad (89)$$

*then their full response vectors determine  $Q, T_1, \dots, T_L$ , and hence the complete structured response, with probability one.*

*Proof.* Express each probe in fixed orthonormal coordinates for  $S \oplus S^\perp$ . Concatenate the  $S^\perp$  input coordinates over all probes into  $X_\perp \in \mathbb{R}^{L \times qp}$ . The corresponding projected responses obey  $Y_\perp = \eta_B Q X_\perp$ . When  $qp \geq L$ , the Gaussian matrix  $X_\perp$  has row rank  $L$  almost surely, so it determines  $Q$ . After subtracting that shared term, the  $q$  input coordinates in  $S$  for any fixed layer form a Gaussian  $K \times q$  matrix. It has row rank  $K$  almost surely when  $q \geq K$ , and therefore determines that layer’s local block. Taking a finite union over layers preserves probability one.  $\square$

At the released dimensions  $(L, K, D) = (12, 4, 787,712)$ , four complete Gaussian probe/response vectors would satisfy equation 89 almost surely if the rank solves were actually retained and checked. The older eight-probe artifacts retain norm comparisons rather than a tomographic reconstruction, so they remain a fixed-alternative test. The original four-probe artifact instead executes the explicit design equation 83, evaluates the analytic full-dimensional response, and records the reconstruction residuals. It does not run four full-model autograd/JVP queries; the analytic formula is validated separately on the declared  $4 \times 32$  extraction. The additive audit in Table 6 measures all four full-dimensional responses by reverse automatic differentiation followed by a forward-mode JVP of the packed factor map. It checks reconstruction on two unseen directions per chart and records deterministic noise-bound diagnostics. These queries use linear objectives on the packed parameters; they do not differentiate the language-model task loss or an AdamW update.

This finite observation result is a structured operator-probing corollary, not an independent central innovation. In particular, Novak et al. (2022, Sec. 3.3, Eqs. (7)–(9)) reconstruct a finite NTK from identity NTK–vector products; generic and structured matrix probing is developed by Chiu & Demanet (2012), Halikias & Townsend (2024), and Amsel et al. (2026). Relative to a generic  $LD$ -dimensional identity reconstruction, Equations 83–86 only exploit the present model’s known support and shared-block structure to compress the queries.

Several boundaries are essential. The packing condition  $D - K \geq L$  is sufficient for the explicit construction, not a lower bound for all finite designs. When  $K = 1$ , the single probe recovers the shared response and the softmax local blocks vanish. Exact common product is required because both charts must share  $S$ ; a small numerical product residual alone gives no row-space perturbation theorem. The implementation accordingly activates the common-design conclusion only for charts produced by an explicit exact invertible gauge, with the floating-point product residual recorded separately. Known  $\eta_B > 0$  is used in equation 84; known  $\eta_Z > 0$  and temperature remain part of the chart theorem. Finally, these interventions are executable through linear checkpoint objectives  $\mathcal{L}_G(\Theta) = \langle G, \Theta \rangle$ , but they need not occur in a natural minibatch trajectory.

**Proposition 8** (Finite Gaussian probes for a fixed response alternative). *Let  $m = LD$ , represent a fixed nonzero response difference by  $D \in \mathbb{R}^{m \times m}$ , and draw independent  $x_1, \dots, x_q \sim \mathcal{N}(0, I_m)$ . For every  $0 < t < 1$ ,*

$$\Pr \left[ \max_{\ell \leq q} \|Dx_\ell\|_2 < t \|D\|_2 \right] \leq [2\Phi(t) - 1]^q. \quad (90)$$

Thus failure probability at most  $\beta$  is obtained when

$$q \geq \frac{\log(1/\beta)}{-\log(2\Phi(t) - 1)}. \quad (91)$$

At  $t = 1/2$ , the coefficient is approximately 1.04. This is a fixed-alternative, absolute-threshold statement. It neither calibrates a relative numerical threshold nor uniformly certifies equality of arbitrary response operators when  $q < m$ .

*Proof.* Let  $v$  be a unit right singular vector for the largest singular value of  $D$ . An SVD gives  $\|Dx\|_2 \geq \|D\|_2 |v^\top x|$ , and  $v^\top x \sim \mathcal{N}(0, 1)$ . Independence across probes proves equation 90; solving the right-hand side for  $q$  gives equation 91.

For the uniform limitation, after any  $q < m$  realized probes there exists a nonzero linear map whose kernel contains their span, so all its observed responses vanish. Conversely,  $m$  noiseless Gaussian probes form an invertible  $m \times m$  probe matrix almost surely and therefore determine an arbitrary response operator. Finally, a reported ratio such as  $\|Dx\|_2 / \max\{\|\mathcal{R}x\|_2, \|\mathcal{R}'x\|_2\}$  has a random denominator; equation 90 does not calibrate it without additional bounds on those response norms.  $\square$

**Corollary 4** (Pointwise probe detection from inverse stability). *Fix two exact product-equivalent charts in one nondegenerate set from Corollary 1, fix them before drawing the probes, and let  $C$  be that set’s common inverse constant. Define*

$$d_{\text{orb}} = \min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2}$$

and  $D = \mathcal{R}_{A,B} - \mathcal{R}_{A',B'}$ . If  $d_{\text{orb}} \geq \Delta > 0$ , then independent standard-Gaussian probes satisfy, for every  $0 < t < 1$ ,

$$\Pr \left[ \max_{\ell \leq q} \|Dx_\ell\|_2 \geq t\Delta/C \right] \geq 1 - [2\Phi(t) - 1]^q. \quad (92)$$

*Proof.* The inverse bound gives  $\|D\|_2 \geq d_{\text{orb}}/C \geq \Delta/C$ . The complement of the event in equation 92 is therefore contained in the small-ball event of Proposition 8, which proves the claim.  $\square$

At  $t = 1/2$  and  $q = 8$ , the upper bound on the miss probability is about  $4.62 \times 10^{-4}$ . This number does not calibrate the released response ratios: the threshold  $t\Delta/C$  is absolute,  $C$  may be loose, the alternative and common nondegenerate set must be fixed before the draws, and the statement is not simultaneous over probe-selected charts. If each observed difference has deterministic error at most  $\epsilon$ , the corresponding observable lower threshold is only  $t\Delta/C - \epsilon$  when positive.

**Why the hypotheses and target matter.** The following examples separate exact theorem hypotheses from broader interpretations.

- Equality of the effective model is indispensable. With  $A' = A$  and  $B' = -B$ , the router Gram and every quadratic matrix  $B^\top C_i^2 B$  are unchanged, but  $A'B' = -AB$ . The response alone does not identify the represented effective matrix.
- Router rank cannot be dropped. For  $K = D = 3$ , let every row of  $A$  be  $\mathbf{1}^\top/3$ , let  $B = I_3$ , and let  $Q$  be a non-permutation orthogonal rotation on  $\mathbf{1}^\perp$  with  $Q\mathbf{1} = \mathbf{1}$ . Then  $A' = AQ = A$ ,  $B' = Q^\top$ , and both the product and response are unchanged. The two factors are not permutation equivalent.
- Basis rank cannot be dropped. Let  $K = D = 3$ ,  $A = 0.3\mathbf{1}\mathbf{1}^\top + 0.1I_3$ , and  $B = \mathbf{1}e_1^\top$ . For a sufficiently small non-permutation orthogonal rotation  $Q$  fixing  $\mathbf{1}$ ,  $A' = AQ$  remains positive and row-stochastic, while  $B' = Q^\top B = B$ . The products and router Gram matrices agree, and  $C_i B = 0$  makes every local softmax response term zero.
- Joint router training is material in general. If  $\eta_Z = 0$  and  $K \geq 3$ , choose any positive full-rank router  $A$  close enough to uniform, a full-row-rank  $B$ , and a small non-permutation orthogonal  $Q$  fixing  $\mathbf{1}$ . Then  $(AQ, Q^\top B)$  is valid and product equivalent, while the remaining response  $\eta_B AA^\top G$  is unchanged.
- Equality on one task gradient is not operator equality. At a stationary point the observed gradient is zero for every parameterization, regardless of whether their response operators differ. There is also a nonzero blind direction away from stationarity. For a non-permutation exact full-rank pair with common row space  $S$ , suppose  $D > K$  and  $L > 2K$ . Choose nonzero  $v \in S^\perp$ . Since  $\text{rank}(AA^\top - A'(A')^\top) \leq 2K < L$ , choose nonzero  $w$  in its kernel and set the rows of  $G$  to  $g_j = w_j v$ . Both local block differences kill  $v$ , while the shared difference kills  $w$ , so  $(\mathcal{R}_{A,B} - \mathcal{R}_{A',B'})(G) = 0$  even though the two response operators differ by Theorem 1. A passive trajectory probes only its realized directions.
- Strict positivity is automatic for finite softmax logits. The exact algebra above in fact only needs each router row to lie in the closed simplex: positive semidefiniteness and  $C_i \mathbf{1} = 0$  make  $H_i$  singular. Thus the formula-defined response theorem extends to boundary rows when all stated factor ranks remain full. A uniform interior margin is material only for a quantitative inverse bound.

These are sufficient conditions, not a claim of a jointly minimal assumption set. In particular, the proof uses  $\eta_B > 0$  to recover the router Gram matrix, but we have not established that  $\eta_B > 0$  is necessary once the complete collection of local softmax terms is available.

**Partial audit of the  $\eta_B = 0$  boundary.** Among the partial subcases closed in the released assumption ledger, we state two simple ones here without promoting them to separate contributions. Write an equivalent chart as  $A' = AM$ ,  $B' = M^{-1}B$ , with  $M\mathbf{1} = \mathbf{1}$ . For a router row written as a column  $p$ , put  $q = M^\top p$ . Equality of the remaining local blocks reduces, after cancelling the full-row-rank basis, to

$$C(q)^2 = M^\top C(p)^2 M. \quad (93)$$

If  $M \geq 0$ , define  $K = \text{Diag}(p)M \text{Diag}(q)^{-1}$ . Then  $\mathbf{1}^\top K = \mathbf{1}^\top$ , and equation 93 is equivalent to

$$(K - p\mathbf{1}^\top)^\top (K - p\mathbf{1}^\top) = (I - q\mathbf{1}^\top)^\top (I - q\mathbf{1}^\top).$$

Thus every two columns of  $K$  are at distance  $\sqrt{2}$ . They also lie in the probability simplex, whose diameter  $\sqrt{2}$  is attained only by distinct vertices. Hence  $K = P$ ; using  $Kq = p$  in its definition gives  $M = P$ . One positive router row therefore rules out every non-permutation nonnegative gauge.

For  $K = 2$ , no sign restriction is needed. Up to a candidate column swap, write  $M = \begin{bmatrix} x & 1-x \\ y & 1-y \end{bmatrix}$  with  $d = x - y > 0$ . A row  $(p, 1-p)$  maps to first coordinate  $q = y + dp$ , and equation 93 becomes  $q(1-q) = dp(1-p)$ . Full router rank supplies two distinct  $p_1 < p_2$ . Subtracting the two equations gives  $q_1 + q_2 = p_1 + p_2 =: S$ ; adding them and using  $q_2 - q_1 = d(p_2 - p_1) = d\Delta$  gives

$$0 = \frac{1-d}{2} \{S(2-S) + d\Delta^2\}.$$

The bracket is positive, so  $d = 1$ ,  $y = 0$ , and  $M = I$ , before undoing the possible column swap. For signed gauges with  $K \geq 3$ , even when both  $A$  and  $AM$  are positive and full rank, compatibility of

the rowwise regular-simplex congruences remains unresolved. Numerical searches that collapse rank, approach a simplex face, or satisfy only one row are not treated as a proof or counterexample. We therefore retain  $\eta_B > 0$  in the main theorem while making no necessity claim.

The theorem is specific to the named Euclidean metric and known block rates. Natural gradient, Adam/AdamW or momentum state, finite optimizer steps, factor-specific penalties, and router auxiliary losses define different response objects. Most importantly,  $(\Theta, \mathcal{R})$  identifies a present optimizer chart, not the factors or graph that historically generated the data.

## B.5 POSITIVE RECOVERY RESULTS

**Lemma 1** (Unique barycentric coordinates). *If  $b_1, \dots, b_K$  are affinely independent, then  $aB = a'B$  for  $a, a' \in \Delta_K$  implies  $a = a'$ .*

*Proof.* Set  $q = a - a'$ . The simplex constraint gives  $q\mathbf{1} = 0$ , and equality of the effective parameters gives  $qB = 0$ . A nonzero  $q$  satisfying both equations would be a nontrivial affine dependence among the rows of  $B$ .  $\square$

*Proof of Proposition 3.* Every effective parameter lies in the convex hull of the bases:  $\{\theta_i\} \subseteq \text{conv}\{b_j\}$ . Separability supplies the reverse containment at the level of convex hulls, because each  $b_j$  occurs as an anchor row of  $\Theta$ . Therefore

$$\text{conv}\{\theta_1, \dots, \theta_L\} = \text{conv}\{b_1, \dots, b_K\}. \quad (94)$$

Affine independence makes the right side a  $(K-1)$ -simplex whose vertices are exactly the unordered set  $\{b_1, \dots, b_K\}$ . Applying the same argument to  $(A', B')$  shows that its bases are the same vertex set. Thus  $B' = P^\top B$  for a permutation matrix  $P$ . Relative to these identified vertices, Lemma 1 gives unique coordinates for every row, hence  $A' = AP$ .

For the one-hot consequence,  $\theta_i = b_{c_i}$ . Affine independence implies that bases are pairwise distinct, so  $\theta_i = \theta_{i'}$  if and only if  $c_i = c_{i'}$ . Full basis usage provides all anchors and makes the basis set observable. Without full usage, an unused basis is unconstrained; with duplicate bases, their assignment labels are observationally interchangeable.  $\square$

Separability must restrict the candidate model class, not merely the planted truth. If alternative bases may lie outside the observed hull and need not appear as anchors, a larger simplex can contain the same  $\theta_i$  and assign them different soft barycentric coordinates.

*Proof of Theorem 2.* If  $c_i = c_{i'}$ , the triangle inequality gives

$$\|\widehat{\theta}_i - \widehat{\theta}_{i'}\|_2 = \|e_i - e_{i'}\|_2 \leq 2\varepsilon. \quad (95)$$

If  $c_i \neq c_{i'}$ , the reverse triangle inequality gives

$$\|\widehat{\theta}_i - \widehat{\theta}_{i'}\|_2 \geq \|b_{c_i} - b_{c_{i'}}\|_2 - \|e_i\|_2 - \|e_{i'}\|_2 \geq \gamma - 2\varepsilon > 2\varepsilon. \quad (96)$$

The threshold graph therefore contains all and only within-class edges, so its connected components recover the partition. A cluster mean equals its basis plus an average of error vectors. By convexity of the norm, that average has norm at most  $\varepsilon$ .  $\square$

The  $4\varepsilon$  condition is a clean sufficient margin, not a necessary one. Stable recovery without some separation condition is impossible: two bases closer than the observation uncertainty can exchange assignments without changing the observed data appreciably.

## B.6 FINITE-SAMPLE RECOVERY BY SHARED-INPUT PROBES

For a pair  $(i, j)$  write

$$X_s^{ij} = \|Y_i(H_s) - Y_j(H_s)\|_2^2, \quad d_{ij}^2 = \mathbb{E}X_s^{ij}, \quad \widehat{d}_{ij}^2 = n^{-1} \sum_{s=1}^n X_s^{ij}. \quad (97)$$

The observation  $Y_i$  may include seeded measurement noise. The population distances and their gap must be computed under that same observation model. Noise that differs systematically between layers can close or reverse the gap for the clean functional partition, in which case the theorem does not apply; it does not silently redefine the recovery target.

*Proof of Theorem 3.* Let  $N_p = \binom{L}{2}$ . Substitute  $u = \delta/4$  into Equation 19. The stated sample-size condition implies for each pair

$$\Pr\left(|\hat{d}_{ij}^2 - d_{ij}^2| > \delta/4\right) \leq \frac{\eta}{N_p}. \quad (98)$$

A union bound gives, with probability at least  $1 - \eta$ ,

$$\max_{i < j} |\hat{d}_{ij}^2 - d_{ij}^2| \leq \delta/4. \quad (99)$$

On this event, every within-class pair obeys

$$\hat{d}_{ij}^2 \leq w + \delta/4 < (w + b)/2, \quad (100)$$

whereas every between-class pair obeys

$$\hat{d}_{ij}^2 \geq b - \delta/4 > (w + b)/2. \quad (101)$$

Thresholding therefore recovers all pairwise equivalence decisions and hence the partition.

For agglomerative clustering, Equation 99 implies that every distance between subsets of one true class is smaller than every distance between subsets of different true classes, for single, complete, or average linkage. While more than  $K_f$  clusters remain, at least one true class is split, and the next merge is necessarily within that class. Induction shows that cutting the dendrogram at  $K_f$  clusters returns the true partition. This class count is a property of the functional equivalence relation and need not equal the number of router bases.  $\square$

**A bounded special case.** If the observations are independent across probe draws  $s$  with the same pairwise expectations, and  $0 \leq X_s^{ij} \leq R^2$  almost surely for every pair, Hoeffding’s inequality gives

$$\Pr(|\hat{d}_{ij}^2 - d_{ij}^2| > u) \leq 2 \exp(-2nu^2/R^4). \quad (102)$$

Thus it is sufficient that

$$n \geq \frac{8R^4}{\delta^2} \log \frac{2\binom{L}{2}}{\eta}. \quad (103)$$

For unbounded probes, Equation 19 must be justified by the probe distribution, module class, and noise model rather than assumed silently. Clipping or normalizing probe responses supplies a directly checkable bounded version at the cost of changing the target metric.

The released paired sweep uses iid Gaussian inputs and  $Y_{is} = r_i(H_s) + \sigma\epsilon_{is}$  with iid Gaussian  $\epsilon_{is}$  and a fixed absolute  $\sigma$ . For its finite MLP and Transformer blocks, LayerNorm with a positive numerical stabilizer bounds the normalized inputs, and the remaining fixed affine, GELU, softmax-attention, and residual-update maps give a bounded clean update. A bounded vector plus Gaussian noise has a sub-exponential squared norm, so Bernstein constants exist for each frozen configuration. The sweep does not estimate uniform numerical values of  $(\nu, c)$  or the population gap  $\delta$ ; consequently its reported empirical gap is descriptive and is not presented as an application of the theorem’s population certificate.

**What the certificate means.** The theorem targets equality under the chosen  $P_H$  and observation model. It cannot certify global functional identity outside the support of  $P_H$ . Nor does it identify a historical generative graph when two distinct teacher modules happen to agree on that support.

## B.7 OPTIMIZATION AND INITIALIZATION EFFECTS

*Proof of Proposition 4.* By the chain rule for  $\Theta = AB$ ,

$$\nabla_A \mathcal{L} = GB^\top, \quad \nabla_B \mathcal{L} = A^\top G. \quad (104)$$

Euclidean gradient flow therefore satisfies  $\dot{A} = -GB^\top$  and  $\dot{B} = -A^\top G$ . Differentiating the product gives

$$\dot{\Theta} = \dot{A}B + A\dot{B} \quad (105)$$

$$= -G(B^\top B) - (AA^\top)G. \quad (106)$$

The right-hand side is not determined by  $AB$  alone. If two factorizations induce different linear operators on  $G$ , choosing  $G$  outside the null space of their difference produces different initial effective velocities. Non-orthogonal gauge transformations generically alter both Gram matrices.  $\square$

The previous calculation optimizes  $A$  directly and is useful because its preconditioning is explicit. The constrained softmax model has the same coordinate dependence. In the next derivation vectors are columns, so  $\theta_i = B^\top a_i$  and  $g_i = \nabla_{\theta_i} \mathcal{L}$ .

**Proposition 9** (Exact softmax effective dynamics). *Let*

$$a_i = \text{softmax}(z_i/\tau), \quad J_i = \frac{1}{\tau} (\text{Diag}(a_i) - a_i a_i^\top). \quad (107)$$

*Without factor-specific regularizers, Euclidean gradient flow in  $Z, B$  obeys*

$$\dot{z}_i = -\eta_Z J_i B g_i, \quad (108)$$

$$\dot{B} = -\eta_B \sum_{r=1}^L a_r g_r^\top, \quad (109)$$

$$\dot{\theta}_i = -\eta_Z B^\top J_i^2 B g_i - \eta_B \sum_{r=1}^L \langle a_i, a_r \rangle g_r. \quad (110)$$

*Proof.* The Jacobian of  $a_i$  with respect to  $z_i$  is the symmetric matrix  $J_i$ . Since  $\nabla_{a_i} \mathcal{L} = B g_i$ , the chain rule gives  $\nabla_{z_i} \mathcal{L} = J_i B g_i$ . Similarly,  $\nabla_B \mathcal{L} = \sum_r a_r g_r^\top$ . Including the block learning rates in the flow therefore gives

$$\dot{a}_i = J_i \dot{z}_i = -\eta_Z J_i^2 B g_i. \quad (111)$$

Substituting  $\dot{a}_i$  and  $\dot{B}$  into  $\dot{\theta}_i = \dot{B}^\top a_i + B^\top \dot{a}_i$  yields Equation 110.  $\square$

*Proof of Proposition 5.* Let  $C_i = \text{Diag}(a_i) - a_i a_i^\top$ , so  $J_i = C_i/\tau$ . The matrix  $C_i$  is positive semidefinite, hence

$$\|C_i\|_2 \leq \text{tr}(C_i) = 1 - \|a_i\|_2^2 \quad (112)$$

$$\leq 1 - (1 - \varepsilon)^2 \leq 2\varepsilon. \quad (113)$$

Using  $\nabla_{z_i} \mathcal{L} = J_i B g_i$  gives

$$\|\nabla_{z_i} \mathcal{L}\|_2 \leq \|J_i\|_2 \|B g_i\|_2 \leq \frac{2\varepsilon}{\tau} \|B g_i\|_2. \quad (114)$$

The exact router velocity additionally obeys

$$\|\dot{a}_i\|_2 \leq \eta_Z \left( \frac{2\varepsilon}{\tau} \right)^2 \|B g_i\|_2. \quad (115)$$

The bound is instantaneous: it accounts for both temperature and the current probability  $1 - \varepsilon$ , while varying  $\tau$  at fixed logits changes  $\varepsilon$  as well. It does not by itself bound cumulative router motion or guarantee preservation of an argmax over a training interval.  $\square$

**Proposition 10** (Symmetric initialization trap). *If every router row is uniform and every basis is initialized identically, then deterministic gradient flow keeps the router uniform and the bases identical for all time for which the solution exists.*

*Proof.* If all basis rows equal  $b$ , then  $B g_i = (b^\top g_i) \mathbf{1}$ . Since  $J_i \mathbf{1} = 0$ , every logit gradient is zero. Under a uniform router, every basis receives the same gradient  $K^{-1} \sum_i g_i$ . Equality of the bases is therefore invariant, which in turn keeps every router gradient zero.  $\square$

The proposition is not a claim that practical floating-point training always remains symmetric: independent random initialization, parameter-specific stochastic perturbations, and implementation nondeterminism can break the equality. Shared minibatch sampling alone preserves this symmetry when every sampled loss depends on the factors only through  $AB$ : the same gradient equalities hold separately for each minibatch. It identifies the source of specialization as an explicit or stochastic symmetry breaker rather than information contained in the initial effective model.

## B.8 DETAILED CLAIM BOUNDARY

The formal results support the following distinctions.

- Router entropy, smoothness, bandedness, cosine similarity, and argmax graphs are gauge-dependent coordinates. They require gauge-counterfactual or initialization-intervention evidence before receiving a structural interpretation.
- The effective-weight equality partition is gauge invariant. It is still only a parameter-space partition: neural permutation and scaling symmetries can make unequal parameters functionally equivalent.
- Shared-input probes target functional equivalence under a named probe distribution. They do not establish equality outside its support and need not recover the historical graph that generated a teacher.
- A generative graph becomes recoverable under explicit conditions such as one-hot routing, full basis usage, affine independence or separation, and an observation regime strong enough to distinguish the bases. Low task loss alone supplies none of these conditions.
- Affine independence is insufficient for an unknown fully soft factorization. It ensures unique barycentric coordinates only after vertices are known; anchors or another valid simplex-identification condition are still required.
- Factor-specific normalization or regularization may choose a useful gauge. It does not retroactively make that coordinate the data-generating graph. Exact equality constraints can reduce the gauge group but require a fresh identifiability analysis.
- The dynamics results demonstrate coordinate-dependent preconditioning, saturation, and a symmetry trap. They do not prove a universal optimization preference for smooth, monotone, periodic, sparse, or low-entropy routers.
- No strict function-class hierarchy, generalization improvement, or causal performance benefit follows from the algebraic factorization alone. Such claims require separate approximation, statistical, and compute-matched arguments.

## C EXTENSIONS TO IMPERFECT PRODUCTS AND NATURAL PROBES

### C.1 JOINT PRODUCT AND RESPONSE PERTURBATIONS

The same-product restriction in Theorem 4 can be relaxed for small product perturbations by constructing an intermediate factorization with exactly the reference product. This is an inverse-property result for valid factors; it does not certify the unconstrained numerical recovery algorithm.

**Corollary 5** (Joint product–response stability). *Let two factor pairs satisfy all hypotheses of Theorem 4 except equal products, with the same constants  $\alpha, s_A, s_B, L_A, L_B$  and known positive rates and temperature. Write*

$$\epsilon_\Theta = \|AB - A'B'\|_F, \quad \epsilon_R = \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}.$$

*Assume  $\epsilon_\Theta \leq \min\{\alpha s_B/2, s_A s_B/4\}$ . Let  $C_*$  be the explicit constant of that theorem evaluated at  $(\alpha/2, s_A/2, s_B/2, 2L_A, 2L_B)$ , and put*

$$\kappa = (s_B^{-2} + 4s_A^{-2})^{1/2}, \quad M = 2L_B, \quad L_R = [(2\eta_B\sqrt{L} + 6\lambda_Z M^2)^2 + (2\lambda_Z M)^2]^{1/2}.$$

*Then the joint Frobenius distance modulo a common permutation satisfies*

$$d_{\text{perm}}((A, B), (A', B')) \leq C_* \epsilon_R + \kappa(1 + C_* L_R) \epsilon_\Theta.$$

*Proof.* Let  $P_A = AA^\dagger$ ,  $\tilde{A} = P_A A'$ , and  $\tilde{B} = \tilde{A}^\dagger AB$ . Since  $A\mathbf{1} = \mathbf{1}$ ,  $P_A\mathbf{1} = \mathbf{1}$  and  $\tilde{A}\mathbf{1} = \mathbf{1}$ . Moreover,

$$(I - P_A)A' = (I - P_A)(A'B' - AB)B'^\dagger, \quad \|\tilde{A} - A'\|_F \leq \epsilon_\Theta/s_B.$$

Thus  $\min \tilde{A} \geq \alpha/2$  and  $\sigma_{\min}(\tilde{A}) \geq s_A/2$ . Its column space is that of  $A$ , so  $\tilde{A}\tilde{B} = AB$ . Also

$$\tilde{B} - B' = \tilde{A}^\dagger P_A(AB - A'B'), \quad \|\tilde{B} - B'\|_F \leq 2\epsilon_\Theta/s_A.$$

Consequently  $\sigma_{\min}(\tilde{B}) \geq s_B/2$ , and both pairs  $(A, B)$ ,  $(\tilde{A}, \tilde{B})$  obey the enlarged domain bounds defining  $C_*$ . The joint displacement from  $(A', B')$  to the intermediate pair is at most  $\kappa\epsilon_\Theta$ .

For simplex rows  $a, a'$ , let  $C(a) = \text{diag}(a) - aa^\top$ . The elementary bounds  $\|a\|_2 \leq 1$ ,  $\|C(a)\|_2 \leq 1$ , and  $\|C(a) - C(a')\|_2 \leq 3\|a - a'\|_2$  imply  $\|C(a)^2 - C(a')^2\|_2 \leq 6\|a - a'\|_2$ . In the response formula, the Gram contribution changes by at most  $2\eta_B\sqrt{L}\|\Delta A\|_F$ . Each local diagonal block changes by at most  $\lambda_Z(6M^2\|\Delta a_i\|_2 + 2M\|\Delta B\|_F)$ . The norm of the direct sum is the maximum block norm; therefore the full response is  $L_R$ -Lipschitz in the joint factor norm on this domain. The response discrepancy between  $(A, B)$  and the intermediate pair is bounded by  $\epsilon_R + L_R\kappa\epsilon_\Theta$ . Apply the same-product theorem and then the triangle inequality; common permutations preserve the joint norm.  $\square$

If two valid feasible pairs each match the same noisy observations within  $e_\Theta, e_R$ , the corollary applies with  $2e_\Theta, 2e_R$ , provided the small-product-error condition holds. It does not establish existence of a feasible pair, an efficient constrained solver, or stability near rank loss. The constants are conservative and not numerical accuracy predictions.

## C.2 A SUFFICIENT CONDITION FOR TOMOGRAPHY WITH NATURAL GRADIENTS

Let  $U \in \mathbb{R}^{K \times D}$  have orthonormal rows spanning the exact product's row space. Write the structured response as  $R^{(p)} = QG^{(p)} + \mathcal{T}(G^{(p)}U^\top)U$ , where  $Q = \eta_B AA^\top$  and  $\mathcal{T}$  applies a  $K \times K$  matrix  $T_i$  to each row coordinate, using column-vector convention. Define  $G_\perp^{(p)} = G^{(p)} - (G^{(p)}U^\top)U$  and similarly  $R_\perp^{(p)}$ .

**Proposition 11** (Rank-checked natural-probe reconstruction). *If  $H = \sum_p G_\perp^{(p)} G_\perp^{(p)\top}$  is invertible and, for every layer  $i$ , the vectors  $(G^{(p)}U^\top)_{i,:}$  span  $\mathbb{R}^K$ , the observations uniquely determine  $Q$  and all  $T_i$ . They therefore determine the response on all gradients within the assumed response family.*

*Proof.* Projection eliminates the local terms, giving  $R_\perp^{(p)} = QG_\perp^{(p)}$ . Hence

$$Q = \left( \sum_p R_\perp^{(p)} G_\perp^{(p)\top} \right) H^{-1}.$$

Subtract  $QG^{(p)}$  and project onto  $U$ ; each layer's remaining linear system has a unique  $T_i$  by its spanning assumption.  $\square$

This is a sufficient algebraic condition, not a guarantee about task gradients or an optimal-query result. The implementation checks numerical ranks and conditioning, and uses fixed pseudoinverse tolerances. Small singular values can amplify measurement noise. The designed probes avoid relying on empirical task-gradient richness; natural probes trade that design for a condition that must be checked at the checkpoint. Measurements here still use controlled Euclidean responses at a fixed checkpoint, not an observed AdamW trajectory.

## D GLOBAL STABILITY FROM FINITE NOISY OBSERVATIONS

This section closes the observation-level gap between an exact-product finite-probe reconstruction and a full-operator inverse bound. It is an explicit measurement-level consequence of Theorem 4 and Corollary 5, not a new general inverse theorem. The conclusion is global over all feasible factors in a specified nondegenerate domain. No numerical recovery algorithm enters its proof.

**Domain, observations, and quantifiers.** Fix  $L \geq K \geq 2$ ,  $D - K \geq L$ , and known  $\eta_B, \eta_Z, \tau > 0$ , with  $\lambda = \eta_Z/\tau^2$ . For positive  $\alpha, s_A, s_B, M$ , let

$$\mathcal{D} = \left\{ (A, B) : \begin{array}{l} A \in \mathbb{R}^{L \times K}, A \mathbf{1}_K = \mathbf{1}_L, A_{ij} \geq \alpha, \sigma_{\min}(A) \geq s_A, \\ B \in \mathbb{R}^{K \times D}, \sigma_{\min}(B) \geq s_B, \|B\|_F \leq M \end{array} \right\}.$$

The margins describe an assumed parameter domain; they are not inferred from one candidate's conditioning. Given any observed product  $\hat{\Theta} \in \mathbb{R}^{L \times D}$ , choose an orthonormal-row matrix  $U \in \mathbb{R}^{K \times D}$  from its top  $K$  right singular vectors, and set  $\hat{P} = U^\top U$ . Choose  $W \in \mathbb{R}^{L \times D}$  with  $WW^\top = I_L$  and  $WU^\top = 0$ ; this is possible since  $D - K \geq L$ . All choices depend only on  $\hat{\Theta}$  and dimensions, with a fixed tie-breaking rule if needed. Writing  $u_p^\top$  for row  $p$  of  $U$ , define

$$G_p = \frac{W + \mathbf{1}_L u_p^\top}{\sqrt{2L}}, \quad p = 1, \dots, K. \quad (116)$$

Orthogonality gives  $\|G_p\|_F = 1$ . The complement code  $W$  is deliberately repeated in every query. Probes are selected before seeing their responses.

For any observed response matrices  $\hat{R}_p$ , define

$$\mathcal{F} = \left\{ (A, B) \in \mathcal{D} : \begin{array}{l} \|AB - \hat{\Theta}\|_F \leq \varepsilon_\Theta, \\ (\sum_{p=1}^K \|\mathcal{R}_{A,B}(G_p) - \hat{R}_p\|_F^2)^{1/2} \leq \varepsilon_R \end{array} \right\}. \quad (117)$$

This is the entire feasible set in the original  $D$ -dimensional basis space, not a fit restricted to  $\hat{P}$ . Noise may be deterministic, adversarial, and dependent on the product observation or probes, provided these norm bounds hold.

**Explicit constants.** Put

$$h = \sqrt{2L}(K^{-1/2} + 1 + \sqrt{L}), \quad \beta = \frac{8\lambda M}{s_A}(h\sqrt{K} + 1), \quad (118)$$

$$\kappa = (s_B^{-2} + 4s_A^{-2})^{1/2}, \quad L_R = \left[ (2\eta_B\sqrt{L} + 24\lambda M^2)^2 + (4\lambda M)^2 \right]^{1/2}. \quad (119)$$

To make the inherited inverse constant fully explicit, define

$$L_M = \max\{1, 4\sqrt{L}/s_A\}, \quad c_G = \frac{4L}{\eta_B s_A^2}, \quad c_H = \frac{8}{\lambda s_B^2}, \quad (120)$$

$$\mu_* = \frac{\alpha}{2} \left( 1 + \frac{1}{4L_M^2} \right), \quad c_J = \frac{L_M^2(c_G + c_H)}{\mu_*}, \quad c_D = \frac{2\sqrt{LK}}{s_A} c_J, \quad (121)$$

$$c_{\text{row}} = [Kc_D^2 + (c_G + Kc_D^2)^2]^{1/2}, \quad \delta_0 = \min \left\{ 1, \frac{1}{2c_G}, \frac{1}{2\sqrt{K} c_{\text{row}}} \right\}. \quad (122)$$

Finally set

$$C_* = \max \left\{ 2\sqrt{L + L_M^2 M^2} \sqrt{K} c_{\text{row}}, \frac{4\sqrt{L + M^2}}{\delta_0} \right\}, \quad (123)$$

$$C_R = 2hC_*, \quad (124)$$

$$C_\Theta = \beta C_* + 2\kappa(1 + C_* L_R), \quad (125)$$

$$\varepsilon_0 = \min\{\alpha s_B/4, s_A s_B/8\}. \quad (126)$$

These constants can be very large. No sharpness or useful numerical accuracy at a particular finite noise level is asserted.

**Theorem 5** (Whole-feasible-set diameter from  $K$  noisy probes). *Under the preceding definitions, if  $0 \leq \varepsilon_\Theta \leq \varepsilon_0$  and  $\varepsilon_R \geq 0$ , then*

$$\text{diam}_{d_{\text{perm}}}(\mathcal{F}) \leq C_\Theta \varepsilon_\Theta + C_R \varepsilon_R, \quad d_{\text{perm}}(x, x') = \min_{\Pi \in \mathfrak{S}_K} (\|A' - A\Pi\|_F^2 + \|B' - \Pi^\top B\|_F^2)^{1/2}. \quad (127)$$

*No smallness restriction on  $\varepsilon_R$  is needed. Empty feasible sets have diameter zero by convention; recovery interpretations require a nonempty feasible set.*

*Proof.* We give the observation-to-operator bridge first, then apply the explicit global factor inverse to the original, unprojected factors.

*Step 1: uniform control outside the observed subspace.* For every  $(A, B) \in \mathcal{F}$ ,  $AB$  is a rank- $K$  competitor in the best rank- $K$  approximation of  $\hat{\Theta}$ . Hence

$$\|\hat{\Theta}(I - \hat{P})\|_F \leq \varepsilon_\Theta, \quad \|AB(I - \hat{P})\|_F \leq 2\varepsilon_\Theta.$$

Since  $A$  has a left inverse with  $\|A^\dagger\|_2 \leq s_A^{-1}$ ,

$$\|B - B\hat{P}\|_F \leq 2\varepsilon_\Theta/s_A. \quad (128)$$

Set  $\bar{B} = B\hat{P}$ . This is an analytical device, not an assumption that the true basis lies in the estimated subspace. For  $S_i = \text{diag}(a_i) - a_i a_i^\top$ ,  $\|S_i\|_2 \leq 1$  and  $\|\bar{B}\|_2, \|B\|_2 \leq M$ , so

$$\|B^\top S_i^2 B - \bar{B}^\top S_i^2 \bar{B}\|_2 \leq 2M\|B - \bar{B}\|_F.$$

The shared Gram term is unchanged; the remaining operator difference is block diagonal. Thus, uniformly over  $\mathcal{F}$ ,

$$\|\mathcal{R}_{A,B} - \mathcal{R}_{A,\bar{B}}\|_{F \rightarrow F} \leq t, \quad t = \frac{4\lambda M}{s_A} \varepsilon_\Theta. \quad (129)$$

No rank property of  $\bar{B}$  is required.

*Step 2: stable decoding on a common support.* Consider the larger linear class  $\mathcal{S}(G) = QG + \mathcal{T}(GU^\top)U$ , where  $Q \in \mathbb{R}^{L \times L}$  and  $\mathcal{T}$  right-multiplies row  $i$  by an arbitrary  $K \times K$  matrix  $T_i$ . Symmetry is unnecessary. For the difference of two such operators, let  $E_p = \Delta\mathcal{S}(G_p)$  and  $E = (\sum_p \|E_p\|_F^2)^{1/2}$ . Set  $c = (2L)^{-1/2}$ . Complement projection gives  $E_p W^\top = c\Delta Q$ , whence

$$\|\Delta Q\|_F \leq \frac{E}{c\sqrt{K}}. \quad (130)$$

Projection onto  $U$  gives, row by row,

$$E_p U^\top = c(\Delta Q \mathbf{1}_L) e_p^\top + c[e_p^\top \Delta T_i]_{i=1}^L.$$

Stacking the  $K$  equations enumerates every row of every  $\Delta T_i$ . The triangle inequality and  $\|\Delta Q \mathbf{1}_L\|_2 \leq \sqrt{L} \|\Delta Q\|_F$  give

$$\left( \sum_i \|\Delta T_i\|_F^2 \right)^{1/2} \leq E/c + \sqrt{K} \|\Delta Q \mathbf{1}_L\|_2 \leq (1 + \sqrt{L})E/c. \quad (131)$$

The shared term has induced norm  $\|\Delta Q\|_2$  and the local direct sum has norm at most  $\max_i \|\Delta T_i\|_2$ . Equations 130–131 therefore imply

$$\|\Delta\mathcal{S}\|_{F \rightarrow F} \leq hE. \quad (132)$$

For a projected factor response, take  $Q = \eta_B A A^\top$  and  $T_i = \lambda U \bar{B}^\top S_i^2 \bar{B} U^\top$ . The lemma applies even if those projected factors fail the domain's rank margins.

*Step 3: return to the original full operators.* Choose any  $x = (A, B), x' = (A', B') \in \mathcal{F}$ . Their observed response stacks differ by at most  $2\varepsilon_R$ . Each unit-norm probe incurs at most  $t$  error under Equation 129; the stack incurs at most  $\sqrt{K}t$  for each pair. Applying Equation 132, then adding the two full-operator compression errors, yields

$$\|\mathcal{R}_x - \mathcal{R}_{x'}\|_{F \rightarrow F} \leq h(2\varepsilon_R + 2\sqrt{K}t) + 2t \quad (133)$$

$$= 2h\varepsilon_R + \beta\varepsilon_\Theta. \quad (134)$$

Every pair uses the same observed  $U, W$  and probes.

*Step 4: the global factor inverse.* The two original factors obey the domain bounds with  $L_A = \sqrt{L}$  and  $L_B = M$ , and  $\|AB - A'B'\|_F \leq 2\varepsilon_\Theta$ . Our threshold ensures  $2\varepsilon_\Theta \leq \min\{\alpha s_B/2, s_A s_B/4\}$ , exactly the product-error condition of Corollary 5. That corollary's  $C_*$  is the constant of Theorem 4 evaluated at  $(\alpha/2, s_A/2, s_B/2, 2\sqrt{L}, 2M)$ , which expands to Equations 122–126. It includes the bounded-domain argument for large response differences, not just a local inverse estimate. Consequently,

$$d_{\text{perm}}(x, x') \leq C_* \|\mathcal{R}_x - \mathcal{R}_{x'}\|_{F \rightarrow F} + \kappa(1 + C_* L_R) \|AB - A'B'\|_F.$$

Substituting Equation 134 proves the claimed bound for an arbitrary pair in  $\mathcal{F}$ , hence for its entire diameter. No connected-component or candidate-neighborhood premise was used.  $\square$

**Recovery interpretation and boundaries.** If the true factors belong to  $\mathcal{D}$  and satisfy the declared noise bounds, they lie in  $\mathcal{F}$ , so every feasible factor estimate obeys the displayed error bound. This does not prove that a particular solver finds a feasible point. It does not verify domain membership from noisy data, establish optimality of  $K$  queries, or cover passive task-gradient queries or unknown AdamW state. Each query returns a full  $L \times D$  response, not one scalar.

No smooth choice of singular-vector basis is needed: the proof holds for every chosen best rank- $K$  right subspace and orthonormal bases. It does not assume a singular-value gap in the observation. For a nonempty feasible set under the stated threshold, separation at the  $K$ th cut also follows from the rank margins and perturbation inequalities, but that fact is not needed in the proof.

For  $K = 1$ , row stochasticity fixes  $A = \mathbf{1}_L$  and the response’s softmax term vanishes. Directly,  $d_{\text{perm}}(x, x') = \|B - B'\|_F \leq 2\varepsilon_{\Theta}/\sqrt{L}$ . Thus the boundary case needs no response information or small-noise threshold. The nontrivial theorem concerns  $K \geq 2$ .

## E OBSERVABLE LOCAL SEPARATION FOR NOISY RECOVERY

This section specializes standard Lipschitz-Jacobian Taylor reasoning (Behling et al., 2018, Section 2, equations (5)–(6)) to the response measurement map. It is not a new generic inverse theorem. The distinction between exact diagonalization and sensitivity of approximate diagonalization is already central to Afsari (2008). Throughout the numerical study,  $\eta_Z = \eta_B = \tau = 1$ .

**Exact objective compression in an estimated subspace.** Let  $U \in \mathbb{R}^{K \times D}$  have orthonormal rows spanning the leading right-singular subspace of the observed product  $\hat{\Theta}$ . Represent candidates by  $B = CU$ . For measured pairs  $(G_p, \hat{R}_p)$ , write  $X_p = G_p U^\top$ ,  $G_p^\perp = G_p - X_p U$ ,  $Y_p = \hat{R}_p U^\top$ , and  $R_p^\perp = \hat{R}_p - Y_p U$ . With  $Q = AA^\top$  and  $S_i = \text{diag}(a_i) - a_i a_i^\top$ , the predicted parallel response is  $QX_p + [C^\top S_i^2 C X_{p,i}]_i$ , and the orthogonal response is  $QG_p^\perp$ . Set

$$H = \sum_p G_p^\perp (G_p^\perp)^\top, \quad T = \sum_p R_p^\perp (G_p^\perp)^\top.$$

When  $H \succ 0$ , least-squares orthogonality gives

$$\sum_p \|QG_p^\perp - R_p^\perp\|_F^2 = \|QH^{1/2} - TH^{-1/2}\|_F^2 + \sum_p \|TH^{-1}G_p^\perp - R_p^\perp\|_F^2, \quad (135)$$

$$\|ACU - \hat{\Theta}\|_F^2 = \|AC - \hat{\Theta}U^\top\|_F^2 + \|\hat{\Theta}(I - U^\top U)\|_F^2. \quad (136)$$

The second terms are candidate independent, but are retained in every reported residual. Dividing product and response blocks by their respective observed Frobenius norms gives an exact small least-squares objective. Candidate-dependent differences are preserved isometrically, including their Jacobian singular values. Numerically singular  $H$  is rejected. Crucially, a subspace estimated from noisy data need not contain the true basis. This restriction is not justified by the local result below.

**An explicit curvature bound.** At a strictly interior candidate  $(A_0, C_0)$ , use  $A = A_0 + s_A ZV^\top$ ,  $C = C_0 + s_C W$ , where  $s_A = \|A_0\|_F$ ,  $s_C = \|C_0\|_F$ , and  $V$  is an orthonormal basis for  $\mathbf{1}^\perp$ . Let  $x = \text{vec}(Z, W)$  and  $r \leq r_0 = \min_{ij} (A_0)_{ij} / (2s_A)$ . All routers in this ball are strictly feasible. Set  $b = s_C(1 + r)$  and  $c = \min\{1, \max_i \|S_i(A_0)\|_2 + 3s_A r\}$ . The inequalities  $\|DS_i[h]\|_2 \leq 3\|h\|_2$  and  $\|D^2 S_i[h, k]\|_2 \leq 2\|h\|_2 \|k\|_2$  follow by differentiating  $\text{diag}(a) - aa^\top$ . Differentiating  $(S_i C)^\top S_i C$  twice then bounds its bilinear Hessian by the largest eigenvalue  $h(r)$  of

$$\begin{pmatrix} (18 + 4c)b^2 s_A^2 & 12cbs_A s_C \\ 12cbs_A s_C & 2c^2 s_C^2 \end{pmatrix}.$$

The matrix is symmetric with nonnegative entries, so its largest eigenvalue equals its spectral norm. For the normalized measurement map  $F$ , a conservative Hessian bound is

$$M(r) = \left[ \left( \frac{2s_A s_C}{\|\hat{\Theta}\|_F} \right)^2 + \left( \frac{(2s_A^2 + h(r))\sqrt{\sum_p \|G_p\|_F^2}}{\|(\hat{R}_p)_p\|_F} \right)^2 \right]^{1/2}.$$

The  $2s_A^2$  term accounts for the Hessian of  $AA^\top G_p$ . The same derivation applies in full coordinates, with  $C$  replaced by  $B$ .

**Proposition 12** (Conditional local feasible-component separation). *Let  $\mu = \sigma_{\min}(DF(0)) > 0$ ,  $\rho = \|F(0) - y\|$ , and suppose  $\|D^2F\| \leq M$  on the closed radius- $r$  ball, with  $\mu - Mr/2 > 0$ . Every  $x$  in this ball satisfies*

$$\|F(x) - F(0)\| \geq (\mu - Mr/2)\|x\|.$$

*If  $\rho \leq \nu$  and  $\rho + \nu < \mu r - Mr^2/2$ , the connected component of  $\{x : \|F(x) - y\| \leq \nu\}$  containing zero lies inside the ball, and each of its points obeys*

$$\|x\| \leq \frac{\rho + \nu}{\mu - Mr/2}.$$

*Proof.* The integral Taylor remainder has norm at most  $M\|x\|^2/2$ . Combining this with  $\|DF(0)x\| \geq \mu\|x\|$  proves the first inequality. No feasible point can lie on the sphere of radius  $r$ , because its measurement difference from  $F(0)$  would exceed  $\rho + \nu$ . The norm image of a connected set containing zero cannot reach beyond  $r$  without meeting  $r$ . The triangle inequality then gives the distance bound.  $\square$

The implementation takes  $r = \min\{r_0, \mu/M(r_0)\}$  and recomputes  $M(r)$ . For known relative observation errors bounded by  $\sigma < 1$  in both blocks,  $\nu = \sqrt{2}\sigma/(1 - \sigma)$  is an observable normalized noise upper bound. The separately recorded  $10^{-10}$  numerical floor affects empirical scores only. At zero declared noise, rounding can prevent a numerical local pass.

**Scope of the conclusion.** This proposition neither excludes distant components nor establishes that the true factors belong to the candidate’s component. It is not a convergence guarantee for the optimizer. Its metric uses candidate norms, whereas benchmark errors use true-factor norms and a common permutation. In a noisy estimated subspace, the true factors can be outside the model class altogether. Floating-point singular values and curvature evaluations are not outward-rounded proofs; we call their successful evaluations *numerical local passes*, not certified global recovery. Sixteen separate small full-coordinate checks place a known feasible point inside the calculated ball and satisfy the distance inequality. These are theorem sanity checks, not sixteen successful noisy-recovery trials.

## F DETAILED PRIOR-ART AUDIT

This appendix records the distinctions and exact variable maps behind the condensed audit in Section 2. It separates uniqueness of a factorization, invariance of a computation or decision under refactorization, and recovery of a historical sharing graph. The literature establishes strong results for the first two and contains a few coordinate-to-action systems, but our audit does not support a broad claim that cross-layer compression papers purport to recover a data-generating sharing graph.

**Simplex factorization: the static ambiguity is prior art.** For  $K \geq 2$ , after transposition, our parameter-space model is exactly the unconstrained simplex-structured matrix factorization (SSMF)

$$X = WH, \quad X = \Theta^\top, \quad W = B^\top, \quad H = A^\top.$$

Section IV-A of Vu Thanh et al. (2023) already proves that unconstrained SSMF is non-identifiable. In their notation,

$$\begin{aligned} W(\delta) &= W \left( (1 + \delta)I - \frac{\delta}{K} \mathbf{1}\mathbf{1}^\top \right), \\ H(\delta) &= \left( \frac{1}{1 + \delta}I + \frac{\delta}{(1 + \delta)K} \mathbf{1}\mathbf{1}^\top \right) H. \end{aligned}$$

Setting  $s = \delta/(1 + \delta)$  makes their coefficient transform  $M_s = (1 - s)I + s\mathbf{1}\mathbf{1}^\top/K$ , so that  $H(\delta) = (AM_s)^\top$  and  $W(\delta) = (M_s^{-1}B)^\top$ . This is exactly our uniformizing gauge after transpose and substitution, not a distinct neural-network theorem. The entropy consequence uses this uniformizing path, which preserves coordinate ordering and argmax assignments. Our local gauge description

and argmax example use the broader family of feasible changes of basis; the argmax example is nonuniform. These specialize known factor ambiguity to router diagnostics, not a new foundational non-identifiability result. Conversely, classical NMF and SSMF work gives recovery under additional support, sufficiently-scattered, facet-coverage, or volume assumptions (Donoho & Stodden, 2003; Fu et al., 2018; Abdolali & Gillis, 2021; Gillis & Rajkó, 2023). These are condition-dependent positive boundaries, not consequences of the simplex constraint alone.

**LoRA gauges, invariant processing, and merge consequences.** The general low-rank gauge is also established prior art. For full-rank LoRA factors, Putterman et al. (2024) show that effective products separate  $GL(r)$  orbits and develop invariant/equivariant processing of the factors. LoRA Done RITE constructs factor-optimization updates invariant to the same change of basis (Yen et al., 2025). LoRA-S equips the full-rank factor quotient with an invariant metric and uses a Sylvester equation to lift preconditioned product-space directions; its invariance comparison uses the same product-space direction in both factor representations (Zheng & Wu, 2026). Thus ordinary factor-gradient chart dependence and quotient-invariant optimizer design are both established precedents. Gauge-aware federated LoRA explicitly shows why raw-factor aggregation is representation dependent and gives an invariant server representation (Chen et al., 2026); W2T obtains gauge-invariant adapter features through product-space QR–SVD processing (Han et al., 2026). Thus neither the existence of the  $GL(r)$  gauge, the use of effective products as invariants, nor the general warning against raw-factor aggregation is new here.

TwistedMerge supplies a particularly important trained-model precedent (Gong & Xu, 2026). On trained rank-four residual adapters, its orthogonal, diagonal, and dense gauge scrambles change naive factor-averaging outcomes, whereas whitened synchronization and deterministic full-update SVD remain stable. It therefore already covers the broad empirical consequence “equivalent trained factors can make naive merging behave differently.” Its action is averaging independently trained adapters in a single residual LoRA-form feature layer, not selecting a directed within-model layer tie from a shared factor; it provides neither an ASLoRA nearest-pair analysis nor the shared-softmax response stabilizer below.

An ICML 2026 workshop poster explicitly studies the  $GL(r)$  gauge and LoRA merging (Karuturi et al., 2026). We verified its official title, authors, and poster record, but could not obtain its primary full text during this audit. We therefore leave its coverage of maximal invariants, nearest-pair selection, Loewner-order tests, merge regret, ASLoRA, and optimizer-response stabilizers adjudicated; no priority claim in this paper is inferred from its title.

**The few concrete coordinate-to-structure actions.** The closest cases differ in what they claim. Savarese & Maire (2019) represent each convolutional layer as a linear combination of global templates, use coefficient cosine geometry as a sharing measure, infer an approximately recurrent pattern, and then evaluate the folded network. This is architectural-discovery language followed by an empirical intervention, not an assertion that the coefficient plot alone authenticates a historical graph. Neural Parameter Allocation Search learns factor coordinates or embeddings and clusters them after early training to make hard parameter groups (Plummer et al., 2022); its claim is automated allocation under a parameter budget, not recovery of a planted sharing history.

ASLoRA is the most direct target for our decision audit. Its arXiv-v2 method shares one input-side LoRA factor  $A$ , ranks layer-specific historical averages of  $B_i$  by raw Euclidean distance, and makes the lower layer use the selected upper layer’s factor (Hu et al., 2024). The paper motivates sharing  $A$  as removing its and its initialization’s influence and making  $B$ -similarity more reasonable and consistent. This is a real coordinate-to-nearest-action rule, but it is not a claim that the selected pair recovers a historical generator. We cite the arXiv version for these equations and algorithm details because the accessible formal journal record supports the broad global- $A$ /selective- $B$  description but did not expose the full text (Hu et al., 2026).

**Recent boundary cases are controls, not offenders.** A full-text audit of three recent systems further narrows the reality claim. E2LoRA computes proxy entropy and relative mutual information from task-subset gradients of the pretrained model *before* adapter fine-tuning, then fixes contiguous sharing intervals and interval ranks (Li et al., 2026). Although it calls the measured patterns intrinsic and reports stable structural decisions across subset seeds, it neither reads a jointly learned factor coordinate nor claims recovery of a historical graph; its seed/noise checks and downstream evaluation

instead make it a positive proxy-to-allocation control. RaSA fixes a global cross-layer rank pool by construction and evaluates reconstruction and task quality (He et al., 2025), so there is no learned partition to interpret. LiSA measures input-dependent attention-score similarity and sensitivity, then trains alignment and difference-compensation modules and directly evaluates the compressed model (Mu et al., 2026); this is nonlinear attention-output reuse, not the linear factor chart studied here. These papers are negative evidence for a prevalence claim and positive examples of validating a structural intervention, not examples of historical-truth misinterpretation.

Most other cross-layer sharing work is not evidence of the alleged misinterpretation. Dynamic Layer Tying searches over explicit copying actions (David-Hay & Wolf, 2024); Basis Sharing and FiPS target compression, reconstruction, and downstream quality (Wang et al., 2025; Üyük et al., 2026); Geo-Sharing selects a low-curvature compression assignment (Zhang et al., 2025). None of these claims a learned soft factor chart recovers a historical truth.

MASA is an especially close mixed boundary case (Zhussip et al., 2026, AAAI pp. 29262–29263; arXiv v2, supplementary p. 14). In its from-scratch path, each attention projection has  $\widehat{W} = DC$ : shared atoms  $D$  and per-layer coefficients  $C$  are jointly learned, each coefficient is stated to lie in  $\mathbb{R}$ , and the paper declines the atom soft constraints it has just described. A block embedding plus three-layer MLP produces  $C$  during training before the MLP is discarded. Thus the retained inference factor obeys  $(D, C) \mapsto (DR, R^{-1}C)$ . Under  $\Theta = \widehat{W}^\top$ , this maps exactly to  $A = C^\top$  and  $B = D^\top$ , but  $A$  is unconstrained rather than row-simplex and the indirect MLP training law is not our Euclidean  $Z/B$  response law.

The paper calls coefficients atom contributions and shared atoms carriers of cross-layer statistical regularities. More strongly, its arXiv v2 supplement uses atom cosine (Equation 18) to describe distinct versus redundant atoms and says raw- $C$  heatmaps reveal specialization, redundancy, and layer-wise adaptivity. These are genuine, but weak, chart-level structural interpretations: the paper defines no historical or generative target, reports no structure-recovery metric, and makes no separate hard action from  $C$ . For pretrained models it instead groups consecutive blocks using KL divergence between vocabulary-space distributions of block outputs, then applies orthonormal matrix PCA, residual refinement, and downstream validation. That action is function-derived rather than coefficient-derived; the PCA constraint removes general scale/shear freedom but still permits an orthogonal change of basis within the retained subspace. MASA therefore shows that descriptive coordinate reading occurs, while remaining negative evidence for the stronger historical-recovery and coordinate-to-action narrative. Likewise, Yeh et al. (2022) is a positive recovery result for an explicitly defined planted equivariance/partition target under stated assumptions, not an example of the error studied here. The defensible empirical claim is therefore narrow: some systems use learned coordinates to make structural actions, and those actions should be audited for representation dependence and downstream consequence.

A separate 2026 preprint supplies a particularly relevant metric-to-decision warning without involving a learned factor chart. In Pythia experiments, Khilar (2026) report that cross-layer subspace coupling improves two-sided Frobenius reconstruction over a per-layer SVD baseline by (37.2%) at 1.4B scale, yet produces WikiText-2 perplexity (159.14) versus (50.17) for that baseline. Their measured per-layer activation error increases by a matching factor of about three. This is direct prior evidence that a weight-space proxy can rank structural compression choices incorrectly and that deployed consequences or activation-matched controls are needed. It is not a result about gauge-equivalent coordinates, a learned router, or the same-function-to-different-action chain isolated in our ASLoRA audit.

**Exact novelty boundary.** Jacobian–Gram response operators are standard tangent-kernel objects (Jacot et al., 2018). More generally, Marcotte et al. (2026) write the lifted Euclidean flow as  $-D\phi(\theta)D\phi(\theta)^\top \nabla f$ ; for  $\phi(U, V) = UV^\top$ , their matrix-factor response acts as  $G \mapsto UU^\top G + GVV^\top$ . Under  $U \leftrightarrow A, V \leftrightarrow B^\top$ , this is the direct non-softmax factor-response precedent, but it contains neither the rowwise  $J(a_i)^2$  blocks nor a common-fiber stabilizer classification. For a single value–softmax output, the derivation in Varre et al. (2026, Lemma D.2) already contains  $\dot{\beta} = (\|s\|^2 I + VC(s)^2 V^\top)r$ , with  $C(s) = \text{Diag}(s) - ss^\top$  and  $r = -\nabla \ell$ . Thus the local squared-covariance response is also prior art. The reduction of a linear-factor  $GL$  gauge to an orthogonal Euclidean isometry and optimizer-compatible softmax permutation/shift symmetries are further precedents (Singh, 2026; Lau & Su, 2026). Gauge-correction work further derives quotient-volume biases

and balanced representatives for stochastic dynamics with matrix-factor and attention symmetries (Aladrah et al., 2026); its target is a stationary implicit-bias law, not equality of the finite-width instantaneous response at two fixed charts.

Two identifiability precedents rule out broader priority language. Tran et al. (2025, Thm. 4.1) prove that, for their ReLU expert class, equality of a dense softmax-MoE function over its input domain, together with pairwise distinct expert functions and pairwise distinct gate-vector differences in each model, identifies its expert functions and gates up to a common gate translation and a permutation. Nguyen et al. (2023, Prop. 1 and Thm. 1) prove translation/label identifiability and a global linear lower bound from conditional-density Hellinger distance to a Voronoi parameter metric for an exact-fitted Gaussian softmax MoE on a compact parameter set. Their observables are respectively an entire MoE function and a conditional density; neither compares two  $AB$  charts on a common exact product fiber through their Euclidean instantaneous effective-gradient response. We therefore claim neither the first softmax permutation/translation identifiability result nor the first global inverse bound for a softmax-gated model. The Gram-to-Procrustes step also has a standard quantitative antecedent (Tu et al., 2016, Lem. 5.4); our quantitative addition is only the complete model-specific chain on the displayed uniform nondegenerate set.

A still closer intermediate step is already known for nonnegative representations. Dorrell et al. (2026, Theorem 2 and Appendix D.2) consider a representation matrix  $Z$ , its sample Gram  $Z^\top Z$ , and orthogonal changes  $Z' = OZ$ . Under two tight-scattering/convex-hull conditions, nonnegativity of  $Z'$  forces  $O$  to be a permutation. With  $Z = A^\top$ ,  $Z' = (A')^\top$ , and  $O = M^\top$ , the Gram step of our proof is exactly in that form. Their scattering hypotheses are additional geometry, however, not consequences of a strictly positive full-rank router. In fact, their first condition requires the convex hull of the representation columns to contain an ellipsoid that touches every coordinate hyperplane, whereas the convex hull of finitely many strictly positive router rows stays a positive distance from every such hyperplane. Our local  $B^\top C_i^2 B$  response blocks, not nonnegativity or the Gram alone, exclude the residual orthogonal ambiguity. The cited theorem supplies no perturbation inverse, and its authors explicitly state that their sufficient scattering conditions are not proved necessary. We therefore credit the orthogonal-to-permutation mechanism while retaining only the response-specific completion and stability chain as candidate additions.

The subsequent diagonal-algebra step is also not a new mechanism. Afsari (2008, Theorem 2.3) proves essential uniqueness of a nonorthogonal exact joint diagonalizer precisely when no two columns of the matrix of diagonal entries are collinear. Once our proof has obtained  $M^\top \text{Diag}(a_i) M = \text{Diag}(a'_i)$  for every layer  $i$ , their diagonal-entry matrix is  $A$ ; full column rank is stronger than their pairwise noncollinearity condition, so their result yields a monomial  $M$  before the row-sum constraint  $M\mathbf{1} = \mathbf{1}$  reduces it to a permutation. Their Section 4 and Theorems 4.1–4.2 further give first-order, cost-function-specific perturbation formulas for stationary joint diagonalizers, with sensitivity controlled by diagonalizer conditioning and a modulus of uniqueness. This directly precedes approximate diagonalization in our Steps 3–4. It does not, however, recover the router Gram and covariance families from a response operator, nor give our global compact-set bound, conditioned on a common exact effective product, from response-operator distance to the simultaneous  $(A, B)$  permutation orbit. The candidate quantitative increment is therefore only that complete model-specific chain, not joint-diagonalization uniqueness or sensitivity.

The same final step also has a global robust tensor precedent. Stack the diagonal family into  $\mathcal{T} = \sum_{k=1}^K e_k \otimes e_k \otimes A'_{:,k}$ . The approximate congruences in our proof compare its decomposition  $[I, I, A']$  with  $[M^\top, M^\top, A]$ . Under our singular-value and norm bounds, the three reference factors have robust Kruskal ranks  $K, K, K$ ; for  $K \geq 2$ , their sum satisfies  $3K \geq 2K + 2$ . Consequently, Bhaskara et al. (2014, Theorem 5) already supplies global stability of these bounded tensor factors up to one permutation and diagonal rescalings. With its conditioning parameters fixed, Theorem 5 is linear in the discrepancy: its required tensor accuracy is the requested factor accuracy divided by a fixed polynomial in those parameters. The linear exponent is therefore not an addition of this paper. This is stronger than viewing Afsari’s result as the only robust precedent for Steps 3–4. It still begins from two close tensor decompositions: it does not derive that tensor comparison from the complete response operator, enforce the common exact effective product, or give the tailored explicit linear end-to-end constant for the simultaneous  $(A, B)$  orbit. We therefore do not claim the diagonal-tensor robust permutation step, its global character, or its linear error dependence as new.

The optimizer metric is essential. van Oostrum et al. (2023, Thms. 1–2) show that the pushforward of a natural parameter gradient is a tangent-space projection and is invariant under diffeomorphic reparameterization. Along a feasible full-rank factor gauge  $A' = AM$ ,  $B' = M^{-1}B$ , the two charts span the same product tangent space because

$$\{XB + AY : X\mathbf{1} = 0\} = \{X'B' + A'Y' : X'\mathbf{1} = 0\}, \quad X' = XM, \quad Y' = M^{-1}Y.$$

Thus this response does not reduce  $GL(K)$  to permutations. Riemannian quotient optimization explicitly treats  $(G, H)$  and  $(GM^{-1}, HM^T)$  as one point under invariant metrics (Mishra et al., 2014). These are exact counter-boundaries to any metric-independent reading of our theorem, which concerns only the stated Euclidean  $Z/B$  flow with common known positive rates and temperature.

Two recent softmax-attention analyses provide additional, architecture-level boundaries. Wang & Wang (2026a) characterize the generic function-preserving group of standard multi-head attention as independent query–key and value–output  $GL$  actions, together with head permutations; Wang & Wang (2026b) study quotient/Fisher geometry and the horizontal natural gradient, assuming no Fisher degeneracy beyond gauge directions. Their factors are Q/K and V/output projection matrices, not a row-simplex shared router and basis bank, and their observable is the Transformer function or quotient geometry rather than equality of two Euclidean effective-gradient response operators. Conversely, Goel et al. (2026) show that a Gaussian softmax self-attention population loss reduces to a weighted matrix-factorization loss and analyze a preconditioned, regularized optimizer. In that model the Gaussian expectation cancels the softmax normalization before the factor loss is formed; the paper does not classify gauges from complete Jacobian–Gram response. These sources therefore strengthen the prior-art boundary without implying Theorem 1.

Finite operator observation is likewise not new. In Novak et al. (2022, Sec. 3.3, Eqs. (7)–(9)), applying NTK–vector products to the  $O$  identity columns reconstructs an  $O \times O$  finite NTK. The variable map here is  $O \mapsto LD$ , their NTK to the learning-rate-weighted response Jacobian Gram, and their identity column to  $\text{vec}(G)$ . More general probing writes an operator in a known linear family and recovers its coefficients from matrix–vector products (Chiu & Demanet, 2012; Halikias & Townsend, 2024; Otto, 2023); the formal COLT result of Amsel et al. (2026) gives broader approximate query bounds, including linear matrix families, with access to both operator directions. Our  $K$ -probe construction is only a response-specific use of the known support and shared-block form that compresses the generic  $LD$  identity probes under  $D - K \geq L$ . It is not claimed as a new matrix-probing principle. We also checked the primary full text of Stupariu & Manolache (2026): its Adam/Muon experiments analyze performance, loss geometry, and ranks, but contain no response-stabilizer, inverse, or tomography theorem; workshop metadata alone is not used to infer priority.

Against those components, our theoretical addition is restricted to the following combination in Theorem 1 and the explicit nondegenerate quantitative inverse in Corollary 1: for a full-rank softmax router and full-row-rank shared bases evaluated under the specified Euclidean  $Z/B$  response law, two charts on one common exact product fiber have equal complete instantaneous response for every effective gradient only when they differ by a basis permutation, while response closeness controls distance to that orbit only on the stated uniformly interior, well-conditioned, norm-bounded set. We did not find this combined stabilizer or its model-specific inverse in the audited full texts; generic inverse-function and perturbation tools are not claimed as new. Corollary 3 further combines the response inverse with a uniform bound outside a noisy product’s estimated row space, giving a whole-feasible-set diameter bound from  $K$  designed queries under the stated domain, dimension, and product-noise conditions. These results identify the current chart relative to a named optimizer metric; they do not identify an AdamW trajectory, a functional-equivalence partition, or a historical graph.

Empirically, we conduct a scoped ASLoRA nearest-action audit. A common  $GL(r)$  change of basis preserves every adapted layer while it can change the raw running- $B$  ranking. We distinguish that historical selection proxy from the current directed weight perturbation, layer-specific activation-weighted disturbance, and full-validation consequence of the candidate tie. This system-specific chain—rather than a claim that gauge dependence or trained merge sensitivity is newly discovered—is an auxiliary empirical contribution.

## G PRETRAINED-LANGUAGE-MODEL PROTOCOL AND FULL ACCOUNTING

### G.1 WHY THE ADDITIVE MODULE IS COVERED BY THE OBSERVATION MODEL

**Proposition 13** (Fixed isometric embedding). *Let  $E$  be a known linear isometry from the product space  $\mathbb{R}^{L \times D}$  to a space of layer-weight perturbations, with  $E^*E = I$ , and let  $W = W_0 + E(AB)$  with fixed known  $W_0$ . For Euclidean updates of  $Z, B$  with the rates and temperature in Section 4.1, the effective weight response to a full-space gradient  $H$  is*

$$-\dot{W} = E \mathcal{R}_{A,B}(E^*H).$$

*Equality of complete full-space response operators is equivalent to equality of the factor response operators. Product and response distances for embedded observations are unchanged. Consequently the identification and stability results apply to this module under their original rank, interior, rate, dimension and noise assumptions.*

*Proof.* The derivative of the affine embedding is  $E$ , so the chain rule pulls  $H$  back to  $G = E^*H$ . The specified factor updates therefore produce  $-\dot{\Theta} = \mathcal{R}_{A,B}(G)$ , and differentiation of the embedding gives the displayed equation. Conversely, any  $G$  is realized by  $H = EG$ , because  $E^*EG = G$ . Applying  $E^*$  to the embedded response recovers  $\mathcal{R}_{A,B}(G)$ , proving both directions of the operator equivalence. The product coordinates are  $E^*(W - W_0) = AB$ . Finally,  $\|EX\|_F = \|X\|_F$  preserves all stated product/query/response norms. These facts transfer the existing results without changing their assumptions or introducing an efficient solver claim.  $\square$

Our  $E$  inserts 64 evenly spaced output rows into each attention output projection, leaving the other rows and the pretrained backbone fixed. All input channels are retained. Thus  $D = 64d_{\text{in}}$ , which need not equal 64 times the model’s hidden width when attention head dimensions differ. The table reports actual projection dimensions. Four FP32 bases and FP32 softmax logits are shared over all  $L$  matching layers. The fixed-row insertion is an isometry in exact arithmetic. Mixed-precision backbone execution supplies measured directions; FP64 factor-map auditing isolates the mathematical response from rounding in the backbone. The proposition does not say that natural task gradients realize every possible query, nor that nonlinear network-output responses are isometries.

### G.2 FROZEN DESIGN, DATA AND ADAPTATION

**Model choice and access.** The nine exact Hugging Face model IDs and revision commits are stored in `research/LLM_ACCESS_INVENTORY.json`; downloaded snapshot file hashes are separately recorded in `LLM_MODEL_BYTES.json`. Pythia and SmolLM2 use the named base models. Qwen3 uses the named dense non-Base releases, so architecture, pretraining and post-training are confounded across families. Initial unauthenticated requests for Llama3.2 1B and 3B returned HTTP 401; both are recorded as unavailable, with the public SmolLM2 fallback chosen before pilot outcomes. There are no Llama results and no coverage of MoE, multimodal models, or models above 8B.

**Development and freeze.** Three representative corrected pilots use Pythia160M, Pythia1B and Qwen3-0.6B, seed90, 20 updates. Earlier development records retain a failed keyword call and an incorrect WikiText article separator; these are excluded from the confirmatory evidence. The separator was corrected to top-level article headings before the final pilots and all main runs. The final protocol, 17 source/config/data files and raw corpus hashes were frozen before calibration and evaluation. The four calibration units are Pythia160M and Pythia1B, each seeds101/102. Evaluation uses seeds0/1/2 for every model; no checkpoint or seed is selected by validation performance. Policy thresholds are fixed before evaluation recovery scoring. Evaluation gradient collection can run concurrently with calibration because it does not inspect recovery quality. The full chronology is retained in `research/LLM_DEVELOPMENT_LOG.md`.

**Data and disjoint roles.** WikiText-2 train supplies adaptation; validation supplies general-text calibration, and test supplies general-text evaluation. GSM8K’s public training JSONL and MBPP’s public JSONL supply held-out math and code text. Dataset repository commits and raw SHA-256 hashes are in `LLM_CORPUS_LOCK.json`. This use of public text is not an official benchmark-accuracy evaluation. We format math as question plus reference answer, and code as comment prompt

Table 2: Adapter-induced change in held-out token NLL (adapted minus frozen backbone), averaged over three adapter seeds. Negative is lower. Domains use 32 fixed blocks per checkpoint. Tokenizers differ, so changes are within-model comparisons; these are not answer-accuracy or code-execution scores.

| Model        | General | Code    | Math    |
|--------------|---------|---------|---------|
| Pythia 160M  | -0.0481 | +0.0303 | +0.1146 |
| Pythia 1B    | -0.4439 | +0.0485 | +0.0812 |
| Pythia 2.8B  | -0.4220 | +0.0301 | +0.0256 |
| Qwen3 0.6B   | -0.7086 | -0.0392 | +0.0036 |
| Qwen3 1.7B   | -0.5666 | -0.1779 | -0.2103 |
| Qwen3 4B     | -0.6006 | -0.1514 | -0.2646 |
| Qwen3 8B     | -0.4193 | -0.1729 | -0.2005 |
| SmolLM2 360M | -0.3016 | +0.0013 | +0.0216 |
| SmolLM2 1.7B | -0.3080 | +0.0071 | +0.0461 |

plus reference program, scoring the entire sequence. We do not execute programs or score generated answers. Public pretraining overlap is unknown.

Within each held-out corpus, alternating documents are assigned to probe and loss roles before tokenization. Each role is EOS-concatenated and divided into nonoverlapping 129-token blocks (128 prediction targets), retaining at most 256 blocks; source document IDs and token hashes are stored. Training draws from up to 4,096 train blocks. Twelve distinct probe blocks are sampled per checkpoint/domain: the first eight are fitting candidates and the last four are reserved for response prediction. Loss evaluation uses the first 32 loss-role blocks, in 16 batches of two. The same documents can contribute multiple probe blocks, so query cases are not independent statistical replicates. Corpus partition names for code/math describe study roles, not official benchmark test splits.

**Adaptation budget.** Each run trains only the  $4D + 4L$  adapter parameters for 256 updates, batch4 and sequence length 128, or 131,072 prediction tokens. AdamW uses basis learning rate  $10^{-3}$  and router rate  $10^{-2}$ , zero weight decay, gradient norm clipping at 1, 16 linear warmup steps, then cosine decay to 0.1 of the initial rate. Initial logits are standard normal, and initial bases are nonzero Gaussian matrices scaled by  $0.01/\sqrt{d_{\text{in}}}$ . The last checkpoint is used. This fixed budget is not a convergence certificate or an optimization comparison. Mean router  $\ell_1$  movement is 0.250–0.850 across checkpoints; minimum router entries are  $6.86 \times 10^{-4}$ –0.0114, minimum router singular values 0.415–2.39, and minimum basis singular values 0.624–6.52. These are measured properties, not acceptance-policy inputs derived from hidden truth. Table 2 includes adverse domain shifts. The noise grid is not restricted to the global theorem’s domain-dependent product-noise threshold, and the numerical solver does not enforce its entire prescribed domain. Small observed errors therefore do not measure the theorem’s universal feasible-set diameter or certify its constants.

### G.3 OBSERVATIONS, ESTIMATORS AND TRANSFER DIAGNOSTICS

**Observation grid and arithmetic.** Each evaluation checkpoint supplies five families, three query budgets and four noise levels: 60 cases, or 1,620 total. Calibration uses only designed, Gaussian and general-text queries, giving 36 cases per checkpoint and 144 total. Designed queries are the repeated-complement unit construction;  $q = 8$  repeats its four-query cycle. Gaussian and natural queries are individually normalized. Designed-query holdouts are independent Gaussian directions; the other families use their last four measured directions. Product and each response receive fixed-relative-Frobenius-norm Gaussian perturbations with common random directions across noise levels. The product row space is recomputed from its noisy observation.

The observation code measures each response by reverse-mode factor differentiation and a forward-mode JVP, rather than by the analytic response formula. Audit temperature and both Euclidean block rates are set to one, independently of the AdamW adaptation learning rates. All formula checks have relative error at most  $1.62 \times 10^{-16}$ . A common factor permutation preserves the response; a fixed non-permutation transform preserves the product to rounding while changing the response by 0.103–0.481 on tested directions. This is a finite witness, not a test of every alternative. Same-

Table 3: Unchanged calibration policies transferred to 1,620 held-out cases per estimator. Coverage includes all 540 insufficient-rank cases at  $q = 1$ ; joint fitting is not run there. Bad means permutation-aligned factor error  $> 0.10$ . An empty accepted set has undefined bad fraction. Raw spectral estimates fail the strict simplex gate in calibration, yielding an all-reject policy.

| Estimator | Score     | Accepted | Coverage | Bad | Bad / accepted |
|-----------|-----------|----------|----------|-----|----------------|
| Spectral  | Residual  | 0        | 0.0%     | 0   | —              |
| Spectral  | Condition | 0        | 0.0%     | 0   | —              |
| Spectral  | Combined  | 0        | 0.0%     | 0   | —              |
| Projected | Residual  | 1037     | 64.0%    | 20  | 1.93%          |
| Projected | Condition | 960      | 59.3%    | 48  | 5.00%          |
| Projected | Combined  | 1056     | 65.2%    | 43  | 4.07%          |
| Joint     | Residual  | 868      | 53.6%    | 0   | 0.00%          |
| Joint     | Condition | 1080     | 66.7%    | 0   | 0.00%          |
| Joint     | Combined  | 1080     | 66.7%    | 0   | 0.00%          |

forward-graph chain checks pass, but separately repeated BF16 backbone backward passes differ by up to 0.0126 relatively; measured directions are treated as fixed inputs afterward. At finite step  $10^{-5}$ , FP64 response errors are  $1.01 \times 10^{-9}$ – $6.88 \times 10^{-9}$ , versus the much larger FP32/BF16 errors reported in the main text. These casts are not an experiment on mixed-precision optimizer steps with FP32 master weights.

**Recovery and scoring.** The three estimators are spectral recovery, simplex-projected spectral recovery with a basis refit, and constrained joint least squares with three starts and at most 80 function evaluations per start. All operate on compressed observations. Compression retains the product component outside the fitted row space and the orthogonal response residual, so an estimated-subspace fit cannot silently discard unexplained energy. Ground truth enters only afterward in the permutation-aligned error

$$e = \min_P \left( \frac{\|\hat{A} - AP\|_F^2}{\|A\|_F^2} + \frac{\|\hat{B} - P^T B\|_F^2}{\|B\|_F^2} \right)^{1/2}.$$

Basis error includes the true basis component outside the estimated subspace. No compression failures occur in the main grid. All 540  $q = 1$  cases fail the sufficient per-layer query rank and skip joint fitting; spectral outputs remain recorded. All 1,080 eligible joint fits return. Raw/projected spectral methods have 580/531 errors above 0.10 over all 1,620 cases; these counts include their insufficient-query outputs. The joint-fit median is conditional on the 1,080 returned estimates.

**Acceptance and uncertainty.** The common gate requires positive simplex entries, row-sum error below  $10^{-8}$  and sufficient query rank. Scores are residual-only, inverse candidate tangent singular value, and a combined residual, conditioning, subspace and competitive-start-disagreement score with a  $10^{-10}$  numerical floor. On calibration, each method/score threshold maximizes accepted count subject to at most 5% bad outputs and at least 20 accepted cases, keeping ties together; no feasible threshold means all reject. Good error is at most 0.05, bad error exceeds 0.10, and intermediate cases stay in the denominators. These are empirical policies. Spectral estimates often have row-sum errors of order  $10^{-8}$ – $10^{-7}$  despite small factor errors, so their strict gate explains the all-reject policy; simplex projection is an explicitly reported separate estimator.

The released summary also includes descriptive intervals from 2,000 bootstrap resamples of the three adapter checkpoints within each fixed model, preserving all query/noise/domain variants within a checkpoint. They are not population intervals over arbitrary LLMs. In particular, resampling zero observed bad joint fits still gives zero; this supplies no upper confidence bound on rare or out-of-distribution failures. Full model/domain breakdowns and all 4,860 method outcomes are retained in `results/llm/recovery-cases.csv`.

#### G.4 SECONDARY CONDITIONING STRESS

A separate prospectively recorded secondary sweep transforms the learned Pythia1B, Qwen3-4B and SmolLM2-1.7B factors at seed 0. For  $t \in \{1, 0.1, 0.01, 0.001\}$  it either contracts routers toward the

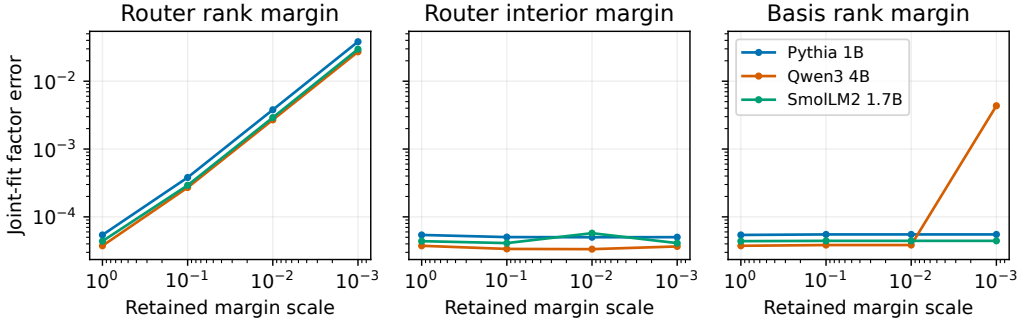


Figure 2: Secondary stress at observation noise  $10^{-4}$ . Shrinking router rank amplifies error from roughly  $4\text{--}5 \times 10^{-5}$  to  $0.027\text{--}0.038$ . The other tested paths do not show comparable monotone growth; the absence of failure on a selected path does not remove the theorem’s nondegeneracy assumptions.

Table 4: Post-recovery NLL minus native-router partition, averaged over three seeds. All actions use four balanced groups, a common effective-tensor refit and 64 matched updates. Both recovered and effective-clustering partitions equal the native partition at all nine checkpoints. Small Pythia discrepancies under equivalent partitions are numerical execution variation, not structural improvements. Per-seed values and immediate losses are retained in the released CSV.

| Model        | Action               | General | Code    | Math    |
|--------------|----------------------|---------|---------|---------|
| Pythia 1B    | Effective clustering | -0.0004 | -0.0005 | -0.0001 |
| Pythia 1B    | Random balanced      | +0.0629 | -0.0196 | -0.0291 |
| Pythia 1B    | Recovered, $10^{-4}$ | -0.0004 | -0.0005 | -0.0002 |
| Pythia 1B    | Recovered, $10^{-2}$ | +0.0001 | -0.0007 | -0.0003 |
| Qwen3 4B     | Effective clustering | +0.0000 | +0.0000 | +0.0000 |
| Qwen3 4B     | Random balanced      | +0.0801 | +0.0096 | +0.0010 |
| Qwen3 4B     | Recovered, $10^{-4}$ | +0.0000 | +0.0000 | +0.0000 |
| Qwen3 4B     | Recovered, $10^{-2}$ | +0.0000 | +0.0000 | +0.0000 |
| SmolLM2 1.7B | Effective clustering | +0.0000 | +0.0000 | +0.0000 |
| SmolLM2 1.7B | Random balanced      | +0.1050 | -0.0041 | -0.0238 |
| SmolLM2 1.7B | Recovered, $10^{-4}$ | +0.0000 | +0.0000 | +0.0000 |
| SmolLM2 1.7B | Recovered, $10^{-2}$ | +0.0000 | +0.0000 | +0.0000 |

uniform row (router-rank stress), interpolates toward balanced one-hot rows (interior stress), or scales the smallest basis singular value by  $t$  (basis-rank stress). Four designed queries, noise0 or  $10^{-4}$ , and unchanged estimators/policies give 72 cases. These are artificial factor modifications, not additionally trained models or real task gradients at the modified checkpoint.

#### G.5 MATCHED STRUCTURAL DECISIONS AND REPEAT CONTROLS

The predeclared decision units are Pythia1B, Qwen3-4B and SmolLM2-1.7B, each at seeds0/1/2. Every action uses four nonempty equal-sized groups. The native and recovered router actions maximize balanced assignment scores; the effective-parameter control uses balanced  $k$ -means with deterministic farthest-point initialization and 50 iterations; the random control uses a fixed seeded balanced partition. Recovered actions use general-text  $q = 4$  joint fits at noise  $10^{-4}$  and  $10^{-2}$ . Every hard group receives the mean of the same canonical FP32 effective tensors, followed by 64 updates of only the four bases with fresh AdamW states, rate  $10^{-4}$ , zero decay, the same batch indices and clipping. All 45 actions complete. Rejected recovery would retain the native action; all 18 recovered decision inputs are accepted by the combined policy.

All recovered and effective-clustering partitions have zero co-membership difference from the native partition. General-text random-minus-native post-recovery NLL is  $0.0383\text{--}0.1098$  across the nine checkpoints; held-out code and math differences have mixed signs. This establishes no advantage

Table 5: Per-checkpoint costs: median over three seeds; maximum allocated GPU memory across training and observation. Training excludes loading, evaluation and gradient collection; observation covers the entire noise/query sweep; recovery is single-thread CPU time elapsed for all three estimators. Timings are local measurements under concurrent jobs.

| Model        | Adapter (M) | Train (s) | Observe (s) | Recover (s) | Peak (GiB) |
|--------------|-------------|-----------|-------------|-------------|------------|
| Pythia 160M  | 0.197       | 3.2       | 14.9        | 10.4        | 0.75       |
| Pythia 1B    | 0.524       | 5.8       | 35.6        | 9.7         | 2.65       |
| Pythia 2.8B  | 0.655       | 11.8      | 45.3        | 34.4        | 6.68       |
| Qwen3 0.6B   | 0.524       | 10.4      | 36.3        | 25.3        | 2.82       |
| Qwen3 1.7B   | 0.524       | 12.4      | 36.1        | 26.7        | 5.26       |
| Qwen3 4B     | 1.049       | 20.0      | 72.4        | 43.3        | 10.95      |
| Qwen3 8B     | 1.049       | 34.3      | 72.4        | 43.3        | 19.08      |
| SmolLM2 360M | 0.246       | 9.9       | 18.9        | 36.0        | 1.50       |
| SmolLM2 1.7B | 0.524       | 9.4       | 35.8        | 19.6        | 4.49       |

over the effective-parameter control. Equivalent partitions on Pythia exhibit small numerical NLL differences under the original BF16 execution. A labeled post hoc diagnostic repeats the *identical* native action three times per Pythia1B seed with identical tensors, labels, batches and optimizer settings. The largest within-seed/domain NLL range is 0.00152, confirming execution variation on the scale of these differences. We do not interpret them as structural improvements. There is no measured compression speedup: the pretrained backbone remains stored and executed in full.

## G.6 COMPUTE, REPLAY AND ARTIFACT SCOPE

The local machine has eight RTX5090 GPUs with about 32GiB each. Model training uses a dedicated environment with Transformers4.57.6 and PyTorch2.7.0a0 (NVIDIA build); the previous project environment is retained. The largest allocated GPU footprint is 19.08GiB. Per-checkpoint timing includes explicit synchronization for training; total training-stage timing additionally includes loading, evaluation and natural-gradient collection. Recovery is performed on CPU, with up to 12 jobs in parallel. The complete environment, source freeze, corpus lock, checkpoint hashes, observations and per-case solver records are retained.

An isolated PyTorch2.7.1 CPU environment replays 81 selected cases: every evaluation checkpoint,  $q = 4$ , designed/noiseless and general-text natural queries at noise  $10^{-4}$  and  $10^{-2}$ . Across 243 estimator outputs, the largest permutation-aligned estimate difference is  $1.39 \times 10^{-10}$ , and all acceptance decisions agree. Three additional full-dimensional natural-query checks compare explicit analytic residuals against the compressed objective for deliberately wrong candidates; absolute discrepancies are below  $3.8 \times 10^{-15}$ . This is independent arithmetic replay, not independent pretrained-model training or external peer review. The source package contains the manuscript; the research artifact separately contains observations, checkpoints, protocols and commands. Upstream model weights and raw gradient caches are not bundled with the manuscript. Scripts regenerate them from pinned inputs.

## H EARLIER SYNTHETIC AND BYTE-LANGUAGE-MODEL ILLUSTRATIONS

This appendix preserves the earlier studies, separately from the pretrained-model study in Section 5. The theory is the main evidence for the global statements. These earlier experiments check the response formula, illustrate finite reconstruction, and expose precision and conditioning limits. They use the exact-product design of Corollary 2 or rank-checked natural/Gaussian probes, *not* the repeated-complement design of Corollary 3. Thus they do not empirically evaluate its whole-feasible-set diameter bound. All numerical tables come from validated result records; protocols, negative results, source hashes, and replay limitations are retained in Appendices I–Q.

**Independent full-dimensional response measurements.** Three soft-router byte-LM checkpoints have  $L = 12$ ,  $K = 4$ , and  $D = 787,712$ . We measure responses by reverse-mode differentiation of a linear effective loss followed by a forward-mode JVP of the packed factor-to-product map. The observation code does not call Equation 7. For each native, permutation, and fixed non-permutation chart,

four designed probes reconstruct the response and two unseen Gaussian probes test its predictions. Table 6 shows numerical agreement with the formula, successful held-out prediction, and separation of the fixed alternative. This finite check does not prove separation of every possible alternative; that quantifier is supplied by the theorem.

Table 6: Full-dimensional autodiff response audit. Errors are relative Frobenius norms. Formula and held-out columns take the maximum over native, permutation, and non-permutation charts; each chart uses four designed and two unseen probes.

| Seed | Formula error | Held-out error | Permutation gap | Non-perm. gap |
|------|---------------|----------------|-----------------|---------------|
| 0    | 3.52e-16      | 4.68e-14       | 9.73e-17        | 0.9734        |
| 1    | 3.37e-16      | 5.07e-14       | 1.05e-16        | 0.9731        |
| 2    | 3.71e-16      | 4.12e-14       | 1.11e-16        | 0.9735        |

**Natural directions and finite precision.** Actual byte-LM cross-entropy gradients with respect to independent effective layer weights supply 16 directions per checkpoint. Controlled autodiff responses are measured at a fixed checkpoint; the first  $q \in \{1, 4, 8, 12\}$  directions fit the structural estimator and the last four test prediction. With the exact product row space, four directions meet the sufficient reconstruction ranks at all six checkpoints in Table 7. Factor errors use permutation alignment only for evaluation. One direction has insufficient local rank and gives errors 0.258–1.34 at the three longer-trained checkpoints.

Table 7: Four natural task-gradient probes with known response structure. Original and new checkpoints use the separately registered 1,500- and 6,000-step configurations. Recovery error is the combined relative factor error after evaluation-only permutation alignment. These are FP64 fixed-checkpoint measurements, not observed training trajectories.

| Configuration | Seed | Max. held-out response error | Factor error          |
|---------------|------|------------------------------|-----------------------|
| Original      | 0    | $9.57 \times 10^{-14}$       | $3.02 \times 10^{-8}$ |
| Original      | 1    | $8.22 \times 10^{-14}$       | $5.37 \times 10^{-9}$ |
| Original      | 2    | $8.63 \times 10^{-14}$       | $1.36 \times 10^{-8}$ |
| New           | 10   | $9.39 \times 10^{-14}$       | $2.31 \times 10^{-8}$ |
| New           | 11   | $6.45 \times 10^{-14}$       | $1.35 \times 10^{-8}$ |
| New           | 12   | $9.95 \times 10^{-14}$       | $1.48 \times 10^{-8}$ |

Finite Euclidean steps reveal an arithmetic limit. At step size  $10^{-5}$ , the longer-trained checkpoints have FP64 response errors  $3.3 \times 10^{-8}$ – $5.2 \times 10^{-8}$ , but FP32 subtraction errors 0.0208–0.0260 and BF16 errors about one. These tests cast factors and arithmetic to the stated precision; they do not test mixed-precision training with FP32 master weights. The step-size sweep is in Appendix P.

**Noisy recovery does not supply an automatic trust certificate.** A separate frozen development/calibration/evaluation protocol compares spectral recovery, simplex projection with a basis refit, and three-start constrained joint least squares. Both the product and measured responses receive relative noise up to  $10^{-2}$ ; the estimated product subspace is recomputed. On 1,920 held-out synthetic variants across 40 seeds, joint fitting reduces the median factor error from  $3.34 \times 10^{-3}$  for projected spectral recovery to  $1.01 \times 10^{-4}$ , and errors above 0.10 from 566 to 227. All 48 variants at four fresh byte-LM checkpoints have joint-fit errors below 0.0107.

Failures remain substantial: the largest joint-fit error is 10.54, and a combined diagnostic falsely accepts an error of 0.82. Its accepted-bad fraction is  $92/1782 = 5.16\%$ , with seed-cluster bootstrap interval  $[4.50\%, 5.79\%]$ ; residual-only rejection yields  $68/1757 = 3.87\%$ . The combined score therefore does not establish superiority, and its point estimate exceeds the 5% calibration target. The candidate-local checks also do not exclude distant solutions or establish truth membership in the fitted subspace. Appendix Q retains all methods, tail errors, failed solver statuses, and independent CPU replay differences. None of these diagnostics is the global certificate above.

**Auxiliary structural decisions.** Identifying coordinates under an optimizer does not make actions derived from them product invariant. In a documented reimplementation of ASLoRA’s first RoBERTa/MRPC merge rule, a fixed grid changes the primary action in 11/30 seed–gauge instances. Only the selected layer indices are carried back to a common canonical model; validation-loss differences have both signs. A separate controlled byte-LM partition audit finds changes at all three longer-trained checkpoints, but matched recovery training greatly reduces the performance gaps. These are bounded decision witnesses, not applications of the softmax theorem to LoRA, universal harm claims, or full-training regret evaluations. Detailed results, failed signed-local searches, and the earlier initialization-dominated experiment remain in Appendices J and P.

## I DETAILED AUDIT PROTOCOLS AND HISTORICAL SETTINGS

This appendix preserves the protocols for the original response and decision audits. The current numerical overview is in Section 5; the later controlled partition experiments and noisy-recovery studies have separate protocols in Appendices P and Q. They use different training configurations and must not be pooled as one replication rate.

### I.1 FROZEN-CHECKPOINT RESPONSE AUDIT

We first audit the distinguishing observation used by the new theorem: the instantaneous Euclidean response of the *current* softmax-router chart. At each of three independently trained soft-router byte-LM checkpoints, we verify the theorem’s assumptions and exact gauge construction. On a predeclared  $4 \times 32$  extraction we check Equation 7 against autograd and finite differences; on the full  $12 \times 787,712$  factor model we evaluate eight seeded Gaussian directions plus a permutation control. Separately, the  $K = 4$  interventions of Corollary 2 reconstruct the structured response. Their full-dimensional outputs are analytic equation evaluations—not four full-model JVP calls—with the extraction supplying the implementation check. The Gaussian test is fixed-alternative; tomography assumes exact common product, rank/dimension conditions, and this response family. Neither is an AdamW-trajectory or historical-graph claim.

An additive full-dimensional audit replaces the analytic observations with reverse-mode factor gradients followed by forward-mode JVPs of the packed effective-parameter map. It retains the same three checkpoints and fixed gauge, uses two unseen response directions per chart to test reconstruction, and records a five-level fixed-norm observation-noise sensitivity grid. Its prospective local protocol, source bindings, and numerical gates are released separately from the original audit; no retraining is involved.

### I.2 ASLoRA FIRST-MERGE DECISION AUDIT

This auxiliary decision experiment audits a documented reimplementation of the accessible ASLoRA-v2 first-merge rule (Hu et al., 2024). We train rank-eight,  $\alpha = 16$  adapters on RoBERTa-base/MRPC to the first scheduled merge opportunity, optimizer step 560. Query and value select separate pairs; a joint pair is sensitivity analysis. Because the source leaves adjacent-versus-all-pairs candidates ambiguous, we report both. A frozen bank contains identity, orthogonal, diagonal, and dense gauges. Each float64 gauge selects only a layer-index action by running- $B$  distance; we then apply the same temporary directed tie in the unchanged canonical model and evaluate the full validation split.

The selection proxy and the executed action are different estimands. Writing  $c = \alpha/r$ , the common change of rank coordinates  $B_{i,Q} = B_i Q$ ,  $A_Q = Q^{-1} A$  leaves  $c B_{i,Q} A_Q = c B_i A$  fixed, but scores a candidate  $p = (\ell \leftarrow u)$  by the order-equivalent squared score

$$s_p(Q) = \|(\bar{B}_\ell - \bar{B}_u)Q\|_F^2. \quad (137)$$

Executing that index action instead replaces the *current*  $B_\ell^t$  by  $B_u^t$ ; its immediate weight perturbation is  $c(B_u^t - B_\ell^t)A$ , and its lower-layer local RMS disturbance is  $\{\mathbb{E}\|c(B_u^t - B_\ell^t)Ax_\ell\|_2^2\}^{1/2}$ . Without assumptions linking historical averages, current factors, and the lower-layer input moment, the running score is not guaranteed to estimate action cost.

Three comparisons separate coordinate dependence from decision quality: current effective-update Frobenius distance is gauge invariant and uses only the prevalidation snapshot; current local-activation

RMS is gauge invariant but post-hoc and label-free; validation loss is a labeled oracle never used for selection. We exhaust query-only, value-only, joint-same-pair, and selected composite actions, not their full Cartesian product. Product invariance, restoration, and canonical-action checks gate the run; passing live-float32 re-execution remains a numerical diagnostic, not the causal estimand.

### I.3 HISTORICAL BYTE-LM SAME-BUDGET PARTITION-FOLD SEARCH

The original three-checkpoint experiment asked whether a controlled downstream consequence found at one checkpoint replicated. At each of the same three byte-LM checkpoints, a predeclared finite router-only search seeks a positive-stochastic gauge that changes the argmax partition at the same four-group budget. Both partitions are refit from the same effective tensors and compared on paired held-out batches; literal factor hardening is only a confounded diagnostic. Fixed-budget failures remain negative results, not proofs of non-existence, so one successful checkpoint establishes possibility rather than prevalence. All three routers meet the protocol’s initialization-dominated flag. Signed-local and same-sorted-group-size searches were unexecuted post-run amendments to this original protocol. They were subsequently executed at three fresh, more extensively trained checkpoints under the separate protocol in Appendix P.

## J DETAILED NUMERICAL RESULTS AND AUXILIARY DECISIONS

The audit and extension tables are generated from validated summary JSON artifacts, including negative seeds and failed searches; exact paths and hashes are recorded in Appendix K.

### J.1 A NON-PERMUTATION CHART IS VISIBLE IN OPTIMIZER RESPONSE

Table 8: Frozen-checkpoint response audit. The effective-product column checks the fixed non-permutation gauge;  $\Delta\mathcal{R}$  compares Euclidean instantaneous response operators on fixed Gaussian effective-gradient probes. The permutation column is the label-swap control.

| Seed | Effective product rel. error | $\min A$              | $\sigma_{\min}(A)$ | $\sigma_{\min}(B)$ | Perm. max rel. $\Delta\mathcal{R}$ | Nonperm. median rel. $\Delta\mathcal{R}$ | Detected probes |
|------|------------------------------|-----------------------|--------------------|--------------------|------------------------------------|------------------------------------------|-----------------|
| 0    | $1.58 \times 10^{-16}$       | $2.88 \times 10^{-4}$ | 1.7291             | 27.445             | $8.82 \times 10^{-17}$             | 0.5045                                   | 8/8             |
| 1    | $1.70 \times 10^{-16}$       | $2.72 \times 10^{-4}$ | 1.7291             | 27.339             | $8.82 \times 10^{-17}$             | 0.5051                                   | 8/8             |
| 2    | $1.63 \times 10^{-16}$       | $2.61 \times 10^{-4}$ | 1.7290             | 27.371             | $8.82 \times 10^{-17}$             | 0.5058                                   | 8/8             |

Table 8 checks the response formula and separates a fixed alternative at frozen checkpoints under Theorem 1’s assumptions. All stated rank conditions hold, the non-permutation reparameterization preserves the effective model to at most  $1.70 \times 10^{-16}$  relative error. On the declared  $4 \times 32$  extraction, the analytic response agrees with autograd within numerical precision, and the worst finite-difference error across native and transformed charts is  $9.57 \times 10^{-13}$ . The maximum permutation-control discrepancy is  $8.82 \times 10^{-17}$  after rounding, whereas all 24/24 checkpoint-probe evaluations detect a change (eight per checkpoint, with overlapping seed windows and 10 unique Gaussian draws; per-seed medians  $[0.5045, 0.5058]$ ). The separate structured audit reduces the all-gradient quantifier to 4 designed analytic response evaluations, which reconstruct all response components with worst relative error  $6.85 \times 10^{-11}$ ; the permutation control stays below  $9.44 \times 10^{-17}$ ; the minimum across seeds of each seed’s maximum fixed non-permutation difference over the four designed probes is 0.9731 after rounding. This does not say that every individual probe reaches that value. These are analytic evaluations of Equation 7, validated on the extraction, not four full-model JVP calls. The eight Gaussian probes remain a fixed-alternative test; the designed certificate assumes exact common product and the stated response family. Neither audit describes AdamW or a historical graph.

**Independent full-dimensional response measurements.** A subsequent audit obtains the responses by differentiating the complete packed map  $(Z, B) \mapsto \text{softmax}(Z/\tau)B$  at all three released checkpoints. Reverse-mode differentiation of a linear probe loss gives the factor gradients; a forward-mode JVP then measures the induced effective-parameter response. The observation code does not call the analytic response formula. Each native, permutation, and fixed non-permutation chart uses four designed probes and two unseen Gaussian probes. Table 6 reports formula agreement and prediction of the unseen responses from the reconstructed operator. The maximum relative formula discrepancy

is  $3.71 \times 10^{-16}$ , and the maximum held-out response error is  $5.08 \times 10^{-14}$ . A five-level observation-noise grid from zero to  $10^{-4}$  relative Frobenius norm satisfies the deterministic component error bounds, with recorded floating-point slack. These measurements concern the full shared-parameter map, not an end-to-end language-model loss or an AdamW step; operator reconstruction does not itself implement noisy factor recovery. The additive records preserve the original analytic audit.

**Exploratory reconstruction of the factors.** The construction in Appendix B.4 recovers factors from the effective product and measured components, without supplying the original factors to the reconstruction. We test it on the native charts of the same three checkpoints, using fresh autodiff observations and one fixed Gaussian weight vector for joint diagonalization. Table 9 reports the errors after a common permutation chosen only for evaluation. The noiseless combined relative errors are below  $3.7 \times 10^{-9}$ . All five noise levels complete without a rank or eigengap failure, but the unprojected noisy estimates need not have exact unit row sums. These are three checkpoint diagnostics with exact effective products, not independent training replications, adversarial-noise tests, or a general stability result for the PSD-clipped recovery algorithm.

Table 9: Exploratory factor recovery from full-dimensional autodiff observations. Orbit error combines relative router and basis errors after one common permutation. Noisy rows use relative observation noise  $10^{-4}$  and no simplex projection.

| Seed | Noiseless orbit error | Noisy orbit error | Noisy row-sum error |
|------|-----------------------|-------------------|---------------------|
| 0    | 3.66e-09              | 1.81e-05          | 6.06e-05            |
| 1    | 3.23e-09              | 1.61e-05          | 6.68e-05            |
| 2    | 3.28e-09              | 1.59e-05          | 5.99e-05            |

## J.2 GAUGE-EQUIVALENT ASLoRA STATES CHANGE THE FIRST MERGE DECISION

Table 10: ASLoRA first-merge decision audit over 10 gauges per seed. Changes are relative to the identity-gauge raw-running- $B$  action. The loss interval is the canonical validation difference (selected action minus identity action), not a gauged float32 re-execution; accuracy and F1 intervals are in Appendix Table 14. The chain column requires a changed action with a nonzero validation consequence in every seed.

| Candidates | Scope      | By seed   | Changed | Seeds | $\Delta$ loss        | Chain |
|------------|------------|-----------|---------|-------|----------------------|-------|
| Adjacent   | per-proj.  | (2, 6, 3) | 11/30   | 3/3   | $[-0.0531, +0.0089]$ | yes   |
| Adjacent   | joint pair | (0, 5, 0) | 5/30    | 1/3   | $[+0.0000, +0.0077]$ | no    |
| All pairs  | per-proj.  | (4, 7, 3) | 14/30   | 3/3   | $[-0.0531, +0.0027]$ | yes   |
| All pairs  | joint pair | (0, 0, 0) | 0/30    | 0/3   | $[+0.0000, +0.0000]$ | no    |

The primary per-projection chain replicates under both readings of the source’s candidate set. Among 30 predeclared gauge instances, the raw-running- $B$  action changes in 11 instances for adjacent candidates and 14 instances for all pairs. Every seed has at least one changed action with a nonzero canonical validation consequence under both candidate readings. The signs in Table 10 go in both directions: the experiment demonstrates representation-dependent decisions, not that every gauge-induced change is harmful or that the identity action is optimal. The joint-same-pair sensitivity does not replicate (one seed for adjacent candidates and none for all pairs), and Appendix Table 15 shows that neither gauge-invariant rule uniformly improves on raw running  $B$  across all seeds under both candidate readings. The causal contrast uses a float64-equivalent snapshot only to select layer indices, then executes the same temporary tie in the unchanged canonical model; factorized-float32 re-execution is a passing secondary diagnostic. Thus the complete chain is concrete but local to one first merge, method, task, and fixed gauge grid.

## J.3 PROSPECTIVE CONTROLS: LEARNED ROUTERS AND RECOVERY TRAINING

The historical witness in Appendix P.1 motivates a separately registered extension; it does not supply its outcomes. We train three fresh width-128, twelve-layer, four-basis byte LMs for a fixed 6,000

Table 11: Prospective equal-size folding audit. Signed BPB differences are gauge-selected minus native-selected folds after matched recovery updates. Each row is one independent training seed. Positive-stochastic search succeeds in 3/3 seeds; signed-local search fails in 3/3, each with 5,000 router-only candidates.

| Seed | Router movement | $\Delta_0$ | $\Delta_{100}$ | $\Delta_{500}$ |
|------|-----------------|------------|----------------|----------------|
| 10   | 0.593           | -0.00112   | +0.00224       | +0.00009       |
| 11   | 0.720           | +0.87112   | +0.03574       | +0.01823       |
| 12   | 0.729           | +0.14873   | +0.00580       | +0.00343       |

steps with random router initialization and a separate router learning rate. Router L1 movement is 0.593–0.729; all three models pass the unigram-improvement and router-movement gates. Both gauge families receive 5,000 router-only trials, now requiring the same sorted group-size multiset as the native partition. Positive-stochastic search finds a first valid candidate in every seed; signed-local search fails in all three. Failures are retained and never replaced by larger searches.

Every fold uses the same native effective tensors. Hard routes remain fixed while all remaining parameters receive identical minibatches, optimizer initialization, update counts, and learning rates. Table 11 shows substantial immediate differences in two seeds, but much smaller differences after 500 recovery updates. Conditional paired 95% bootstrap intervals at that stage are  $[-0.00094, 0.00112]$ ,  $[0.01654, 0.01989]$ , and  $[0.00244, 0.00437]$  BPB, respectively; they describe the fixed evaluation blocks, not uncertainty over training populations. The same-size effective-parameter local-search baseline selects the native partition in two seeds and performs slightly worse after recovery in the third. Thus removing the earlier initialization and size confounds strengthens the decision-sensitivity evidence, without establishing persistent large harm or general superiority of an invariant rule. This is one architecture and corpus, with a fixed training budget rather than a convergence certificate.

#### J.4 NATURAL TASK GRADIENTS AND FINITE ARITHMETIC

We differentiate actual byte-LM cross-entropy with respect to independent layer-effective weights, holding other parameters fixed. A separate forward and chain-rule check agrees with the original model. At each of the three original and three new checkpoints, 16 fixed training minibatches supply natural directions. Controlled reverse/forward autodiff measures the response; the first  $q \in \{1, 4, 8, 12\}$  directions fit the estimator, and the last four test prediction. The estimator uses the exact product row space and known response structure, not the true factors.

At  $q = 4$ , all six checkpoints meet the rank conditions of Proposition 11. The largest held-out response error is below  $10^{-13}$  and the factor error below  $3.1 \times 10^{-8}$ . This is stronger than separation of a single alternative chart, but remains conditional on the known Euclidean response family and precise measurements. A single direction has insufficient local rank; at the new checkpoints its factor errors range from 0.258 to 1.34. Generic linear prediction from the observed gradient span also differs from the structural estimator. These directions are measurements at fixed checkpoints, not observations of an uncontrolled training trajectory.

Explicit finite Euclidean steps expose a numerical boundary. At step size  $10^{-5}$ , the new checkpoints’ FP64 response errors are  $3.3 \times 10^{-8}$ – $5.2 \times 10^{-8}$ , whereas FP32 subtraction errors are 0.0208–0.0260 and BF16 errors are about one. Here factors and arithmetic are cast to the stated precision; this is not a test of BF16 training with FP32 master weights. Appendix Figure 7 reports the step-size tradeoff and the separate joint-noise conditioning diagnostic.

#### J.5 NOISY RECOVERY AND THE LIMITS OF OBSERVABLE REJECTION

We fix a prospective development/calibration/evaluation split before measuring the final outcomes. Forty held-out synthetic seeds each contribute 48 correlated noise/probe/regime variants; four freshly trained byte-LM checkpoints each contribute 12. Both product and autodiff response observations receive relative noise from zero to  $10^{-2}$ , and the rank- $K$  product row subspace is re-estimated from the noisy product. Three fixed methods use the same observations: spectral recovery, simplex projection with a basis refit, and three-start constrained joint least squares. The estimator never receives the true factors. Appendix Q gives splits, budgets, calibration, and all diagnostic comparators.

Table 12: Frozen held-out noisy recovery. Each synthetic instance contributes 48 correlated variants; each language checkpoint contributes 12. Acceptance uses the calibrated empirical score. Bad means factor error  $> 0.10$ ; good means  $\leq 0.05$ . Local passes concern only a numerical restricted-subspace component.

| Domain    | Method    | Median error | Bad/total | Accept | Bad/accept | Local |
|-----------|-----------|--------------|-----------|--------|------------|-------|
| Synthetic | Spectral  | 0.0041       | 564/1920  | 196    | 0/196      | 0     |
| Synthetic | Projected | 0.0033       | 566/1920  | 1396   | 56/1396    | 0     |
| Synthetic | Joint fit | 0.0001       | 227/1920  | 1782   | 92/1782    | 140   |
| Byte-LM   | Spectral  | 0.00017      | 1/48      | 0      | —          | 0     |
| Byte-LM   | Projected | 0.0002       | 1/48      | 48     | 1/48       | 0     |
| Byte-LM   | Joint fit | 3.3e-05      | 0/48      | 48     | 0/48       | 2     |

Joint fitting reduces the synthetic median factor error from  $3.34 \times 10^{-3}$  for projected spectral recovery to  $1.01 \times 10^{-4}$ , and the number of errors above 0.10 from 566 to 227 out of 1920. At the four byte-LM checkpoints, all 48 joint-fit errors are below 0.0107, including the 1% observation-noise cases. These are controlled Euclidean measurements at checkpoints, not noisy AdamW trajectory observations.

The more demanding question is whether an observable diagnostic can tell when to trust a result. With its threshold fixed on separate calibration data, the combined residual/conditioning/subspace/multi-start score accepts 1782/1920 synthetic outputs, including 92 bad outputs. Its 5.16% accepted-bad fraction has a seed-cluster bootstrap interval of [4.50%, 5.79%] and has a point estimate above the 5% calibration target; the interval includes that target. Residual-only rejection accepts 1757/1920 with  $68/1757 = 3.87\%$  bad. Thus, the combined score does not establish superiority. The full evaluation retains a joint-fit error of 10.54, and a falsely accepted error of 0.82. These failures prevent a general reliability claim.

The explicit local separation bound in Proposition 12 addresses a different question. Only 140/1920 synthetic and 2/48 language outputs pass its numerical local-component conditions. These passes neither establish truth membership in the restricted subspace nor rule out distant solutions.

## K EXPERIMENTAL DETAILS

### K.1 ORIGINAL AUDIT INVENTORY

Table 13 records the original audit units. The three response checkpoints and three byte-LM decision checkpoints are paired by training seed. The ASLoRA runs are a separate RoBERTa-base/MRPC experiment.

All numerical entries in the three original audit tables are generated directly from `results/response_summary.json`, `results/aslora_summary.json`, and `results/router_decisions_summary.json` by `experiments/generate_paper_audit_tables.py`. Each generated file records the SHA256 of its source summary, retaining negative seeds and failed searches in the reported aggregates; the same script emits the appendix metric and search-detail tables. The additive tomography numbers are generated separately from `results/response_tomography_summary.json` by `experiments/generate_response_tomography_latex.py`; this leaves the source-attested older response table unchanged. The subsequent full-dimensional autodiff and factor-recovery tables are generated by their separate summarizers from `results/autograd_tomography/` and `results/factor_recovery/`. Both validate generation-source and checkpoint hashes. They reuse the same three checkpoints and therefore do not increase the number of independent training seeds.

Table 13: Original primary audit inventory. Each audit has three seeds; counts include failed searches and controls. Additive audits reuse the response checkpoints, as detailed below.

| Audit                  | Per-seed protocol                                                               | Required checks                                    |
|------------------------|---------------------------------------------------------------------------------|----------------------------------------------------|
| Optimizer response     | 8 Gaussian and 4 designed probes; fixed non-permutation and permutation control | Rank, formula, exact-gauge product, reconstruction |
| ASLoRA first merge     | 10 gauges; adjacent/all pairs and per-projection/joint actions                  | Coverage, canonical action, restoration            |
| Byte-LM partition fold | 5,000 trials; four-group, same-effective re-fits                                | Fixed search and paired evaluation                 |

The response audit uses a fixed gauge strength of 0.35, unit Euclidean learning rates for both parameter blocks, and probe seeds determined before evaluation. Each result records its checkpoint and source hashes. The best finite-difference scale on a predeclared four-row, 32-coordinate checkpoint extraction is selected only to validate the closed-form response; the eight fixed Gaussian directions on the full  $12 \times 787,712$  packed model provide the separate non-permutation detection test. The additive four-probe audit constructs the Corollary 2 design from  $\Theta$  and reconstructs the response components. Its full-dimensional responses are evaluations of the already checked analytic formula, not four additional full-model autograd/JVP calls. Its version-2 artifacts separately record the reference rank/dimension gate, the floating-point product residual, and exact common-product validity from the explicit invertible-gauge construction.

The subsequent autodiff audit instead differentiates the packed factor map. For each checkpoint it measures four designed and two unseen responses in each of three charts: native, permutation, and a fixed non-permutation gauge. Its five-level observation-noise diagnostic uses the native chart. The factor-recovery pilot makes four fresh native-chart measurements, reuses one fixed diagonalization weight seed across checkpoints, and reports all five noise levels without selecting a favorable weight vector. True factors enter the observation oracle and post-reconstruction error calculation, but not the reconstruction routine. Neither experiment differentiates the language-model task loss or simulates an AdamW trajectory.

The ASLoRA protocol follows the paper’s defined configuration: 30 epochs, effective batch size 16, rank eight,  $\alpha = 16$ ,  $T_s = 320$ , and merge interval 240. Appendix Table 5 of the source additionally records an update ratio  $\lambda = 0.5$ , but the accessible method text and Algorithm 1 define no operation involving that quantity. We record  $\lambda$  in provenance as `update_ratio_record_only` and deliberately do not invent an implementation for it. With the one-indexed rule  $t > T_s$  and  $(t - T_s) \bmod 240 = 0$ , the first audit point is  $t = 560$ . Running  $B$  is observed after every optimizer update. The three artifacts contain 204, 208, and 200 canonical action evaluations, respectively: all 66 query-only, 66 value-only, and 66 joint-same-pair actions plus every additional composite selected by a declared baseline or gauge. There are no missing or undeclared candidates. Every temporary tie is restored, the trainable digest and parameter identities agree before and after the audit, and no merge is committed.

The amendment chronology in `research/ASLoRA_EMPIRICAL_PROTOCOL.md` records the superseded nondeterministic 12/30 and 16/30 batch, its source hashes, the outcome-blind deterministic protocol lock, and the canonical replacement by the 11/30 and 14/30 results reported here.

For transparency, the ASLoRA artifacts contain two numerical paths. The scientific path transforms the frozen running factors in float64, chooses a layer-index action, and evaluates that action in the canonical model. All canonical gates pass. The diagnostic path re-executes the live factorized float32 model and also passes its tolerance in all three deterministic runs. No main-paper metric is taken from that diagnostic path. Both candidate interpretations are retained because Algorithm 1 and the prose in the source differ on whether all or only adjacent layer pairs are sorted.

Table 14: Canonical validation accuracy and F1 differences for the four ASLoRA candidate/scope cells in Table 10. Intervals range over all 30 selected gauge actions in each cell, including unchanged selections, and are selected-action minus identity-action differences.

| Candidates | Scope          | $\Delta$ accuracy    | $\Delta$ F1          |
|------------|----------------|----------------------|----------------------|
| Adjacent   | per-projection | $[-0.0025, +0.0098]$ | $[-0.0022, +0.0057]$ |
| Adjacent   | joint pair     | $[-0.0049, +0.0000]$ | $[-0.0051, +0.0000]$ |
| All pairs  | per-projection | $[-0.0025, +0.0098]$ | $[-0.0022, +0.0057]$ |
| All pairs  | joint pair     | $[+0.0000, +0.0000]$ | $[+0.0000, +0.0000]$ |

Table 15: Declared alternatives for the primary per-projection ASLoRA decision. Effective-update and local-activation rules are gauge invariant. Their cells give the selected action and canonical validation-loss difference from the identity raw- $B$  action (negative is lower; exact zero is a tie). Across the six rows, effective has 2 negative/2 positive/2 zero differences and local activation has 2 negative/1 positive/3 zero differences; neither rule uniformly improves across all seeds under both candidate readings. Validation minima are separate query-only and value-only enumerations; they are not a full Cartesian per-projection oracle.

| Seed | Candidates | Raw running $B$      | Current effective                              | Local activation                              | Single-target minima   |
|------|------------|----------------------|------------------------------------------------|-----------------------------------------------|------------------------|
| 0    | Adjacent   | $Q : 0 \leftarrow 1$ | $Q : 0 \leftarrow 1$                           | $Q : 0 \leftarrow 1$                          | $Q : 6 \leftarrow 7$   |
|      |            | $V : 0 \leftarrow 1$ | $V : 10 \leftarrow 11$<br>$\Delta L = -0.0336$ | $V : 0 \leftarrow 1$<br>$\Delta L = +0.0000$  | $V : 10 \leftarrow 11$ |
| 0    | All pairs  | $Q : 0 \leftarrow 1$ | $Q : 0 \leftarrow 1$                           | $Q : 0 \leftarrow 1$                          | $Q : 6 \leftarrow 9$   |
|      |            | $V : 0 \leftarrow 1$ | $V : 1 \leftarrow 11$<br>$\Delta L = -0.0075$  | $V : 0 \leftarrow 11$<br>$\Delta L = -0.0008$ | $V : 10 \leftarrow 11$ |
| 1    | Adjacent   | $Q : 0 \leftarrow 1$ | $Q : 0 \leftarrow 1$                           | $Q : 0 \leftarrow 1$                          | $Q : 4 \leftarrow 5$   |
|      |            | $V : 2 \leftarrow 3$ | $V : 10 \leftarrow 11$<br>$\Delta L = +0.0018$ | $V : 0 \leftarrow 1$<br>$\Delta L = +0.0001$  | $V : 3 \leftarrow 4$   |
| 1    | All pairs  | $Q : 0 \leftarrow 2$ | $Q : 0 \leftarrow 1$                           | $Q : 0 \leftarrow 1$                          | $Q : 4 \leftarrow 11$  |
|      |            | $V : 0 \leftarrow 2$ | $V : 10 \leftarrow 11$<br>$\Delta L = +0.0028$ | $V : 0 \leftarrow 11$<br>$\Delta L = -0.0005$ | $V : 3 \leftarrow 5$   |
| 2    | Adjacent   | $Q : 0 \leftarrow 1$ | $Q : 0 \leftarrow 1$                           | $Q : 0 \leftarrow 1$                          | $Q : 8 \leftarrow 9$   |
|      |            | $V : 0 \leftarrow 1$ | $V : 0 \leftarrow 1$<br>$\Delta L = +0.0000$   | $V : 0 \leftarrow 1$<br>$\Delta L = +0.0000$  | $V : 8 \leftarrow 9$   |
| 2    | All pairs  | $Q : 0 \leftarrow 1$ | $Q : 0 \leftarrow 1$                           | $Q : 0 \leftarrow 1$                          | $Q : 8 \leftarrow 11$  |
|      |            | $V : 0 \leftarrow 1$ | $V : 0 \leftarrow 1$<br>$\Delta L = +0.0000$   | $V : 0 \leftarrow 1$<br>$\Delta L = +0.0000$  | $V : 8 \leftarrow 11$  |

The byte-LM search makes 5,000 predeclared positive-stochastic trials per checkpoint and preserves a four-group action budget. Failed searches retain their attempted, invertible, condition-passing, simplex-passing, and group-passing counts. The successful contrast refits both partitions from common effective tensors. Literal hardening is excluded from the primary comparison because it changes basis coordinates and the partition jointly. All three routers satisfy the initialization-dominated flag. Only seed 0 reaches decision evaluation: its k-means and Ward effective-tensor baselines both have BPB  $2.1753145$  and centroid SSE  $5.46896 \times 10^{-5}$ . Seeds 1 and 2 store no folds or baseline evaluation because their searches exit first. Signed-local and same-sorted-group-size searches were not executed in that historical audit; they were post-run, non-primary protocol amendments. In particular, the observed group-size change  $3/3/3/3 \rightarrow 5/3/3/1$  remains a balance confound. The reported interval resamples the 256 paired evaluation-batch differences 10,000 times. It describes evaluation variability conditional on this checkpoint and selected contrast; it does not cover training-seed or gauge-search uncertainty.

The protocol defines the initialization-dominated flag as mean row-wise router  $L_1$  movement from initialization strictly below 0.01. The three observed movements round to 0.000313–0.000401, so every checkpoint meets that declared threshold; this flag describes training movement, not recovery of a historical graph.

Table 16: Byte-LM checkpoint and finite-search diagnostics omitted from the compact main table. The funnel is attempted/invertible/condition-pass/simplex-pass/group-pass/partition-change.

| Seed | Soft BPB | Mean row router $L_1$ move | Search funnel A/I/C/S/G/P  |
|------|----------|----------------------------|----------------------------|
| 0    | 2.186    | $3.35 \times 10^{-4}$      | 5000/5000/4586/4586/3762/1 |
| 1    | 2.188    | $3.13 \times 10^{-4}$      | 5000/5000/4601/4601/3790/0 |
| 2    | 2.168    | $4.01 \times 10^{-4}$      | 5000/5000/4581/4581/3769/0 |

Table 17: Geometry and gauge-invariant comparator details for the sole completed byte-LM witness. The k-means and Ward folds coincide here; the unigram baseline is included only as a scale reference.

| Seed | $\kappa(Q)$ | Group sizes ref. $\rightarrow$ cand. | Disagreement | k-means/Ward BPB | Centroid SSE          | Unigram BPB |
|------|-------------|--------------------------------------|--------------|------------------|-----------------------|-------------|
| 0    | 7.93        | 3/3/3/3 $\rightarrow$ 5/3/3/1        | 0.121        | 2.175/2.175      | $5.47 \times 10^{-5}$ | 4.624       |

## L SECONDARY PLANTED-DATA AND PROBE SANITY CHECKS

The experiments below were the original TYINGBENCH and SHAREPROBE suite. They remain useful implementation and theorem sanity checks, but they do not establish that router misinterpretation is prevalent in deployed systems, and they are not the paper’s primary empirical contribution.

### L.1 TYINGBENCH PROTOCOLS

**Exact factorization.** We generate  $K$  orthogonal basis vectors with controlled condition number, choose one of five depth graphs (cycle, contiguous, palindrome, balanced random, or imbalanced), and expose the resulting effective layer vectors. We jointly fit soft bases and router coefficients from neutral, random, correct-pattern, and deliberately wrong-pattern initializations. Hard straight-through, vertex-regularized,  $k$ -means, and oracle estimators provide comparisons. This setting removes sampling error: failure is factorization or optimization ambiguity, not insufficient task data.

**Functional distillation.** A planted one-hot teacher and a jointly trained soft student share the same architecture and basis budget. We use an eight-layer residual MLP and an eight-layer decoder-only Transformer. The Transformer mixes parameters before applying each block, evaluates one block per layer, and uses an explicit upper-triangular causal mask. Students observe teacher outputs but not the teacher graph. Test MSE or KL measures function fit; ARI measures recovery of the planted graph. These labels are legitimate for the planted teacher, not for the natural-language experiment.

**Language-model stability arm.** We train a 12-layer, four-basis causal Transformer on WikiText-2 bytes. The task has no privileged sharing graph, so this arm has no recovery label. BPB, router movement, and cross-run graph agreement diagnose path dependence only. It is separate from the original same-budget partition-fold audit.

Table 18: Secondary sanity-sweep sizes. Every listed seed is crossed with the conditions defined by the corresponding executable sweep.

| Family                          | Depth | Bases | Width | Seeds | Budget / grid                                       |
|---------------------------------|-------|-------|-------|-------|-----------------------------------------------------|
| Factorization                   | 12    | 4     | 128   | 10    | 2,000 steps; 5 graphs                               |
| Residual MLP distillation       | 8     | 4     | 64    | 5     | 1,500 steps; 4 graphs                               |
| Causal Transformer distillation | 8     | 4     | 64    | 5     | 1,000 steps; 4 graphs                               |
| Functional probes               | 12    | 4     | 64    | 10    | $n \in \{1, 2, 4, 8, 16, 32, 64\}$ ; 6 noise levels |
| WikiText-2 stability            | 12    | 4     | 256   | 3     | 1,500 steps; context 128                            |

All models use parameter-space mixing: each effective tensor is formed once from the same layer router row, and each block is evaluated once. This avoids the compute mismatch of an output-space mixture that evaluates all  $K$  basis blocks. Teacher graphs and datasets are deterministic functions

of recorded seeds. The causal Transformer test changes future tokens while asserting that all earlier logits are unchanged.

## L.2 SHAREPROBE AS A CONDITIONAL DIAGNOSTIC

Native activations can confound a layer transition with the distribution created by preceding layers. SHAREPROBE draws  $H_1, \dots, H_n \sim P_H$  once and evaluates every residual transition  $r_\ell$  on the same inputs. In the revised paired protocol, observations satisfy  $Y_{\ell s} = r_\ell(H_s) + \sigma \epsilon_{\ell s}$  with iid standard-Gaussian noise at a fixed absolute scale  $\sigma$ . Its signature and empirical squared distance are

$$\phi_\ell = n^{-1/2}[Y_{\ell 1}; \dots; Y_{\ell n}], \quad \hat{d}_{\ell m}^2 = \|\phi_\ell - \phi_m\|_2^2. \quad (138)$$

Counterfactual and native estimators use the same initial probes, noise array, and optional random projection. The native trajectory advances only with clean residual updates. Thus each method has the stated marginal observation model and their difference uses common random numbers; unlike the superseded implementation, noise is not rescaled by a realized batch standard deviation. For example, layers using  $f(h) = h$  and  $g(h) = -h$  in the true pattern  $(f, g, f, g)$  yield the wrong native-output groups  $\{1, 2\} \mid \{3, 4\}$  on inputs  $(1, -1, -1, 1)$ , whereas the common input  $h = 1$  recovers  $\{1, 3\} \mid \{2, 4\}$ . This is a counterexample to an unqualified native-trajectory interpretation, not evidence that native signatures are generally inferior.

Indeed, Table 27 shows no stable empirical advantage for the named probe. Shared-input versus native mean ARI is 0.6764 versus 0.6751 for the MLP and 0.5006 versus 0.5009 for the Transformer. The paired win/tie/loss counts are 65/1570/45 and 61/1558/61, respectively, and the seed-averaged difference changes sign (six of ten MLP seeds but only three of ten Transformer seeds favor shared inputs). A simple baseline clusters the concatenated visible effective weights and biases of every mixed linear module; it obtains ARI 1.000 in every planted cell. That baseline requires parameter access and parameter distance is *not* functional equivalence in a neural network with parameter symmetries. It nevertheless shows that the probe is unnecessary in this benchmark when effective block tensors are visible.

We therefore retain SHAREPROBE only as a conditional functional diagnostic. Its graph is invariant to factor coordinates and targets equivalence on  $P_H$ , but the probe distribution must cover the claimed domain,  $K$  must be supplied to the clusterer, and the observed planted-partition gap becomes a certificate only under the population-separation and uniform-tail assumptions of Appendix B.6. The experiment does not estimate those population quantities.

## M SECONDARY SANITY RESULTS

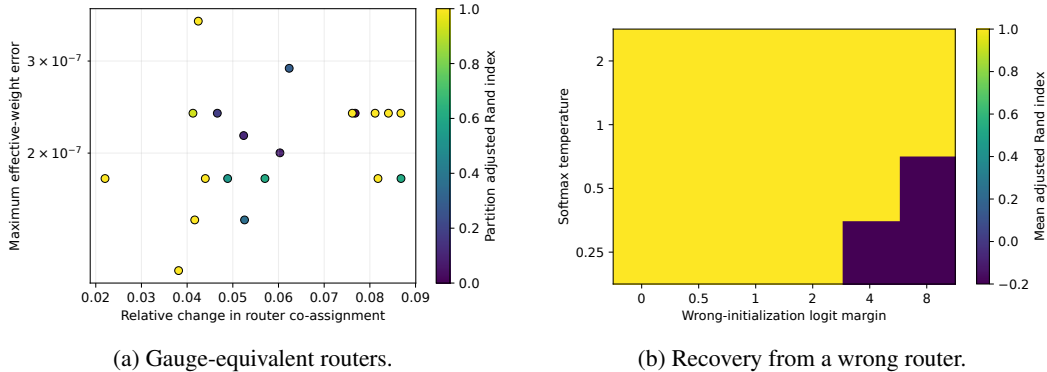


Figure 3: Synthetic theory checks. Effective weights remain equal at numerical precision while router summaries can change; saturation can suppress recovery from a deliberately wrong initialization.

The exact-factorization and distillation sweeps show that, for the tested planted teachers and budgets, similar reconstruction or task loss need not select the same router graph. They are possibility and path-dependence checks, not estimates of real-world frequency. Discrete clustering succeeds when

its one-hot model matches the teacher, which is a positive control rather than a general recovery guarantee.

Table 19: Gauge counterfactuals. Values are mean  $\pm$  sample standard deviation across trials.

| Metric                                  | Value                                         | $n$ |
|-----------------------------------------|-----------------------------------------------|-----|
| Effective max error                     | $2.11 \times 10^{-7} \pm 5.41 \times 10^{-8}$ | 20  |
| Argmax-partition ARI                    | $0.686 \pm 0.380$                             | 20  |
| Co-assignment relative change           | $0.059 \pm 0.019$                             | 20  |
| Post-step effective relative difference | $0.002 \pm 9.90 \times 10^{-4}$               | 20  |

Table 20: Softmax-saturation sweep. Values are aggregated across seeds for each temperature and wrong-initialization margin.

| Temperature | Margin | Initial grad.                                   | Final MSE                                      | ARI                               | $n$ |
|-------------|--------|-------------------------------------------------|------------------------------------------------|-----------------------------------|-----|
| 0.250       | 0      | $0.008 \pm 4.18 \times 10^{-5}$                 | $8.07 \times 10^{-8} \pm 2.96 \times 10^{-10}$ | $1.000 \pm 0$                     | 5   |
| 0.250       | 0.500  | $0.002 \pm 1.04 \times 10^{-10}$                | $7.88 \times 10^{-8} \pm 2.08 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 0.250       | 1.000  | $4.62 \times 10^{-5} \pm 1.63 \times 10^{-12}$  | $6.00 \times 10^{-8} \pm 1.86 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 0.250       | 2.000  | $1.55 \times 10^{-8} \pm 7.94 \times 10^{-16}$  | $1.94 \times 10^{-8} \pm 9.27 \times 10^{-14}$ | $1.000 \pm 0$                     | 5   |
| 0.250       | 4.000  | $9.14 \times 10^{-16} \pm 1.16 \times 10^{-22}$ | $0.021 \pm 2.28 \times 10^{-9}$                | $-0.222 \pm 3.10 \times 10^{-17}$ | 5   |
| 0.250       | 8.000  | $0 \pm 0$                                       | $0.021 \pm 2.28 \times 10^{-9}$                | $-0.222 \pm 3.10 \times 10^{-17}$ | 5   |
| 0.500       | 0      | $0.004 \pm 8.64 \times 10^{-6}$                 | $2.52 \times 10^{-7} \pm 6.31 \times 10^{-10}$ | $1.000 \pm 0$                     | 5   |
| 0.500       | 0.500  | $0.004 \pm 2.65 \times 10^{-10}$                | $2.68 \times 10^{-7} \pm 4.13 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 0.500       | 1.000  | $0.001 \pm 5.21 \times 10^{-11}$                | $2.76 \times 10^{-7} \pm 2.96 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 0.500       | 2.000  | $2.31 \times 10^{-5} \pm 8.13 \times 10^{-13}$  | $2.49 \times 10^{-7} \pm 3.02 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 0.500       | 4.000  | $7.74 \times 10^{-9} \pm 3.97 \times 10^{-16}$  | $1.44 \times 10^{-7} \pm 9.62 \times 10^{-14}$ | $1.000 \pm 0$                     | 5   |
| 0.500       | 8.000  | $4.57 \times 10^{-16} \pm 5.80 \times 10^{-23}$ | $0.021 \pm 2.28 \times 10^{-9}$                | $-0.222 \pm 3.10 \times 10^{-17}$ | 5   |
| 1.000       | 0      | $0.002 \pm 1.73 \times 10^{-6}$                 | $7.25 \times 10^{-7} \pm 7.43 \times 10^{-10}$ | $1.000 \pm 0$                     | 5   |
| 1.000       | 0.500  | $0.002 \pm 2.85 \times 10^{-10}$                | $7.17 \times 10^{-7} \pm 4.67 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 1.000       | 1.000  | $0.002 \pm 1.33 \times 10^{-10}$                | $7.58 \times 10^{-7} \pm 5.92 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 1.000       | 2.000  | $5.41 \times 10^{-4} \pm 2.60 \times 10^{-11}$  | $7.66 \times 10^{-7} \pm 3.52 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 1.000       | 4.000  | $1.16 \times 10^{-5} \pm 4.07 \times 10^{-13}$  | $6.68 \times 10^{-7} \pm 6.33 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |
| 1.000       | 8.000  | $3.87 \times 10^{-9} \pm 1.99 \times 10^{-16}$  | $4.91 \times 10^{-7} \pm 1.19 \times 10^{-12}$ | $1.000 \pm 0$                     | 5   |
| 2.000       | 0      | $9.76 \times 10^{-4} \pm 1.81 \times 10^{-7}$   | $2.13 \times 10^{-6} \pm 4.74 \times 10^{-10}$ | $1.000 \pm 0$                     | 5   |
| 2.000       | 0.500  | $9.90 \times 10^{-4} \pm 8.23 \times 10^{-11}$  | $2.10 \times 10^{-6} \pm 1.15 \times 10^{-12}$ | $1.000 \pm 0$                     | 5   |
| 2.000       | 1.000  | $0.001 \pm 1.43 \times 10^{-10}$                | $2.09 \times 10^{-6} \pm 1.19 \times 10^{-12}$ | $1.000 \pm 0$                     | 5   |
| 2.000       | 2.000  | $9.09 \times 10^{-4} \pm 6.64 \times 10^{-11}$  | $2.19 \times 10^{-6} \pm 2.68 \times 10^{-12}$ | $1.000 \pm 0$                     | 5   |
| 2.000       | 4.000  | $2.71 \times 10^{-4} \pm 1.30 \times 10^{-11}$  | $2.17 \times 10^{-6} \pm 3.48 \times 10^{-12}$ | $1.000 \pm 0$                     | 5   |
| 2.000       | 8.000  | $5.78 \times 10^{-6} \pm 2.03 \times 10^{-13}$  | $1.82 \times 10^{-6} \pm 9.90 \times 10^{-13}$ | $1.000 \pm 0$                     | 5   |

Table 21: Exact-factorization results for cycle teachers. Values are mean  $\pm$  sample standard deviation across seeds.

| Method           | Init relation | Raw init   | Rel. error                                    | ARI                               | Assignment acc.   | $n$ |
|------------------|---------------|------------|-----------------------------------------------|-----------------------------------|-------------------|-----|
| k-means          | neutral       | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| oracle           | neutral       | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| soft             | matched       | cycle      | $0.003 \pm 8.36 \times 10^{-4}$               | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| soft             | mismatched    | contiguous | $0.004 \pm 7.92 \times 10^{-4}$               | $0.636 \pm 0.158$                 | $0.725 \pm 0.142$ | 10  |
| soft             | neutral       | neutral    | $0.004 \pm 9.08 \times 10^{-4}$               | $0.894 \pm 0.171$                 | $0.925 \pm 0.121$ | 10  |
| soft             | random        | random     | $0.005 \pm 0.002$                             | $0.894 \pm 0.171$                 | $0.925 \pm 0.121$ | 10  |
| straight-through | matched       | cycle      | $0.004 \pm 0.001$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | mismatched    | contiguous | $0.009 \pm 0.025$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | neutral       | neutral    | $8.44 \times 10^{-4} \pm 9.05 \times 10^{-4}$ | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | random        | random     | $0.006 \pm 0.015$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| vertex-hard      | matched       | cycle      | $0.005 \pm 0.002$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| vertex-hard      | mismatched    | contiguous | $0.854 \pm 0.001$                             | $-0.222 \pm 2.93 \times 10^{-17}$ | $0.333 \pm 0$     | 10  |
| vertex-hard      | neutral       | neutral    | $4.93 \times 10^{-4} \pm 3.89 \times 10^{-5}$ | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| vertex-hard      | random        | random     | $0.475 \pm 0.271$                             | $0.523 \pm 0.333$                 | $0.783 \pm 0.172$ | 10  |

Table 22: Exact-factorization results for contiguous teachers. Values are mean  $\pm$  sample standard deviation across seeds.

| Method           | Init relation | Raw init   | Rel. error                                    | ARI                               | Assignment acc.   | $n$ |
|------------------|---------------|------------|-----------------------------------------------|-----------------------------------|-------------------|-----|
| k-means          | neutral       | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| oracle           | neutral       | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| soft             | matched       | contiguous | $0.003 \pm 8.04 \times 10^{-4}$               | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| soft             | mismatched    | cycle      | $0.004 \pm 6.87 \times 10^{-4}$               | $0.636 \pm 0.158$                 | $0.725 \pm 0.142$ | 10  |
| soft             | neutral       | neutral    | $0.005 \pm 0.001$                             | $0.929 \pm 0.150$                 | $0.950 \pm 0.105$ | 10  |
| soft             | random        | random     | $0.006 \pm 0.003$                             | $0.929 \pm 0.150$                 | $0.950 \pm 0.105$ | 10  |
| straight-through | matched       | contiguous | $0.004 \pm 0.001$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | mismatched    | cycle      | $0.009 \pm 0.025$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | neutral       | neutral    | $0.006 \pm 0.016$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | random        | random     | $0.004 \pm 0.005$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| vertex-hard      | matched       | contiguous | $0.005 \pm 0.002$                             | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| vertex-hard      | mismatched    | cycle      | $0.855 \pm 0.002$                             | $-0.222 \pm 2.93 \times 10^{-17}$ | $0.333 \pm 0$     | 10  |
| vertex-hard      | neutral       | neutral    | $5.21 \times 10^{-4} \pm 6.09 \times 10^{-5}$ | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| vertex-hard      | random        | random     | $0.492 \pm 0.220$                             | $0.525 \pm 0.297$                 | $0.792 \pm 0.153$ | 10  |

Table 23: Exact-factorization results for palindrome teachers. Values are mean  $\pm$  sample standard deviation across seeds.

| Method           | Init relation    | Raw init   | Rel. error                                    | ARI                              | Assignment acc.   | $n$ |
|------------------|------------------|------------|-----------------------------------------------|----------------------------------|-------------------|-----|
| k-means          | neutral          | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$                    | $1.000 \pm 0$     | 10  |
| oracle           | neutral          | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$                    | $1.000 \pm 0$     | 10  |
| soft             | neutral          | neutral    | $0.007 \pm 0.006$                             | $0.940 \pm 0.126$                | $0.967 \pm 0.070$ | 10  |
| soft             | random           | random     | $0.005 \pm 0.002$                             | $0.910 \pm 0.145$                | $0.950 \pm 0.081$ | 10  |
| soft             | structured-other | contiguous | $0.004 \pm 6.92 \times 10^{-4}$               | $0.970 \pm 0.095$                | $0.983 \pm 0.053$ | 10  |
| soft             | structured-other | cycle      | $0.004 \pm 7.73 \times 10^{-4}$               | $0.940 \pm 0.126$                | $0.967 \pm 0.070$ | 10  |
| straight-through | neutral          | neutral    | $9.97 \times 10^{-4} \pm 5.35 \times 10^{-4}$ | $1.000 \pm 0$                    | $1.000 \pm 0$     | 10  |
| straight-through | random           | random     | $0.046 \pm 0.128$                             | $0.969 \pm 0.098$                | $0.975 \pm 0.079$ | 10  |
| straight-through | structured-other | contiguous | $0.010 \pm 0.026$                             | $1.000 \pm 0$                    | $1.000 \pm 0$     | 10  |
| straight-through | structured-other | cycle      | $0.002 \pm 9.25 \times 10^{-4}$               | $1.000 \pm 0$                    | $1.000 \pm 0$     | 10  |
| vertex-hard      | neutral          | neutral    | $0.092 \pm 0.201$                             | $0.936 \pm 0.163$                | $0.958 \pm 0.106$ | 10  |
| vertex-hard      | random           | random     | $0.604 \pm 0.136$                             | $0.359 \pm 0.237$                | $0.708 \pm 0.132$ | 10  |
| vertex-hard      | structured-other | contiguous | $0.907 \pm 0.001$                             | $-0.243 \pm 0$                   | $0.333 \pm 0$     | 10  |
| vertex-hard      | structured-other | cycle      | $0.795 \pm 4.82 \times 10^{-4}$               | $0.139 \pm 2.93 \times 10^{-17}$ | $0.500 \pm 0$     | 10  |

Table 24: Exact-factorization results for random balanced teachers. Values are mean  $\pm$  sample standard deviation across seeds.

| Method           | Init relation    | Raw init   | Rel. error                                    | ARI                | Assignment acc.   | $n$ |
|------------------|------------------|------------|-----------------------------------------------|--------------------|-------------------|-----|
| k-means          | neutral          | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$      | $1.000 \pm 0$     | 10  |
| oracle           | neutral          | neutral    | $0 \pm 0$                                     | $1.000 \pm 0$      | $1.000 \pm 0$     | 10  |
| soft             | neutral          | neutral    | $0.005 \pm 0.002$                             | $0.858 \pm 0.183$  | $0.900 \pm 0.129$ | 10  |
| soft             | random           | random     | $0.007 \pm 0.003$                             | $0.856 \pm 0.254$  | $0.900 \pm 0.175$ | 10  |
| soft             | structured-other | contiguous | $0.004 \pm 9.89 \times 10^{-4}$               | $0.818 \pm 0.311$  | $0.875 \pm 0.212$ | 10  |
| soft             | structured-other | cycle      | $0.003 \pm 6.11 \times 10^{-4}$               | $0.965 \pm 0.112$  | $0.975 \pm 0.079$ | 10  |
| straight-through | neutral          | neutral    | $5.97 \times 10^{-4} \pm 2.50 \times 10^{-4}$ | $1.000 \pm 0$      | $1.000 \pm 0$     | 10  |
| straight-through | random           | random     | $0.105 \pm 0.209$                             | $0.919 \pm 0.173$  | $0.942 \pm 0.125$ | 10  |
| straight-through | structured-other | contiguous | $0.004 \pm 0.007$                             | $1.000 \pm 0$      | $1.000 \pm 0$     | 10  |
| straight-through | structured-other | cycle      | $0.002 \pm 0.001$                             | $1.000 \pm 0$      | $1.000 \pm 0$     | 10  |
| vertex-hard      | neutral          | neutral    | $5.00 \times 10^{-4} \pm 4.46 \times 10^{-5}$ | $1.000 \pm 0$      | $1.000 \pm 0$     | 10  |
| vertex-hard      | random           | random     | $0.493 \pm 0.279$                             | $0.497 \pm 0.343$  | $0.775 \pm 0.162$ | 10  |
| vertex-hard      | structured-other | contiguous | $0.796 \pm 0.061$                             | $-0.039 \pm 0.172$ | $0.475 \pm 0.125$ | 10  |
| vertex-hard      | structured-other | cycle      | $0.785 \pm 0.050$                             | $-0.019 \pm 0.127$ | $0.492 \pm 0.100$ | 10  |

Table 25: Exact-factorization results for imbalanced teachers. Values are mean  $\pm$  sample standard deviation across seeds.

| Method           | Init relation    | Raw init   | Rel. error                      | ARI                               | Assignment acc.   | $n$ |
|------------------|------------------|------------|---------------------------------|-----------------------------------|-------------------|-----|
| k-means          | neutral          | neutral    | $0 \pm 0$                       | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| oracle           | neutral          | neutral    | $0 \pm 0$                       | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| soft             | neutral          | neutral    | $0.006 \pm 0.004$               | $0.997 \pm 0.010$                 | $0.992 \pm 0.026$ | 10  |
| soft             | random           | random     | $0.004 \pm 0.002$               | $0.960 \pm 0.086$                 | $0.958 \pm 0.044$ | 10  |
| soft             | structured-other | contiguous | $0.004 \pm 9.57 \times 10^{-4}$ | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| soft             | structured-other | cycle      | $0.289 \pm 2.68 \times 10^{-5}$ | $0.413 \pm 0.028$                 | $0.617 \pm 0.043$ | 10  |
| straight-through | neutral          | neutral    | $0.008 \pm 0.013$               | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | random           | random     | $0.216 \pm 0.150$               | $0.698 \pm 0.275$                 | $0.800 \pm 0.181$ | 10  |
| straight-through | structured-other | contiguous | $0.030 \pm 0.050$               | $1.000 \pm 0$                     | $1.000 \pm 0$     | 10  |
| straight-through | structured-other | cycle      | $0.289 \pm 1.44 \times 10^{-4}$ | $0.440 \pm 0.017$                 | $0.658 \pm 0.026$ | 10  |
| vertex-hard      | neutral          | neutral    | $0.662 \pm 0.070$               | $0.090 \pm 0.101$                 | $0.458 \pm 0.044$ | 10  |
| vertex-hard      | random           | random     | $0.658 \pm 0.064$               | $0.078 \pm 0.098$                 | $0.500 \pm 0.039$ | 10  |
| vertex-hard      | structured-other | contiguous | $0.722 \pm 0.003$               | $-0.031 \pm 7.31 \times 10^{-18}$ | $0.500 \pm 0$     | 10  |
| vertex-hard      | structured-other | cycle      | $0.696 \pm 0.002$               | $0.026 \pm 0$                     | $0.417 \pm 0$     | 10  |

Table 26: Full functional-distillation results by architecture, teacher pattern, and raw initialization. Values are mean  $\pm$  sample standard deviation.

| Student / init relation                         | Raw init           | Test loss                                     | ARI                | $n$ |
|-------------------------------------------------|--------------------|-----------------------------------------------|--------------------|-----|
| MLP: contiguous / matched                       | pattern_contiguous | $0.006 \pm 8.56 \times 10^{-5}$               | $1.000 \pm 0$      | 5   |
| MLP: contiguous / mismatched                    | pattern_cycle      | $0.006 \pm 7.00 \times 10^{-5}$               | $-0.167 \pm 0$     | 5   |
| MLP: contiguous / neutral                       | neutral            | $0.012 \pm 1.99 \times 10^{-4}$               | $0.283 \pm 0.082$  | 5   |
| MLP: cycle / matched                            | pattern_cycle      | $0.005 \pm 6.21 \times 10^{-5}$               | $1.000 \pm 0$      | 5   |
| MLP: cycle / mismatched                         | pattern_contiguous | $0.006 \pm 7.65 \times 10^{-5}$               | $-0.167 \pm 0$     | 5   |
| MLP: cycle / neutral                            | neutral            | $0.012 \pm 1.85 \times 10^{-4}$               | $-0.226 \pm 0.056$ | 5   |
| MLP: palindrome / neutral                       | neutral            | $0.011 \pm 2.24 \times 10^{-4}$               | $-0.077 \pm 0.062$ | 5   |
| MLP: palindrome / structured_other              | pattern_contiguous | $0.005 \pm 1.27 \times 10^{-4}$               | $-0.189 \pm 0$     | 5   |
| MLP: palindrome / structured_other              | pattern_cycle      | $0.005 \pm 9.60 \times 10^{-5}$               | $0.075 \pm 0$      | 5   |
| MLP: random_balanced / neutral                  | neutral            | $0.012 \pm 1.91 \times 10^{-4}$               | $-0.053 \pm 0.130$ | 5   |
| MLP: random_balanced / structured_other         | pattern_contiguous | $0.006 \pm 9.98 \times 10^{-5}$               | $-0.108 \pm 0.130$ | 5   |
| MLP: random_balanced / structured_other         | pattern_cycle      | $0.006 \pm 6.30 \times 10^{-5}$               | $-0.050 \pm 0.261$ | 5   |
| Transformer: contiguous / matched               | pattern_contiguous | $4.03 \times 10^{-4} \pm 9.10 \times 10^{-5}$ | $1.000 \pm 0$      | 5   |
| Transformer: contiguous / mismatched            | pattern_cycle      | $4.09 \times 10^{-4} \pm 9.89 \times 10^{-5}$ | $-0.167 \pm 0$     | 5   |
| Transformer: contiguous / neutral               | neutral            | $5.24 \times 10^{-4} \pm 1.30 \times 10^{-4}$ | $0.205 \pm 0.050$  | 5   |
| Transformer: cycle / matched                    | pattern_cycle      | $3.90 \times 10^{-4} \pm 6.52 \times 10^{-5}$ | $1.000 \pm 0$      | 5   |
| Transformer: cycle / mismatched                 | pattern_contiguous | $4.04 \times 10^{-4} \pm 7.90 \times 10^{-5}$ | $-0.167 \pm 0$     | 5   |
| Transformer: cycle / neutral                    | neutral            | $4.98 \times 10^{-4} \pm 9.52 \times 10^{-5}$ | $-0.032 \pm 0.122$ | 5   |
| Transformer: palindrome / neutral               | neutral            | $4.99 \times 10^{-4} \pm 8.18 \times 10^{-5}$ | $-0.091 \pm 0.032$ | 5   |
| Transformer: palindrome / structured_other      | pattern_contiguous | $4.20 \times 10^{-4} \pm 6.29 \times 10^{-5}$ | $-0.189 \pm 0$     | 5   |
| Transformer: palindrome / structured_other      | pattern_cycle      | $3.93 \times 10^{-4} \pm 5.14 \times 10^{-5}$ | $0.075 \pm 0$      | 5   |
| Transformer: random_balanced / neutral          | neutral            | $4.72 \times 10^{-4} \pm 1.11 \times 10^{-4}$ | $0.007 \pm 0.159$  | 5   |
| Transformer: random_balanced / structured_other | pattern_contiguous | $3.91 \times 10^{-4} \pm 7.13 \times 10^{-5}$ | $-0.108 \pm 0.130$ | 5   |
| Transformer: random_balanced / structured_other | pattern_cycle      | $3.76 \times 10^{-4} \pm 7.10 \times 10^{-5}$ | $-0.050 \pm 0.261$ | 5   |

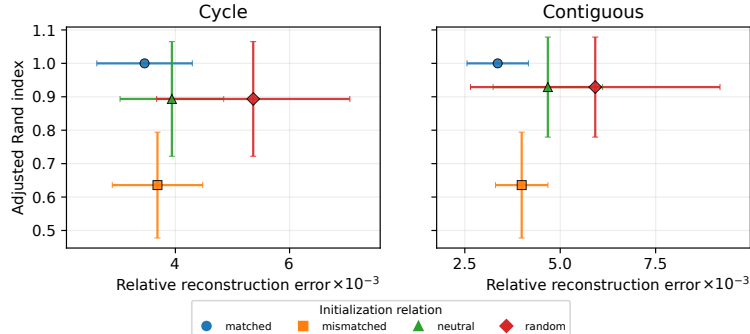


Figure 4: Soft-router exact-factorization runs for cycle and contiguous teachers. Reconstruction error and planted-graph recovery are separate axes.

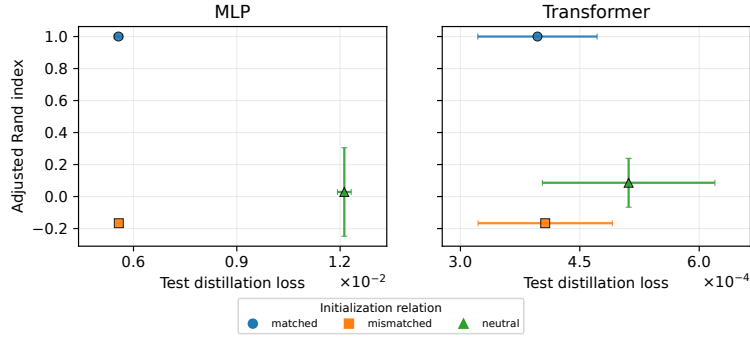


Figure 5: Distillation loss versus planted-graph ARI for the residual MLP and causal Transformer sanity checks.

Table 27: Paired probe recovery aggregated over probe counts, fixed additive-noise levels, graphs, and seeds. Functional modes share inputs, noise, and projection. The visible-effective-parameter baseline uses neither observation noise nor projection and is not a test of functional equivalence. Values are mean  $\pm$  sample standard deviation.

| Architecture | Signature           | ARI               | Empirical gap                   | $n$  |
|--------------|---------------------|-------------------|---------------------------------|------|
| MLP          | shared input        | $0.676 \pm 0.432$ | $-27.192 \pm 66.232$            | 1680 |
| MLP          | native              | $0.675 \pm 0.434$ | $-27.293 \pm 66.173$            | 1680 |
| MLP          | effective parameter | $1.000 \pm 0$     | $0.008 \pm 5.82 \times 10^{-5}$ | 1680 |
| Transformer  | shared input        | $0.501 \pm 0.471$ | $-88.706 \pm 190.527$           | 1680 |
| Transformer  | native              | $0.501 \pm 0.470$ | $-88.758 \pm 190.490$           | 1680 |
| Transformer  | effective parameter | $1.000 \pm 0$     | $0.009 \pm 2.08 \times 10^{-5}$ | 1680 |

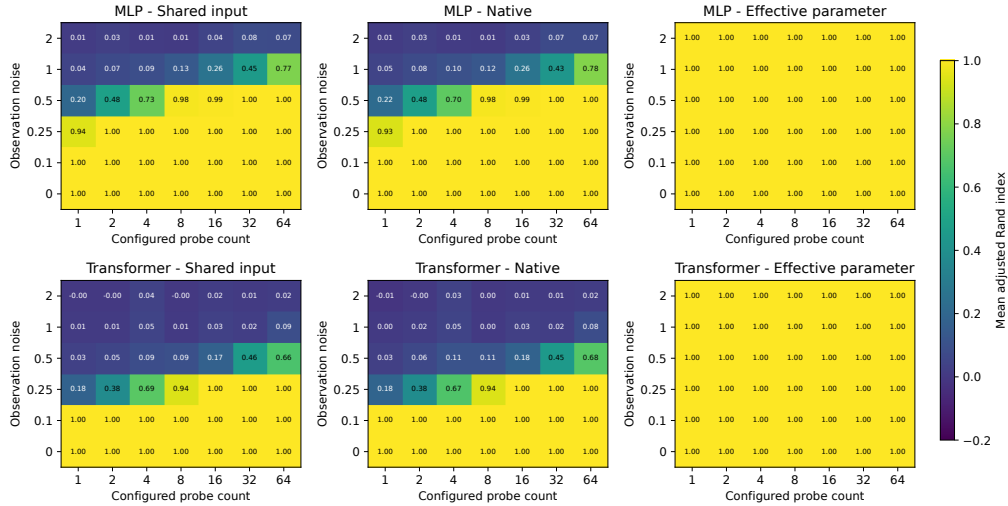


Figure 6: Shared-input, native-signature, and visible-effective-parameter recovery as probe count and fixed additive observation noise vary. The parameter baseline is repeated across the grid only for comparison and is not a functional-equivalence test. This is a planted-partition sanity check, not evidence of superiority on natural systems.

Functional-signature recovery generally improves with probe count and degrades with observation noise. The empirical within/between gap is computed against the planted partition, rather than against the predicted clusters, but remains only a finite-sample diagnostic unless the theorem’s population gap and uniform concentration assumptions are justified for this observation model.

## N LANGUAGE-MODEL STABILITY RESULTS

Table 28: Language-model stress test. BPB is estimated from 100 seeded, sampled test batches. Values are mean  $\pm$  sample standard deviation across seeds; pairwise ARI descriptively summarizes the three dependent run pairs.

| Initialization     | Sampled test BPB  | L1 movement                                   | Pairwise ARI       | $n$ |
|--------------------|-------------------|-----------------------------------------------|--------------------|-----|
| neutral            | $2.415 \pm 0.009$ | $0.012 \pm 0.003$                             | $0.099 \pm 0.157$  | 3   |
| pattern_contiguous | $2.111 \pm 0.008$ | $3.10 \times 10^{-4} \pm 1.35 \times 10^{-5}$ | $1.000 \pm 0$      | 3   |
| pattern_cycle      | $2.181 \pm 0.011$ | $3.50 \times 10^{-4} \pm 4.60 \times 10^{-5}$ | $1.000 \pm 0$      | 3   |
| random             | $2.319 \pm 0.015$ | $0.065 \pm 0.008$                             | $-0.045 \pm 0.112$ | 3   |

Distinct structured initializations can retain their partitions while reaching competitive BPB. With no privileged natural-language graph, this is evidence of initialization dependence, not evidence that either retained partition is false or historically correct. The one-of-three partition-fold witness in Table 29 is reported separately and does not convert this stability arm into a prevalence result.

## O REPRODUCIBILITY CHECKLIST

The response summary validates source hashes, checkpoint hashes, rank assumptions, analytic/native formula agreement, the permutation control, and all fixed probe outcomes. The separate tomography summary validates three newly generated raw artifacts, exact-gauge/common-design gates, rank and dimension conditions, component reconstruction, and the permutation and non-permutation controls without changing the older response generator. The ASLoRA summary validates a common protocol hash, generation-time source hashes, complete declared candidate coverage, the canonical equivalence gate, and exact restoration for every seed and gauge. The router-decision summary retains both exhausted searches, their full funnel counts, checkpoint hashes, and the paired successful-seed evaluation; unlike the other three summaries, it does not itself contain a generation-source-hash attestation, so we do not claim equal provenance strength.

The artifact additionally includes deterministic seed handling, raw JSON records, executable summarizers, the script that generates the three original audit LaTeX tables and their appendix details from validated summaries, tests for summary-gate failures, and unit tests for causal masking, permutation-invariant metrics, gauge equivalence, response formulas, structured response reconstruction and its approximate-product rejection boundary, restoration, and probe recovery. Dataset download is separated from training and accesses no private data.

## P PROSPECTIVE ENHANCEMENT PROTOCOLS AND NUMERICAL LIMITS

### P.1 HISTORICAL BYTE-LM WITNESS

Table 29: Finite same-group-budget partition-fold search at trained, initialization-dominated byte-LM checkpoints. “Not found” means the predeclared finite search exhausted; it is not a non-existence claim. Fold BPB compares partitions refit from the same effective tensors. The sole measurable consequence was not replicated under the same search (one witness in three seeds).

| Seed | Search    | Fold BPB ref. $\rightarrow$ cand. | $\Delta$ BPB [95% CI]   | Replicated? |
|------|-----------|-----------------------------------|-------------------------|-------------|
| 0    | found     | 2.175 $\rightarrow$ 4.815         | +2.640 [+2.629, +2.651] | no (1/3)    |
| 1    | not found | —                                 | —                       | no (1/3)    |
| 2    | not found | —                                 | —                       | no (1/3)    |

The original fixed-budget search succeeds at one of three checkpoints; all three routers are initialization-dominated. The successful candidate changes group sizes from 3/3/3/3 to 5/3/3/1, so its large BPB difference does not isolate membership from balance. Failed searches store no

downstream fold or baseline evaluation. The detailed historical protocol and conditional interval interpretation remain in Appendix K. The new equal-size experiments below are separately registered evidence.

**Training and matched decision budgets.** The extension uses fresh seeds 10–12, width 128, MLP width 512, 12 layers, four bases, eight heads, context 128 and batch size 16. AdamW uses base learning rate  $3 \times 10^{-4}$ , router multiplier ten, 300 warmup steps, cosine decay to one tenth, weight decay 0.1 for weights and zero for routers, and BF16 autocast. There is no routing regularizer or temperature annealing. Only the fixed final step 6,000 is audited; a separate 100-step seed-90 validation-only pilot checked implementation and timing. This configuration differs from the original checkpoints and is not a controlled comparison of training lengths. Router movement and validation curves do not alone prove convergence or task-essential routing.

The positive-stochastic and signed-local families retain the previously specified distributions and condition-number bound 30; the new legality gate requires identical sorted group sizes. Selection sees router probabilities only. Test evaluation uses 64 disjoint batches of eight 128-token sequences, with identical stored starts for every action. Recovery uses 500 updates at learning rate  $10^{-4}$ , fresh AdamW state, weight decay 0.1 and clipping norm one, evaluated at steps 0,100,500. The hard routing assignments are frozen; inactive stored slots are retained, so no latency or realized storage saving is claimed. The fixed-size effective-parameter comparator uses 16 seeded restarts and at most 50 improving pair swaps per restart to minimize centroid SSE. It is a local search, not a global optimum or a task-loss optimizer. Unconstrained Ward is included descriptively, since its group sizes may differ.

**Natural-gradient measurement.** Each checkpoint contributes 16 training minibatches of two 32-token sequences. The effective-weight forward matches the native model, and its gradients, pulled back through the factor map, match the native factor gradients. For numerical conditioning, each direction and its measured response receive the same normalization. The complement Gram and local linear systems in Appendix C.2 use a fixed relative pseudoinverse cutoff  $10^{-12}$ . Factor recovery retains the original fixed diagonalization weights. All ranks, spectra, component errors, held-out predictions, unsuccessful low-probe reconstructions and finite-step measurements are stored. The last four directions are held out for every  $q$ ; larger fits are nested, so these measurements are not independent training replicates.

**Joint noise and degeneracy.** A prospective 450-cell FP64 synthetic grid uses ten seeds with  $L = 6$ ,  $K = 3$ ,  $D = 16$ , interior factors and contractions toward router rank loss or simplex vertices. Relative product and response noise levels are 0,  $10^{-8}$ ,  $10^{-6}$ ,  $10^{-4}$ ,  $10^{-2}$ ; common noise directions couple levels within each cell. The observed product is truncated to rank  $K$ , its row space supplies the designed probes, and the true response to those probes is measured before adding response noise. This tests geometry estimation as well as response error. All runs return factors, but this does not imply accurate or feasible recovery: near rank loss, relative factor errors can exceed 60 already at  $10^{-6}$  noise. Corollary 5 requires valid, uniformly nondegenerate factors and does not certify these unconstrained outputs.

**Artifacts and scope.** Versioned protocols and source-bound raw records are supplied under `research/` and `results/enhancement/`, including checkpoints, seeds and corpus hashes. Each seed’s paired bootstrap uses 10,000 resamples of its evaluation blocks; intervals are conditional and do not substitute for independent training seeds. The new evidence supplements, rather than overwrites, the original raw records and their negative findings.

## Q FROZEN NOISY-RECOVERY AND REJECTION STUDY

**Prospective splits and observations.** The local protocol and source freeze are `research/TRUSTED_RECOVERY_PROTOCOL.md` and `research/TRUSTED_RECOVERY_FREEZE.json`. They are timestamped execution records, not an independent preregistration. Development uses synthetic seeds 0–3 and existing byte-LM seed 10. Calibration uses synthetic seeds 100–119 and newly trained byte-LM seeds 20–21; evaluation uses synthetic seeds 200–239 and new byte-LM seeds 22–25. The six fresh checkpoints use the

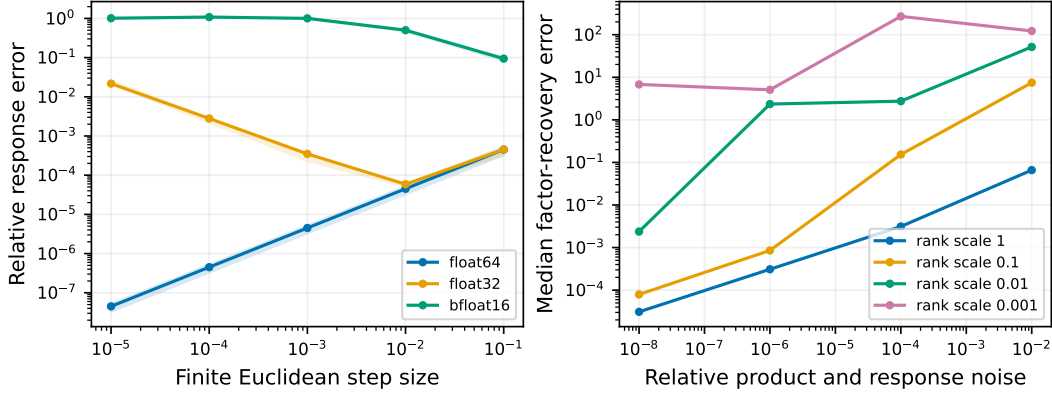


Figure 7: Two numerical limits. Left: finite-step response error at the three new checkpoints; curves show medians and shading the observed seed range. Smaller steps reduce FP64 truncation error but amplify low-precision subtraction error. Right: median factor error over ten synthetic seeds as both product and response observations are perturbed. The rank scale contracts the router toward a rank-one uniform matrix; it is not a noise level. These finite grids are diagnostic rather than worst-case guarantees.

unchanged width-128,  $L = 12$ ,  $K = 4$ , 6000-step configuration of the preceding controlled training study. There is no checkpoint selection on recovery outcomes.

Synthetic factors have  $L = 6$ ,  $K = 3$ ,  $D = 16$ , random-normal bases, and four router regimes: interior softmax, contractions toward the uniform router by factors 0.1 and 0.01, and a contraction toward balanced one-hot rows by 0.01. Each instance supplies 16 normalized Gaussian gradient directions. Language directions are actual effective-parameter cross-entropy gradients from 16 training-split minibatches of size  $2 \times 32$ , sampled with fixed seed offset 3100000. Four or twelve initial directions fit the estimator; the last four assess held-out response fit. Response observations come from automatic differentiation of the factor map, without calling its closed-form response formula.

Product and each response receive independently drawn Gaussian perturbation directions rescaled to relative Frobenius norms  $\sigma \in \{0, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}\}$ . Directions are shared across noise levels within an instance. The supplied bound  $\sigma$  is known; the true perturbation vector and factors are not supplied to recovery. The estimated rank- $K$  product row subspace is used by all methods, and evaluation includes basis error outside it. This observation-noise experiment does not model biased, adversarial, quantized, or unknown optimizer-state errors.

**Fixed estimators and observable scores.** The projected spectral baseline clips router entries below  $10^{-8}$ , normalizes rows, and refits the basis by least squares. Joint fitting uses simplex logits in orthonormal zero-sum coordinates and normalized basis coordinates, with SciPy trust-region reflective least squares and exact forward-mode Jacobians. Three starts are fixed: projected spectral, a 0.1 logit perturbation, and a random simplex router. Each has a budget of 80 function evaluations; the smallest observed training residual selects the output. There is one developed solver variant, with no tuning on calibration or evaluation factors.

All acceptance rules require a numerically feasible router and finite score. The combined score is the maximum of (i) training plus held-out residual plus twice the normalized noise budget, divided by the simplex-tangent Jacobian’s smallest singular value; (ii) an observed product-subspace perturbation ratio; and (iii) permutation-aligned disagreement among competitive starts for the joint fit. A separately disclosed  $10^{-10}$  numerical floor enters the empirical noise budget. The comparison scores use only residual or inverse Jacobian singular value. Their implementation and thresholds are frozen, and no score uses truth at prediction time.

For each method/score pair, one threshold is chosen on pooled calibration data to maximize accepted count, subject to at least 20 accepted synthetic outputs and at most 5% bad among all accepted outputs. Ties are accepted together. An infeasible threshold would reject all and be reported as a

Table 30: Synthetic held-out diagnostic comparison (percent). Coverage is the accepted fraction; bad rejection is recall on errors  $> 0.10$ ; good rejection is false rejection on errors  $\leq 0.05$ ; accepted bad is the bad fraction among accepted outputs. Each threshold was frozen using calibration only.

| Method    | Score     | Coverage | Bad reject | Good reject | Accepted bad |
|-----------|-----------|----------|------------|-------------|--------------|
| Spectral  | Combined  | 10.21    | 100.00     | 85.04       | 0.00         |
| Spectral  | Residual  | 10.21    | 100.00     | 85.04       | 0.00         |
| Spectral  | Condition | 10.26    | 100.00     | 84.96       | 0.00         |
| Projected | Combined  | 72.71    | 90.11      | 0.00        | 4.01         |
| Projected | Residual  | 72.66    | 92.23      | 0.00        | 3.15         |
| Projected | Condition | 21.30    | 98.06      | 70.48       | 2.69         |
| Joint fit | Combined  | 92.81    | 59.47      | 0.18        | 5.16         |
| Joint fit | Residual  | 91.51    | 70.04      | 0.18        | 3.87         |
| Joint fit | Condition | 24.32    | 91.63      | 72.73       | 4.07         |

failure. The same threshold transfers to both evaluation domains. Good errors are at most 0.05, bad errors exceed 0.10, and intermediate errors are reported separately. This empirical calibration is not a distribution-free risk guarantee.

**Results, including unsuccessful diagnostics.** All 1920 synthetic joint fits return; there are 1639 good, 54 intermediate, and 227 bad outputs. Projected spectral recovery has three failed outputs, all retained in the 1920-case denominator. Seventy selected joint-fit runs exhaust their 80-evaluation budget; a returned estimate is not a convergence claim. Competitive-start agreement is also not a uniqueness test, especially if only one start meets the residual cutoff. Joint fitting improves error in 1916 of the 1917 paired successful synthetic comparisons and all 48 language comparisons. This comparison counts correlated variants, not independent replications.

The combined joint-fit diagnostic rejects 59.47% of bad synthetic outputs, with 0.18% false rejection of good outputs and 92.81% coverage. Seed-cluster bootstrap 95% intervals, resampling the 40 independent seeds and retaining all their variants, are respectively [54.79%, 63.82%], [0, 0.49%], and [92.08%, 93.54%]. Residual-only rejection instead rejects 70.04% of bad outputs with the same good false-rejection rate, at 91.51% coverage. Condition-only rejection discards 72.73% of good outputs. The raw spectral method’s feasibility gate rejects almost all noisy candidates: its high rejection recall is not useful broad coverage.

Failures concentrate near router rank collapse. At contraction 0.01, joint fitting has 165/480 bad outputs; the combined score accepts 352, including 37 bad outputs. At contraction 0.1, it accepts 473 of 480, including 55 bad outputs. Thus aggregate calibration masks substantial regime-specific error rates. Interior and near-boundary regimes have no errors above 0.10 on this grid. The largest joint-fit error is 10.54; the largest falsely accepted error, 0.820, occurs at seed 210, contraction 0.01,  $\sigma = 10^{-3}$ ,  $q = 4$ . These cases remain in the released records and were not used to retune the primary thresholds.

All 48 language joint fits are good and accepted. Maximum error at seeds 22, 23, 24, and 25 is respectively 0.00752, 0.01063, 0.01002, and 0.00650. There are only four independent training runs, and no bad joint-fit output in this cohort: bad-output rejection recall is undefined, not demonstrated to be perfect. The projected baseline has one bad language output and accepts it. The condition-only joint-fit rule rejects every language output. An all-reject outcome is explicitly uninformative.

**Numerical local separation and reproducibility.** The 140 synthetic local passes occur in the interior (60) and near-boundary (80) regimes; there are none in the two rank-contracted regimes. Only two language outputs pass, both at seed 22. The pass concerns an estimated-subspace feasible component and has no unconditional truth-error interpretation. Raw observations, estimates, starts, failures, diagnostics, source hashes, calibration thresholds, and per-seed summaries are released under `results/trusted_recovery/`. Full-coordinate theorem checks are distinct from the recovery experiment. A separate CPU environment replays the frozen observations; its results are recorded separately because floating-point optimization need not be bitwise identical. Across all 1968 cases, good/intermediate/bad classifications agree for every method. Combined-score acceptance changes in four spectral and two joint-fit cases; the largest joint-fit error difference is 0.0171. Thus environment

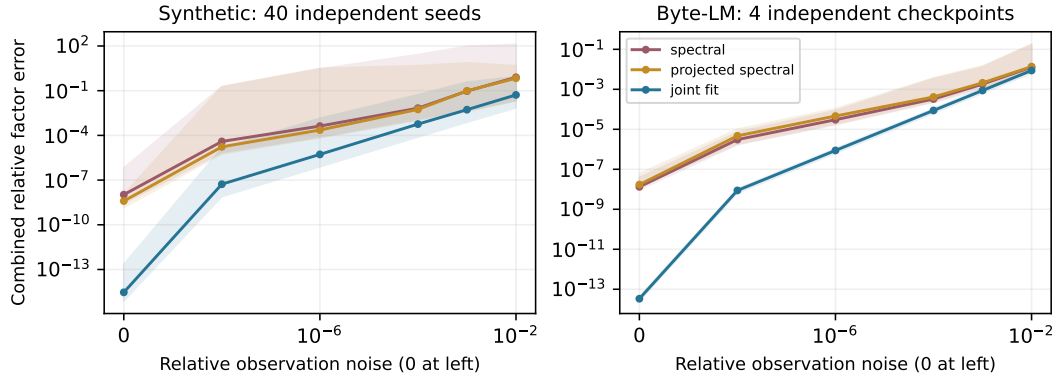


Figure 8: Held-out recovery under simultaneous product and response noise. Lines show medians and shading the 10th–90th percentiles of the fixed grid, not confidence intervals. Synthetic aggregation includes all four regimes and both probe counts; language aggregation includes four checkpoints and both probe counts. Zero noise is placed at the leftmost plotting coordinate. Tail failures and diagnostic errors are reported separately.

agreement is assessed explicitly, not inferred from tests. The release verification rebuilds the reported tables and checks source and threshold bindings.